

PNP-SPT Platform

A Specialized Personnel Training Platform for the Philippine
National Police

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Systems Analysis and Detailed Design for IT and CS
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Executive Summary

The Philippine National Police (PNP) faces a significant operational challenge in equipping its nationwide workforce of approximately 250,000 personnel with foundational Artificial Intelligence (AI) literacy. The organization's existing Police Officer Academy (POA) e-learning platform, while effective for course enrollment and examinations, was not designed to support the highly interactive, multimedia-driven training environment that effective AI literacy instruction requires. Compounding this, the traditional classroom-based instruction currently used to deliver AI training is logistically unsustainable at the scale demanded by a nationwide law enforcement workforce.

This documentation presents the planning and design phase of the PNP Specialized Personnel Training (PNP-SPT) Platform, a proposed interoperable web application built on Laravel 11, PHP 8.2, and MySQL, matching the existing POA's technology stack to support future integration between the two systems upon handover. Rather than replacing the POA, the PNP-SPT Platform is designed to operate independently, maintaining its own dedicated login and registration system and database, with its data scope limited strictly to training-related records.

The platform centers on a standardized 40-hour Program of Instruction, divided into seven self-paced, multimedia-driven modules that replace static content delivery with dynamic, interactive learning. An approved third-party AI API is integrated directly into the learning environment as a real-time virtual tutor, providing personnel with contextual guidance restricted to approved educational materials, without transmitting any personally identifiable information to the external provider. Upon completing and passing each module's final assessment, personnel automatically receive a scannable, verifiable digital certificate, eliminating the manual administrative overhead of tracking training completion.

The system is designed around four user roles: PNP Personnel, who complete the training curriculum; Supervisors, who manage training content and monitor workforce progress; Admins, who maintain platform security, account access, and the AI integration's technical health; and the AI Chatbot, which supports learners throughout their coursework.

This iteration of the project encompasses the requirements analysis, use case modeling, prototype wireframes, feasibility study, and product backlog needed to guide development. Supporting deliverables include a Project Lean Canvas, a SWOT analysis, a defined release plan, and documented consultations with the PNP ITMS Office that shaped the project's final scope — particularly its pivot toward an interoperable, rather than standalone, architecture in response to the technical and policy constraints identified by the client. The next phase of the project will proceed with actual platform development, guided by the team's eight-stage product roadmap and conducted in coordination with the PNP ITMS Office.

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Introduction

Project Context

The Philippine National Police (PNP) requires a high degree of technological literacy, specifically in Artificial Intelligence (AI), to optimize field operations and institutional governance. The organization faces the massive operational challenge of training approximately 250,000 personnel on these emerging tools.

Currently, the PNP utilizes the Police Officer Academy (POA) e-learning platform for course enrollment and exams. However, this legacy system is not designed to handle the highly interactive environments required for effective AI training such as a real-time AI-powered virtual tutor that learners can consult mid-lesson for immediate, context-aware guidance, multimedia-driven modules that replace static PDF and PowerPoint materials, and per-topic knowledge checks that assess understanding progressively rather than through a single end-of-course exam. Furthermore, relying on traditional classroom-based instruction to bridge this gap is logistically unsustainable at the scale required for the entire force.

To navigate these constraints, this project proposes the development of the PNP Specialized Personnel Training (PNP-SPT) Platform. Rather than functioning as a replacement for the existing POA platform, the PNP-SPT Platform will be developed as an interoperable web application built on Laravel 11, PHP 8.2, and MySQL, matching the existing POA technology stack to support future integration between the two systems. The platform will maintain its own dedicated user registration and database, allowing PNP personnel to create accounts directly on the platform to access training. This design keeps the PNP-SPT Platform's data scope limited to training-related records only, without touching or storing sensitive PNP personnel or intelligence data managed by the POA. By leveraging approved third-party AI APIs, the platform will replace static modules with dynamic, self-paced learning tracks, culminating in the automated generation of scannable, verifiable certificates to officially track personnel upskilling.

Statement of the Problem

This study addresses the critical challenges facing the Philippine National Police (PNP) in deploying scalable and effective AI literacy training to its nationwide workforce of over 250,000 personnel. Specifically, this study focuses on the following problems:

1. Mass training deployment is logistically unsustainable, as traditional classroom-based instruction cannot be scaled to meet the demands of a nationwide law enforcement workforce.
2. The existing Police Officer Academy (POA) platform lacks the interactive capabilities required for effective AI literacy training, relying on static content delivery that does not support modern, dynamic learning environments.
3. The PNP's existing AI training materials are designed for classroom delivery and are not structured for a self-paced, interactive digital learning environment.

4. PNP Personnel have no access to real-time interactive guidance during training, limiting their ability to clarify concepts and engage deeply with the course material.
5. Tracking training completion and issuing certificates manually for a large-scale workforce creates significant administrative overhead and institutional bottlenecks.

Objectives

The main objective of this study is to develop the PNP Specialized Personnel Training (PNP-SPT) Platform, an interoperable web application designed to deliver standardized, interactive AI literacy training scalable to the Philippine National Police's full workforce of approximately 250,000 personnel.

Specifically, this study aims to:

1. To develop a self-paced, web-based training platform that eliminates dependence on classroom-based instruction, designed to scale to the PNP's full 250,000-personnel workforce, with core platform functionality completed and validated through pilot testing.
2. To design and implement an interactive, multimedia-driven training interface across all 7 curriculum modules, replacing static content delivery, with each module's functionality validated through documented test cases (e.g., lesson progression, AI assistant guidance, and progress preservation) achieving a 100% pass rate prior to deployment.
3. To research and finalize a standardized 40-hour Program of Instruction divided into 7 progressive modules, adapting existing PNP AI training materials, with the finalized curriculum structure submitted to the client for formal validation and approval prior to platform handover.
4. To integrate an approved third-party AI API as a real-time virtual tutor across all 7 modules, with all test cases confirming zero transmission of personally identifiable information to the external provider and correct decline of out-of-scope questions.
5. To implement an automated progress-tracking and certificate-generation system that issues a scannable, verifiable certificate immediately upon a personnel member passing a module's final assessment, with full test-case coverage confirming correct function across all 7 modules.

Significance of the Project

The implementation of the PNP Specialized Personnel Training (PNP-SPT) Platform provides substantial benefits to several key groups within the organization:

1. **PNP Personnel.** These individuals will gain access to a highly interactive, self-paced training platform equipped with a dedicated AI learning assistant. This enables them to

efficiently complete the standardized 40-hour AI curriculum, directly improving their technological literacy and modern operational readiness without the limitations of traditional classroom schedules.

2. **PNP Administrators and IT Department.** Administrators will significantly reduce their manual workload through the system's automated progress tracking and verifiable certificate generation tools. Furthermore, since the SPT Platform is built using the same technology stack as the existing POA platform, PNP's IT team retains the option to integrate or synchronize the two systems in the future without requiring a full redevelopment, reducing long-term maintenance overhead.
3. **Philippine National Police.** The organization as a whole will achieve massive scalability in its training deployment while adhering to ICTMS policies and the Data Privacy Act of 2012. The platform's dedicated training-records database and integration of an approved third-party AI assistant ensure that the force is rapidly upskilled in modern technology, with the system's data scope limited strictly to training-related records and never extending to sensitive PNP personnel or intelligence data.

Furthermore, this solution directly aligns with and helps achieve several United Nations Sustainable Development Goals (SDGs):

Contributes to:

- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure
- SDG 16: Peace, Justice and Strong Institutions

Indirectly aids:

- SDG 4: Quality Education
- SDG 17: Partnerships for the Goals

Scope and Limitations

The scope of this study focuses on the development of the PNP Specialized Personnel Training (PNP-SPT) Platform, an interactive web application designed specifically for the 250,000 personnel of the Philippine National Police. The training content is strictly standardized into a 40-hour Program of Instruction encompassing seven core modules. The project will include a dedicated user registration and account system, the integration of an approved third-party AI assistant to guide users through the coursework, and the development of an automated, scannable, verifiable certification system upon completion.

There are notable boundaries to this project. The SPT Platform will not replace the existing Police Officer Academy (POA) e-learning platform, and the two systems will operate independently upon handover, with the PNP IT Department retaining the option to pursue further

integration in the future. The platform's database will contain only training-related records created through its own registration process; it will not store, host, or process sensitive PNP personnel records or intelligence data managed by the POA. The scope of the AI integration is limited solely to educational guidance within the standardized curriculum and does not extend to live operational AI deployment for the PNP.

Review of Related Literature / Systems

This chapter presents an analysis of existing systems and platforms that serve as the foundational benchmarks for this research. By examining these adjacent works, the study identifies key functionalities, User Interface (UI) configurations, and User Experience (UX) strategies that inform the development of the proposed output. Specifically, this review focuses on different educational web applications to establish a standard for usability and learning impact. The systems reviewed are categorized thematically into role-based architectures, gamified environments, artificial intelligence integration, and large-scale organizational deployment to highlight their relevance to the proposed Philippine National Police (PNP) platform.

United Nation Inspira LMS

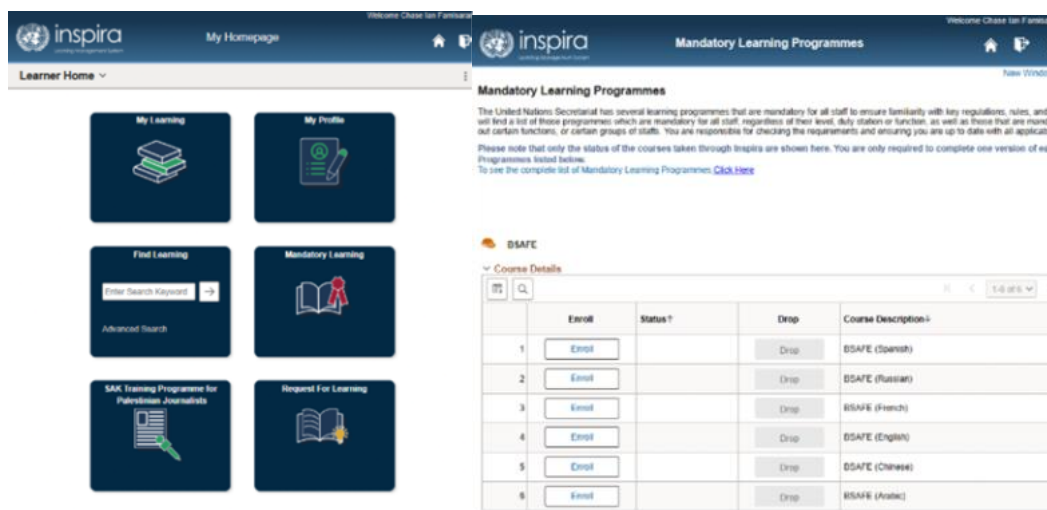


Figure 1: United Nation Inspira LMS

The United Nations Inspira serves as one of the primary Learning Management Systems (LMS) for the UN's global workforce and partners. It is a robust platform designed to manage mandatory training, professional development, and compliance tracking for hundreds of thousands of users. The system organizes courses into mandatory and elective modules, providing a blueprint for the curriculum organization of the proposed system.

A defining characteristic of the UN Inspira LMS is its implementation of Structured Learning Paths. Inspira utilizes mandatory prerequisites where learning progression is sequential, requiring users to complete foundational content before advancing to subsequent modules. [2]

Boot.dev



Figure 2: Boot.dev

Boot.dev is an online learning platform that uses gamified and interactive environment. Where there is a interactive sandbox to encourage "learning-by-doing," designed the platform like a Role-Playing Game (RPG) to maintain high student engagement and completion rates. The platform implements an engagement system where learners earn Experience Points (XP), gain levels, and unlock achievements.

This gamification strategy serves as a potential reference for the research, particularly in motivating users to complete challenging modules. Furthermore, a core technical feature of Boot.dev utilizes a browser-based coding development environment is integrated, which allows users to write and test code directly in the browser without local setup. This serves an providing instant feedback and reducing technical barriers for beginner users. By framing the learning journey as a "quest" with a character-driven narrative, Boot.dev creates a unique User Experience (UX) that reduces "tutorial fatigue," informing one way to visually show progress in this research. [3]

IBM z Xplore learning platform

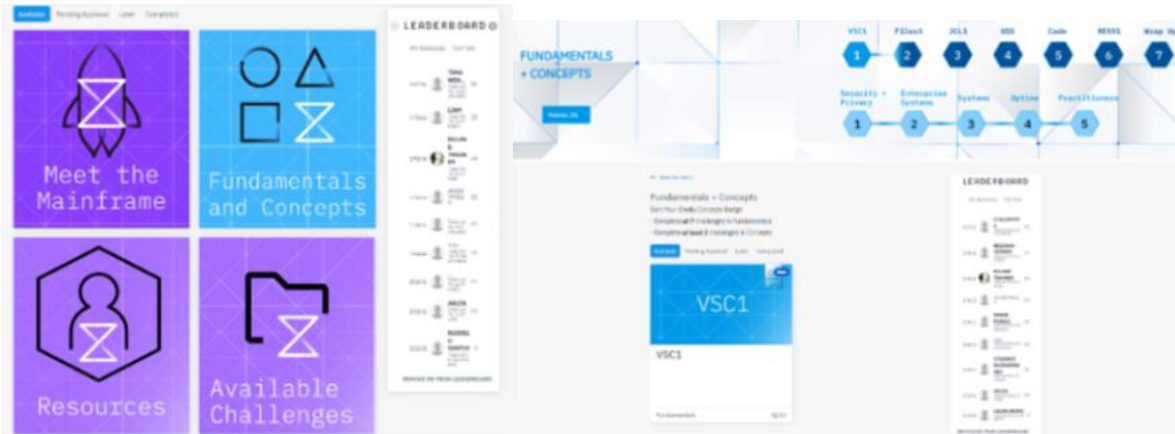


Figure 3: IBM z Xplore learning platform

IBM z Xplore is a learning platform designed to provide hands-on experience with enterprise computing and mainframe technology. It has structured, cloud-accessible environment where learners can gain proficiency in coding, data management, and operating systems through progressive challenges. The courses are structured in tiers Fundamentals, Concepts, and Advanced. Which ensure a steady increase in difficulty. This tiered architecture is a primary reference for the staged difficulty scaling intended for the proposed system's curriculum.

A standout feature of IBM z Xplore is its use of a VS Code extension (Zowe) to interact with actual mainframe resources. It shows one one the ways specifically in how the UI can bridge the gap between a learning interface and real-world tools for the research output. With each completion of the course, the learners earn digital badges to show a proof of completion and evidence of achievement in learning. This serves as point of reference for a credentialing system of the research [4]

Litmos



Figure 4: Litmos

Litmos is an AI-powered LMS that has an AI Assistant that helps learners find the right courses by simply typing what they need, and AI Playlists that let users build their own learning tracks based on their goals. What makes it stand out is that AI is not just a background feature; it actively supports the learner throughout the whole training process, from finding content to tracking progress. This is a key reference for the proposed PNP-SPT Platform, where the AI assistant is built directly into the module experience to guide each personnel through lessons and assessments step by step. Unlike Litmos where the AI mainly recommends what to take, the SPT Platform embeds the AI assistant directly within a standardized 7-module curriculum, ensuring that every personnel receives consistent, guided instruction throughout the entire training program. [5]

360Learning

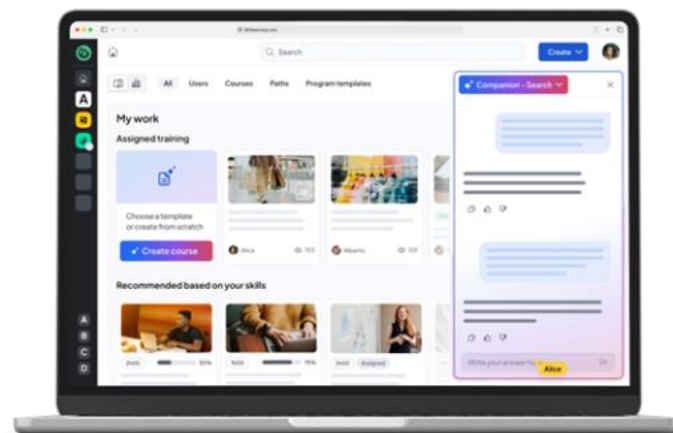


Figure 5: 360Learning

360Learning is an AI-powered learning platform that handles role-based training, onboarding, and compliance all in one place. Its AI looks at how each learner is performing and adjusts the content being suggested to better match their needs and goals over time. What makes it a useful reference for the proposed PNP LMS is that its AI is not a separate add-on but is built into how the whole system works, which is exactly the approach the research output aims to follow. This is a key reference for the proposed PNP-SPT Platform, where the AI assistant is built directly into the module experience to guide each personnel through lessons and assessments step by step. Unlike Litmos where the AI mainly recommends what to take, the SPT Platform embeds the AI assistant directly within a standardized 7-module curriculum, ensuring that every personnel receives consistent, guided instruction throughout the entire training program. [6]

Microsoft Learn

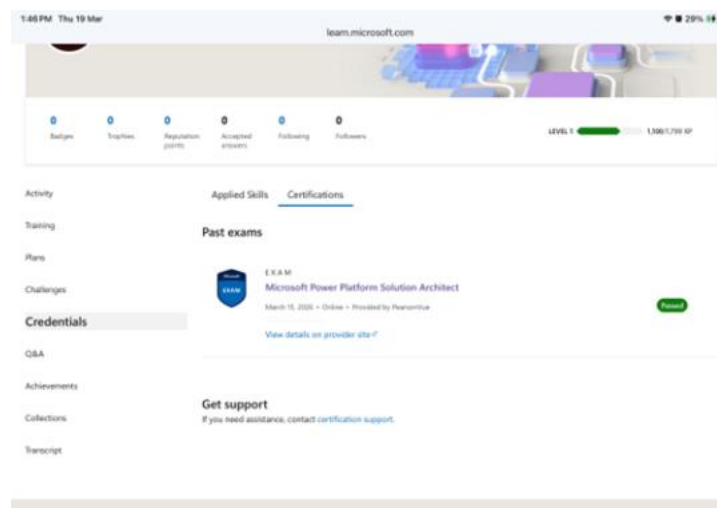
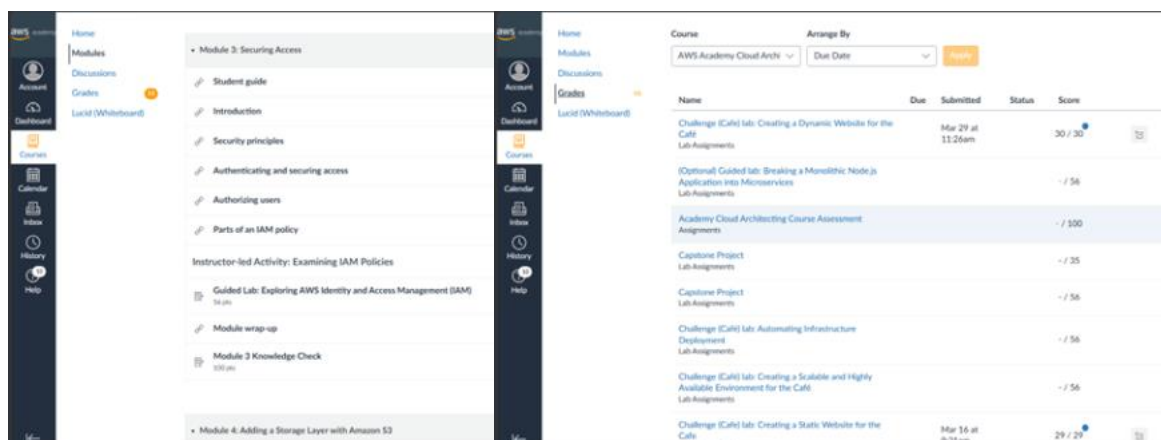


Figure 6: Microsoft Learn

Microsoft Learn is a free learning platform by Microsoft that arranges its content into role-based modules and learning paths, guiding users from basic to advanced topics in a clear and organized way. Each path is built around a specific role or skill, so learners are always working through content that is relevant to where they are headed. When a learning path is completed, users receive an official certification as proof of what they have learned and achieved. This structure is a direct reference for the proposed PNP-SPT Platform, where the 7 modules are organized sequentially and must be completed progressively, ensuring a clear training progression from one module to the next. The certification system of Microsoft Learn also serves as a reference for how the SPT Platform automatically awards official barcode-verifiable certificates upon completing the entire 7-module curriculum. [7]

AWS Academy



The screenshot displays the AWS Academy user interface. On the left, a sidebar contains navigation links: Home, Modules, Discussions, Grades, Lucid (Whiteboard), Courses, Calendar, Inbox, History, and Help. The main content area is divided into two panels. The left panel shows the course structure for 'Module 3: Securing Access', including a Student guide, Introduction, Security principles, Authenticating and securing access, Authorizing users, Parts of an IAM policy, an Instructor-led Activity (Examining IAM Policies), a Guided Lab (Exploring AWS Identity and Access Management (IAM)), a Module wrap-up, and a Module 3 Knowledge Check. The right panel shows a 'Grades' table for the 'AWS Academy Cloud Arch' course, sorted by 'Due Date'.

Name	Due	Submitted	Status	Score
Challenge (Café) lab: Creating a Dynamic Website for the Café Lab Assignments	Mar 29 at 11:26am			30 / 30
(Optional) Guided lab: Breaking a Monolithic Node.js Application into Microservices Lab Assignments				- / 56
Academy Cloud Architecting Course Assessment Assignments				- / 100
Capstone Project Lab Assignments				- / 35
Capstone Project Lab Assignments				- / 56
Challenge (Café) lab: Automating Infrastructure Deployment Lab Assignments				- / 56
Challenge (Café) lab: Creating a Scalable and Highly Available Environment for the Café Lab Assignments				- / 56
Challenge (Café) lab: Creating a Static Website for the Café Lab Assignments	Mar 16 at 9:21am			29 / 29

Figure 7: AWS Academy

The AWS Academy platform provides a structured curriculum for learning cloud computing through sequential modules. Each section ends with a Knowledge Check, which serves as the primary model for the exams in the PNP LMS. This setup ensures that learners understand the material before moving forward.

The SPT Platform builds on this model by integrating an approved third-party AI assistant that actively guides personnel through each module, providing real-time support beyond static knowledge checks. The platform also includes an automated certification process that generates official barcode-verifiable certificates upon successful completion of the standardized 7-module curriculum. [8]

COMPASS Training Center

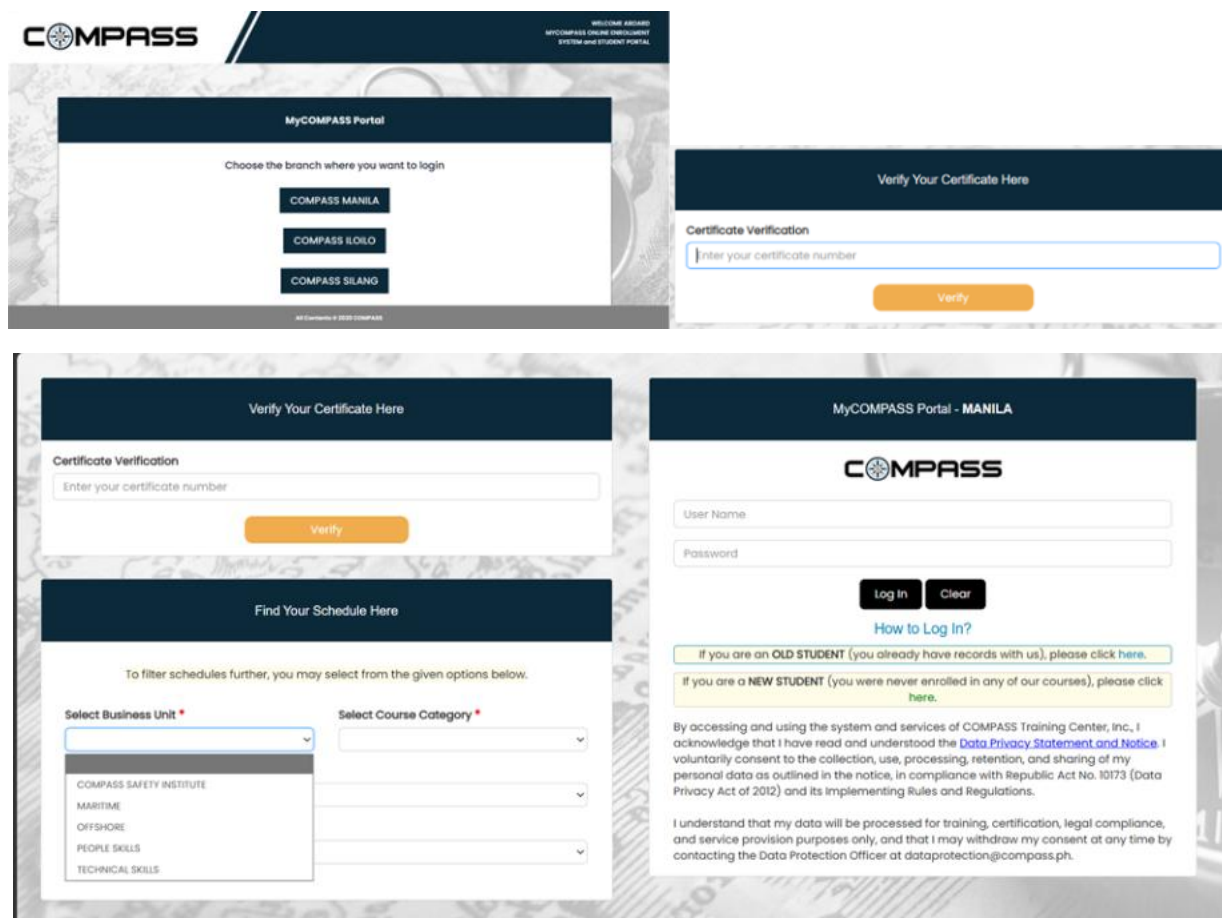


Figure 8: COMPASS Training Center

COMPASS Training Center is a specialized platform built for Filipino seafarers to manage their professional training. It organizes courses into different levels based on specific job positions and required skills. A key feature of the system is its "MyCOMPASS Portal," which requires users to select their specific branch (such as Manila, Iloilo, or Silang) before logging in. This ensures that the training and resources are tailored to the user's specific location and hub.

The structured approach of COMPASS serves as a reference for how the SPT Platform organizes its training pathway. By following how COMPASS delivers standardized training to a large professional workforce, the SPT Platform ensures that all PNP personnel complete a consistent, standardized 7-module AI curriculum regardless of their location or assignment. [9]

DataCamp

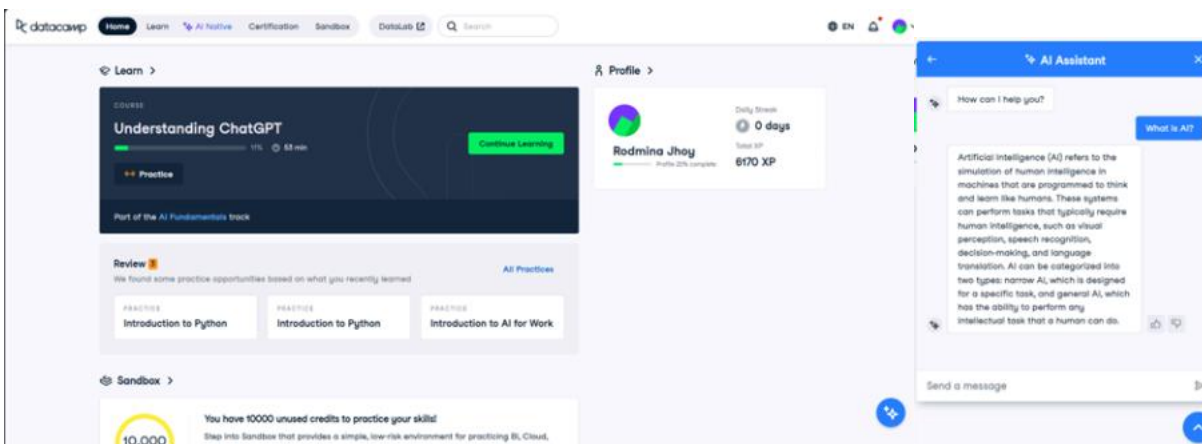


Figure 9: DataCamp

DataCamp is an interactive online learning platform that incorporates gamification elements such as experience points and activity streaks to sustain learner motivation and consistent engagement. The platform features a browser-based practice environment where users can complete hands-on tasks directly on the page and receive immediate feedback without any local setup. While DataCamp does include an AI assistant, its role is largely limited to troubleshooting support rather than serving as an active component of the learning experience.

The proposed PNP-SPT Platform builds upon this foundation by repositioning AI from a peripheral help tool into the core of the learning experience itself. Rather than simply responding to errors or questions, the integrated AI assistant in the proposed system actively guides personnel through each module in real time, providing contextual educational support throughout the entire training process. This approach ensures that the AI serves a more purposeful and instructional role, directly aligned with the platform's goal of delivering dynamic, self-paced AI literacy training to the PNP workforce. [10]

EdX.org

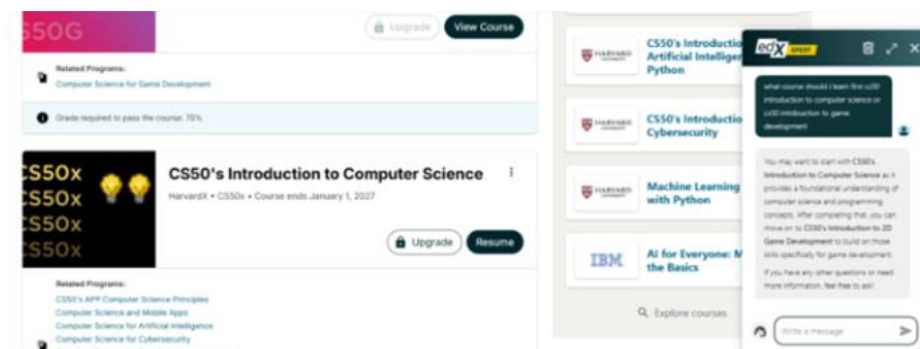


Figure 10: EdX.org

EdX is a large-scale LMS platform widely adopted by Ivy League universities and other institutions for delivering online courses. It offers structured learning paths, graded assessments, and certificates of completion. A notable feature is its AI chatbot assistant, which learners can query for course-related guidance. This provides a reference for integrating conversational AI into the proposed PNP LMS, ensuring that personnel can ask contextual questions about their training modules and receive immediate support. Unlike EdX, which limits the chatbot to course-related queries, the SPT Platform's AI assistant is embedded directly within each module of the standardized curriculum, providing contextual guidance specific to the AI literacy content being covered at that point in the training. [11]

Khan Academy

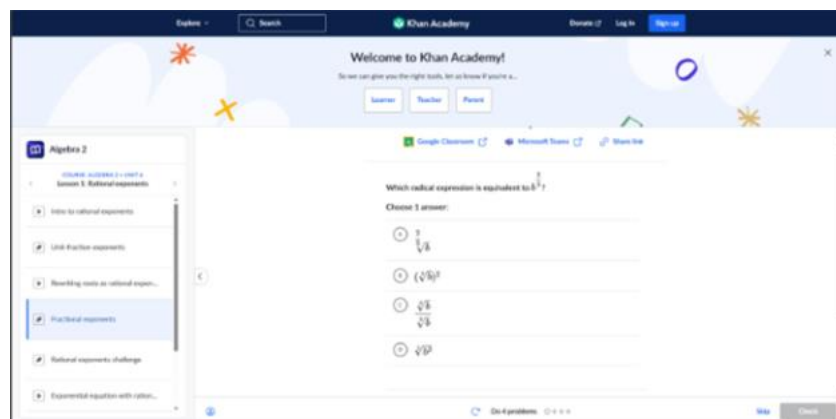


Figure 11: Khan Academy

Khan Academy is a nonprofit educational platform known for its approachable interface and clear, self-paced learning format. It provides units, exercises, quizzes, and tests across a wide range of subjects, emphasizing accessibility and learner autonomy. The platform's design demonstrates how simplicity and clarity can enhance engagement, making it a useful reference for structuring modules in the PNP LMS. By adopting Khan Academy's model of self-paced progression and interactive exercises, the SPT Platform ensures that PNP personnel can learn at their own pace while still being evaluated through per-topic examinations across the standardized 7-module AI literacy curriculum. [12]

W3Schools

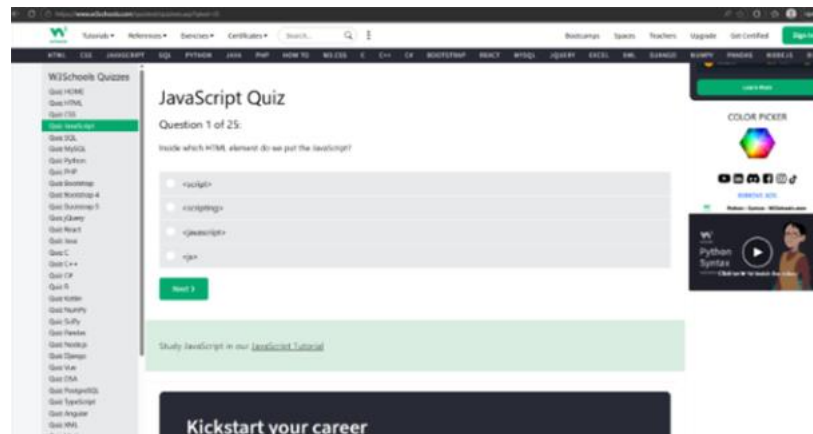


Figure 12: W3Schools

W3Schools is an interactive LMS focused on web development and programming. It features a practice sandbox environment where learners can directly test code, alongside multiple-choice quizzes for reinforcement. Its straightforward format displays all lessons with a visible course roadmap, allowing learners to easily track their progress. This roadmap and sandbox approach inform the proposed PNP LMS by showing how transparency in module progression and hands-on practice can improve user engagement and comprehension. For PNP personnel, this model ensures that training modules are not only interactive but also provide clear visibility of progress, reducing confusion and enhancing motivation. [13]

Philippine Public Safety College LMS

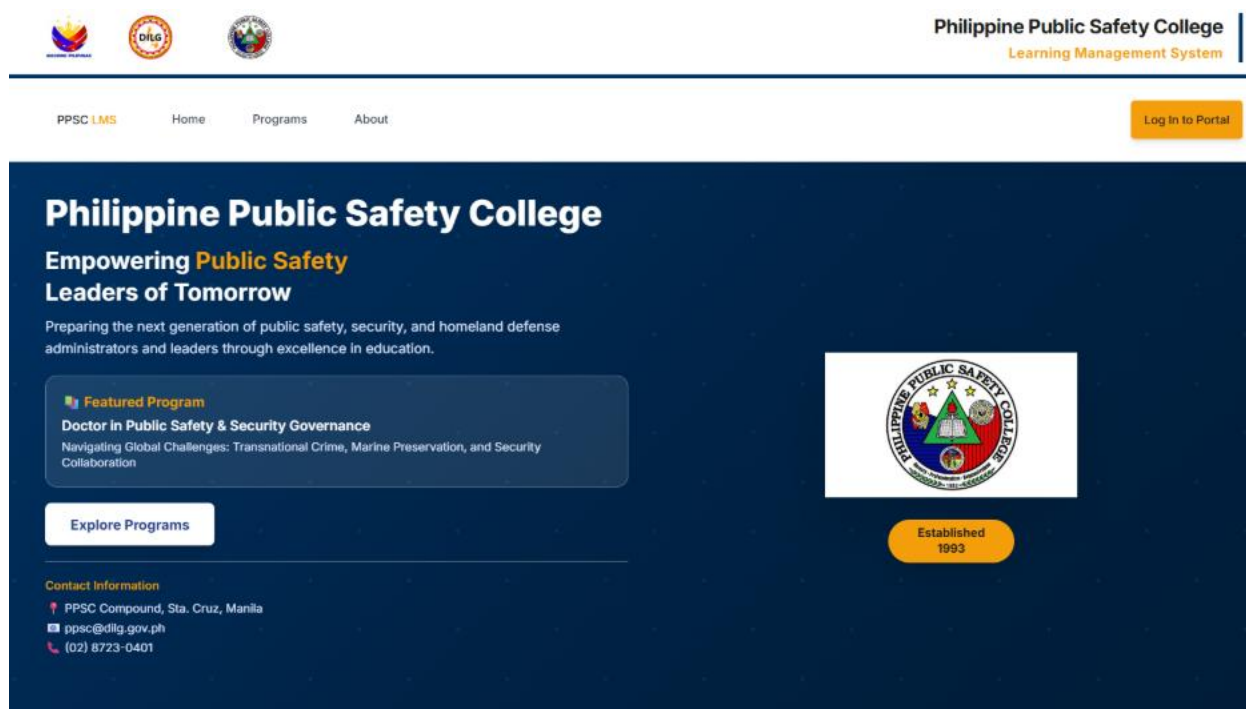


Figure 13: Philippine Public Safety College LMS

The Philippine Public Safety College Learning Management System (PPSC-LMS) is an institutional training portal focused on the mandatory career schooling and accreditation of Philippine tri-bureau personnel, Which includes the PNP. It utilizes a standardized Canvas-based infrastructure to host digital modules aligned with the Training Lineal List (TLL) for rank advancement and official certification. Its centralized format allows for the systematic tracking of thousands of officers across various regions, ensuring that all training completion data is routed for official promotion eligibility. This institutional model informs the proposed PNP-SPT Platform by demonstrating how a centralized, accredited environment maintains organizational standards. For PNP personnel, adopting this benchmark ensures that the new platform remains compliant with existing national training protocols while upgrading the traditional static experience into a more specialize. [14]

Police Open Academy

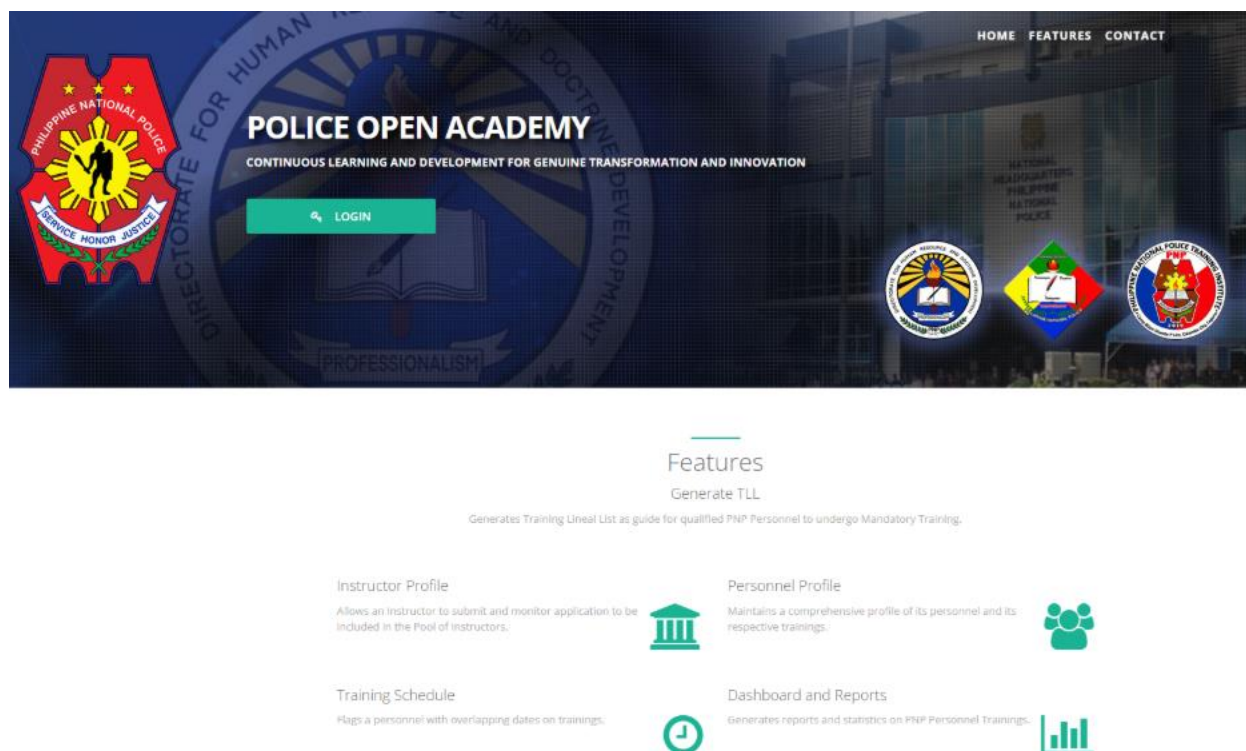


Figure 14: Police Open Academy

The Police Open Academy is a specialized training portal designed to deliver targeted education on emerging technologies and technical operational skills to law enforcement personnel. Built on a standardized PHP, Laravel, and SQL architecture, it provides streamlined features including modular content delivery, fundamental user progress tracking, and structured knowledge pathways.

For the proposed PNP-SPT Platform, the Police Open Academy serves as the most direct structural reference, as it is the existing system the standalone platform is designed to connect with upon deployment. By acknowledging its established technology stack, the SPT Platform is built to be compatible with the POA's infrastructure, allowing the PNP IT Department to establish a secure tokenized redirect without disrupting its existing database or requiring duplicate user management. This approach allows the standalone platform to focus entirely on delivering the specialized 7-module AI curriculum, integrating an approved third-party AI assistant, and automating barcode-verifiable certificate generation upon completion. [15]

Technical Background

This section presents the technical foundation of the PNP Specialized Personnel Training (PNP-SPT) Platform. It begins with an overview of the current system's hardware, software, and network environment, establishing the baseline the proposed platform is designed to improve upon. It then discusses the hardware, software, peopleware, and network components of the proposed system, detailing how these elements work together to deliver standardized AI literacy training at scale. The section closes with the methodology followed in developing this platform, outlining the steps already completed and the steps planned for the remainder of the project.

Current System

The existing educational infrastructure relies on the central Police Officer Academy (POA) e-learning platform, a centralized system that acts as the primary gateway for digital training across the organization. This platform is built on an existing technology stack consisting of Laravel 11 as the core framework, running on PHP 8.2 with a MySQL 9.1 database. While this infrastructure effectively manages baseline course delivery and centralized user authentication, specialized training for AI literacy remains entirely nonexistent within the current digital environment. Consequently, any instruction regarding advanced technological standards must be conducted through traditional, onsite classroom setups. This operational framework requires personnel to be manually scheduled in separate daily clusters, creating severe organizational bottlenecks, high logistical overhead, and a distinct lack of self-paced learning opportunities that ultimately limits the agency's ability to efficiently scale modern training to its entire workforce.

Proposed System

The proposed solution, the PNP Specialized Personnel Training (PNP-SPT) Platform, is engineered as an interoperable web application built to match the Police Officer Academy (POA)'s existing technology stack. The back end is built on Laravel 11 running on PHP 8.2, which provides routing, authentication, role-based middleware, and the data layer. The front end is built with Vue 3 and delivered through Inertia, so that each page is rendered as a single-page application while routing and authorization remain under the control of the Laravel back end. Tailwind CSS and Vite are used for styling and asset compilation. Because the platform shares the POA's core stack, the PNP IT Department retains the option to pursue integration or synchronization with the POA after handover without a full redevelopment.

The platform maintains its own dedicated registration and login system and its own database, limited strictly to training-related records. Personnel register directly on the platform with their name, rank, Police Regional Office, and unit or office, which are later used for certificate generation. Three roles are supported: Trainees, who complete the curriculum; Supervisors, who author and maintain training content and monitor trainee progress; and Admins, who additionally manage accounts, staff, certificate logs, and analytics. Access to each function is enforced at the route level by role middleware rather than by hiding navigation alone.

During development, the application runs against a local MySQL-compatible database (MariaDB), and the deployed prototype uses TiDB Cloud Serverless, a MySQL-compatible managed database, so the schema remains portable to the client's MySQL environment. Uploaded module files and generated certificate PDFs are stored in a private Amazon S3 bucket with public access fully blocked; files are streamed through the application only after ownership and validity checks. Transactional email, such as password reset links, is sent through an SMTP relay (Brevo). The prototype is packaged as a multi-stage Docker image serving the application through Nginx and PHP-FPM and is hosted on Render. This third-party cloud deployment is a deliberate testing-phase choice; migration of the application and database to PNP-controlled infrastructure is planned for handover, so that all operational data remains under PNP ownership in line with the client's data privacy requirements.

Within the application, the traditional instructor-led framework is replaced by a self-paced, multimedia-driven curriculum organized as courses, modules, and lessons, designed to accommodate the client's 40-hour Program of Instruction across seven modules once the finalized POI is received. Lesson content is authored in a block-based editor that supports headings, paragraphs, bulleted lists, uploaded videos, and YouTube embeds, and is stored as structured JSON rather than raw HTML, which prevents stored cross-site scripting by any compromised authoring account. Existing PowerPoint materials can be imported as text into lesson blocks, while module files are limited to browser-native PDF and MP4 formats with server-side verification of the actual file type. Each lesson ends with a knowledge check; lesson progress and exam attempts are saved automatically, so a trainee can leave the platform and resume from the last completed lesson. A final examination supporting multiple-choice, true/false, matching, and identification questions, with a configurable time limit and passing threshold, concludes each course.

An AI learning assistant is planned as a backend-mediated virtual tutor that will appear as a sidebar within each lesson to clarify, summarize, or expand on the approved materials. Following the client's security requirements, the assistant will be served by a locally hosted open-source language model rather than an external cloud API, and all requests will pass through the platform's back end so that only course-related educational content, and no personally identifiable information, is ever transmitted to the model. This component is scheduled for the next iteration.

Finally, the administrative layer automates progress tracking and certificate generation. Upon passing the final examination, the system immediately generates a certificate bearing a unique serial number, a training control number, and a SHA-256 verification hash, renders it as a PDF with a scannable QR code, and stores it in private storage. Supervisors and the public can verify a certificate by scanning the QR code or entering its serial number on a verification page, establishing a secure validation loop that eliminates manual supervisory overhead while maintaining institutional verification standards. Supervisor and Admin dashboards aggregate completion status, examination results, and certificates issued across trainees, units, and regions.

Methodology

The development of the PNP-SPT Platform followed an iterative, client-consultative methodology, structured around continuous engagement with the PNP ITMS Office and the project's academic adviser to ensure the proposed system remained technically feasible and policy-compliant throughout its design.

The process began with a review of related systems, examining thirteen existing learning platforms, ranging from large-scale institutional systems such as the UN Inspira LMS and EdX to AI-integrated platforms such as Litmos and 360Learning, to establish a benchmark for interactivity, AI integration, and large-scale organizational deployment. This review directly informed the design decisions behind the platform's 7-module curriculum structure, its embedded AI learning assistant, and its certificate-generation system.

Requirements gathering was conducted through four formally documented consultations with the client and academic adviser, held on March 11, March 13, April 10, and May 11, 2026. These consultations were critical in shaping the project's technical direction, most notably the pivot from an initially proposed standalone Learning Management System toward an interoperable web application built to match the existing Police Officer Academy (POA)'s technology stack, in direct response to the technical and policy constraints raised by the PNP ITMS Office regarding duplicate systems and data privacy compliance.

Based on these requirements, the team developed a product backlog of user stories organized into epics, a use case diagram identifying the system's four actors and their interactions, and fully dressed use case descriptions detailing the flow of activities and exception conditions for each core function. Test cases were then derived directly from these fully dressed use cases to validate that each function behaves as specified once implemented.

For the remainder of the project, development will proceed according to the team's eight-stage product roadmap, covering technical stack setup, front-end design, backend API and logic implementation, core feature development, AI chatbot integration, pilot testing, feedback and iteration, and final deployment preparation, carried out in close coordination with the PNP ITMS Office, which has committed on-site workspace and internal programming support for the implementation phase.

Requirements Analysis

Product Backlog / User Stories

PNP-SPT Platform Product backlog				
ID	As a...	I want to be able to...	So that...	Priority

1	PNP Personnel	register and log in directly on the SPT Platform	I can access the specialized training through my own dedicated account	Must
2	Admin	use the platform's backend to securely manage user registration, authentication, and account credentials	personnel identities are validated and their accounts are properly secured within the platform's own database	Must
3	PNP Personnel	engage in a self-paced, multimedia-driven 7-module curriculum	I can fulfill the required 40-hour Program of Instruction at my own convenience.	Must
4	PNP Personnel	process my text inputs through a real-time virtual AI tutor panel	I can receive immediate, context-aware educational guidance on the coursework materials.	Must
5	Admin	handle all third-party AI requests through the platform's backend layer, which only has access to approved educational materials	no personally identifiable information is ever transmitted to the external AI provider, ensuring accuracy of responses within the scope of the topic at hand.	Must
6	Admin	limit the AI tutor's responses strictly to the current module's approved educational materials	the virtual assistant never responds to questions outside the scope of the training curriculum.	Should
7	PNP Personnel	track and save my course progression asynchronously in the background	I can safely exit the application and resume my modules exactly where I left off without losing progress.	Must
8	Admin	isolate curriculum configurations and learning metrics within the platform's dedicated separate database	the system keeps its data footprint minimal, preventing server strain and optimizing performance for massive traffic spikes.	Must
9	PNP Personnel	Receive a digital certificate featuring a unique, barcode-verifiable cryptographic hash immediately upon final module completion	I receive instant, official proof of course completion without waiting for manual processing or administrative sign-offs.	Must
10	Supervisors	cross-reference an encrypted barcode hash against the platform's completion logs	I can instantly verify a personnel member's credentials and completely	Must

			eliminate manual document review overhead.	
11	Supervisors	aggregate and display workforce completion rates on an automated dashboard	I can monitor real-time organizational compliance metrics without needing to manually collect or grade physical submissions.	Should
12	Admin	securely synchronize internal platform transaction logs directly with the client's localized production infrastructure	all active operational data remains strictly under PNP ownership and complies with institutional data privacy mandates.	Must

Table 1: Product Backlog

Use Case Diagram

Use-Case Diagram of Specialized Personnel training Platform (SPT-Platform)

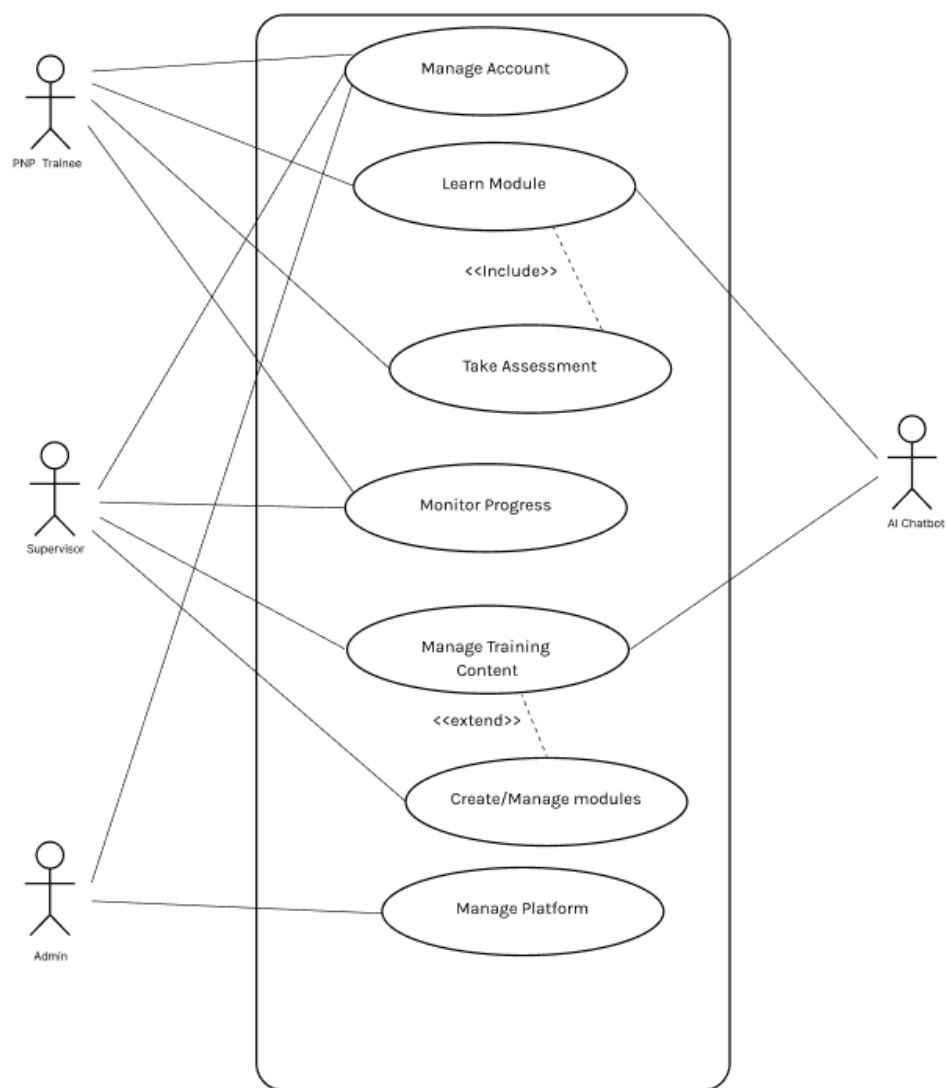


Figure 15: Use Case Diagram

Use Case Full Description/Fully Dressed Use Case

Use Case Name:	Manage Account
Scenario:	A PNP Trainee, Supervisor, or Admin registers a new account (including setting up two-factor authentication via Google Authenticator), logs into an existing one, updates their account

	information (e.g., contact details, office/unit) or password, resets/recovers a forgotten password, or deletes their account. Separately, an Admin reviews a specific account — for example, locking it after suspicious activity or reinstating one that was mistakenly disabled.
Triggering Event:	User visits the SPT Platform for the first time (registration, including 2FA setup) or returns to access it (login, including 2FA verification); user requests a password reset/recovery or wants to update their account information; or a specific account needs Admin attention — a security concern, a user request to be reinstated, or a routine account audit.
Brief Description:	Allows PNP Trainee, Supervisor, and Admin to create a new account by registering with a verified email, required details (such as office/unit), and a linked Google Authenticator app for two-factor authentication; log into an existing account using their credentials and a time-based 2FA code; update their account information (e.g., name, contact number, office/unit) independently of their password; reset/recover a forgotten password; or delete their own account. Also allows the Admin to review, lock, disable, or reinstate individual user accounts to keep access to the platform secure.
Actors:	PNP Trainees, Supervisor, Admin
Related Use Cases:	Monitor Progress (progress is tracked against the account managed here); Manage Platform (base use case for Admin's account-management actions)
Stakeholders:	PNP Trainee, Supervisor, and Admin (need reliable, secure access and control over their own account details), IT Department (responsible for maintaining registration, authentication, and account security), PNP Organization (needs accurate trainee/unit records tied to training data, with access properly controlled)
Preconditions:	User has a valid, accessible email address. For returning users, an existing account must already be registered, and the Google Authenticator app must already be linked to that account. For updates to account information, password changes, or deletion, the user must be logged in (or verified via the recovery flow). For administrative actions, the Admin must be logged in with account-management permissions and the account in question must already exist.
Postconditions:	A new account is created, its email is verified/approved, and 2FA is linked (Create); the user's identity and 2FA code are verified and they land on their role-appropriate dashboard (Read); the user's account information is successfully updated, independent of any password change (Update — Info); the user's password is

	successfully updated (Update — Password); the account is permanently removed from the SPT database (Delete); or, where an Admin has taken action, the account's status (active, locked, disabled, or reinstated) is updated and the affected user's access changes accordingly.	
Flow of Activities:	Actor	System
	Create / Read (Register & Login) 1. User selects Register (new user) or Login (returning user). 3. New user enters email, password, and required details (e.g., office/unit); returning user enters existing credentials for login. 5. New user checks their email and clicks the verification link (only if new register). 7. New user scans the QR code and enters the generated 6-digit code to confirm the link; returning user enters the current 6-digit code from their Google Authenticator app.	2. System displays the login/registration screen. 4. System validates the input. 6. System confirms the email, activates the account, and prompts the new user to set up two-factor authentication by displaying a QR code to scan with the Google Authenticator app (only if new). 8. System verifies the 2FA code. 9. System creates an authenticated session. 10. System loads the role-appropriate dashboard (Trainee curriculum, Supervisor content tools, or Admin panel). 11. System logs the successful authentication for audit purposes.

	Update (Edit Details / Password Reset & Recovery) 12. User navigates to account settings and edits their account information (e.g., name, contact number, office/unit). 13. User sets a new password via the reset link (if resetting) or saves their edited details. Delete (Account Deletion) 16. User requests account deletion and confirms. Admin Account Management 18. Admin searches for or selects a specific user account.	14. System validates the changes, or sends a password reset link to the registered email. 15. System updates the account record and confirms the change. 17. System permanently removes the account and associated session data, and logs the deletion for audit purposes. 19. System displays the account's details and current status.
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	20. Admin chooses an action: lock, disable, or reinstate the account. 22. Admin confirms.	21. System asks for confirmation. 23. System updates the account status and applies the access change immediately. 24. System logs the action for audit purposes.
Exception Conditions:		
Duplicate or invalid email	System rejects the registration and prompts the user to correct the email.	
Unverified email	User cannot log in until the verification link is confirmed; system prompts them to check their inbox or resend the link.	
Incorrect login credentials	System denies access and displays an error, allowing the user to retry or reset their password.	
Invalid or expired 2FA code	If the code entered from Google Authenticator is incorrect or expired, the system denies login and allows the user to re-enter a new code.	
Lost access to Authenticator app	If the user can no longer generate a valid code (e.g., lost or reset device), the system directs them to an account-recovery process verified through their registered email before 2FA can be re-linked.	
Expired/invalid reset link	If the password reset link has expired or is invalid, the system denies the reset and prompts the user to request a new one.	
Deletion of account with active data	If the user attempts to delete an account with in-progress or incomplete training records, the system warns of the impact before allowing the deletion to proceed.	
Account not in database	If the Admin tries to modify an account that no longer exists or was already removed, the system shows an error and cancels the action	

Table 2: Fully Dressed Use Case: Manage Account

Use Case Name:	Learn Module	
Scenario:	A PNP Trainee learns through a lesson inside one of the SPT Platform — reading content, viewing multimedia, and consulting the AI Chatbot when a concept needs clarifying.	
Triggering Event:	PNP Trainee launches a module from "My Learning" or continues an in-progress module.	
Brief Description:	Allows PNP Trainee to view and progress through the self-paced, multimedia-driven lessons of a training module, with the AI Chatbot available for real-time guidance.	
Actors:	PNP Trainees (primary); AI Chatbot (supporting — provides contextual guidance during the lesson)	
Related Use Cases:	take assessment (include)	
Stakeholders:	PNP Trainees (need to understand the material), Supervisor (relies on Trainees completing lessons accurately), PNP Organization (needs a consistently trained workforce)	
Preconditions:	PNP Trainee has been redirected into the SPT Platform and is enrolled in the module.	
Postconditions:	The lesson is marked as viewed/completed, and progress is saved for that module.	
Flow of Activities:	Actor	System

	1. PNP Trainee selects a lesson within the module. 3. PNP Trainee reads/views the material and may ask the AI Chatbot a question. 5. PNP Trainee marks the lesson complete and moves to the next one.	2. System loads the lesson content (text, multimedia). 4. System (via AI Chatbot) responds with guidance limited to the current module's approved content. 6. System saves the lesson as completed and updates the module progress bar. 7. System unlocks the next lesson or, if all lessons are done, unlocks the module's final assessment.
Exception Conditions:		
Connectivity interruption	If the PNP Trainee loses connection mid-lesson, the system preserves their last completed step so they can resume without losing progress.	
AI Chatbot out-of-scope question	If the question falls outside the current module's approved educational materials, the AI Chatbot declines to answer and redirects the Trainee back to the lesson's content.	

Table 3: Fully Dressed Use Case: Learn Module

Use Case Name:	Take Assessment
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Scenario:	After finishing all lessons in a module, a PNP Trainee answers the module's final assessment questions to demonstrate understanding before the module is marked passed.	
Triggering Event:	PNP Trainee completes all lessons in a module and proceeds to its final assessment.	
Brief Description:	A module-specific action within Learn Module, where PNP Trainee is evaluated on the topic before receiving credit and, ultimately, a certificate.	
Actors:	PNP Trainees	
Related Use Cases:	Learn Module (base use case)	
Stakeholders:	PNP Trainees (need to pass to receive credit), Supervisor (relies on assessment scores to verify readiness), PNP Organization (needs assurance that training was actually absorbed, not just viewed)	
Preconditions:	PNP Trainee has completed all lessons within the module and the final assessment has been unlocked.	
Postconditions:	The assessment is scored, and if passed, the module is marked complete and a barcode-verifiable certificate becomes available.	
Flow of Activities:	Actor	System
	1. PNP Trainee starts the module's final assessment. 3. PNP Trainee answers each question and submits. 5. PNP Trainee finishes and confirms submission.	2. System presents the assessment questions one at a time. 4. System records each answer as it's submitted. 6. System instantly grades the assessment against the answer key. 7. System marks the module "Passed" and unlocks certificate generation or marks it "Failed" and allows a retake.

Exception Conditions:		
Failed assessment	If the PNP Trainee does not meet the passing score, the system informs them of the result and allows them to review the material before retaking the assessment.	
Incomplete submission	If the Trainee exits before answering all required questions, the system does not grade the attempt and prompts them to complete it later.	

Table 4: Fully Dressed Use Case: Take Assessment

Use Case Name:	Monitor Progress
Scenario:	A PNP Trainee checks their own completion status across the 7-module curriculum, while a Supervisor monitors the progress of all registered trainees, either as a whole or narrowed by batch, unit, or module, for example to identify who has not completed required modules ahead of a compliance deadline.
Triggering Event:	PNP Trainee opens their own progress view, or Supervisor opens the progress dashboard for a routine check of all trainees.
Brief Description:	Allows a PNP Trainee to view their own module completion status, and allows a Supervisor to view the asynchronously tracked completion data of all registered trainees, either in full or filtered by batch, unit, or module, without manual submission collection.
Actors:	PNP Trainees (primary — view own progress); Supervisor (primary — view trainee progress)

Related Use Cases:	login/register (progress is tied to the account established at login)	
Stakeholders:	PNP Trainees (want visibility into their own standing), Supervisor (responsible for monitoring workforce readiness), PNP Organization (needs visibility into compliance)	
Preconditions:	User is logged in; progress data exists for the account(s) being viewed.	
Postconditions:	The user has an up-to-date view of module completion status.	
Flow of Activities:	Actor	System
	1. PNP Trainee/Supervisor opens the progress tracking view. 3. Supervisor may view all trainees at once or filter by batch, unit, module, or individual trainee. 5. User reviews the data or exports a report.	2. System retrieves saved progress data (own record for Trainee; all trainee records for Supervisor). 4. System displays completion percentages, passed/failed assessments, and certificate status (an overall summary with per-trainee details for the Supervisor). 6. System generates the requested view or report.
Exception Conditions:		
No progress data available	If no module has been started yet, the system displays a "No activity" status instead of an error.	

Table 5: Fully Dressed Use Case: Monitor Progress

Use Case Name:	Manage Training Content	
Scenario:	System rejects the registration and prompts the user to correct the email.	
Triggering Event:	The Supervisor decides that a module needs to be added, updated, or removed (e.g., new AI policy released, feedback from trainees, scheduled curriculum refresh).	
Brief Description:	Allows a Supervisor to oversee and update the overall training content within the platform, ensuring the curriculum stays accurate and current.	
Actors:	Supervisor (primary); AI Chatbot (indirect — draws its answers from whatever content is currently active)	
Related Use Cases:	create/manage modules (extend)	
Stakeholders:	Supervisor (responsible for content accuracy), PNP Trainees (rely on content being correct and up to date), PNP Organization (needs training to stay compliant and relevant)	
Preconditions:	Supervisor is logged into the platform with content-management permissions.	
Postconditions:	The training content is updated, and any changes are reflected in what Trainees see and what the AI Chatbot can reference.	
Flow of Activities:	Actor	System

	1. Supervisor logs in and goes to the content management section. 3. Supervisor selects what to add, update, or remove. 5. Supervisor makes the changes and saves	2. System displays the current list of training content. 4. System opens the content editor. 6. System validates the changes and saves them. 7. System updates the active content so Trainees and the AI Chatbot see the latest version.
Exception Conditions:		
Invalid/incomplete changes	If the Supervisor's changes are incomplete or invalid (e.g., missing required fields), the system rejects the save and shows an error message asking the Supervisor to correct it.	

Table 6: Fully Dressed Use Case: Manage Training Content

Use Case Name:	Create/Manage Modules
Scenario:	A Supervisor builds a specific training module from scratch — adding lessons, activities, and a final assessment for one topic (e.g., "Module 3: Responsible AI, Ethics, and Human Rights").
Triggering Event:	A new module needs to be created, or an existing module needs its structure (lessons, order, assessment) changed.
Brief Description:	A more detailed, module-specific action within Manage Training Content, where the Supervisor builds out or restructures an individual module.
Actors:	Supervisor

Related Use Cases:	Manage Training Content (base use case)	
Stakeholders:	Supervisor (author of the module), PNP Trainees (will go through the finished module), PNP Organization (needs a complete, well-structured 7-module curriculum)	
Preconditions:	Supervisor is logged in and inside the Manage Training Content section.	
Postconditions:	The module is created or updated and becomes available for Trainees to enroll in and complete.	
Flow of Activities:	Actor	System

	1. Supervisor chooses to create a new module or edit an existing one. 3. Supervisor adds lessons, activities, and the module's final assessment questions 5. Supervisor arranges the order of lessons and confirms the module is complete.	2. System opens the module builder. 4. System saves each part as it's added. 6. System checks that the module has all required parts (lessons + assessment). 7. System publishes the module, making it visible to Trainees.
Exception Conditions:		
Missing/incomplete content publish	If the Supervisor tries to publish a module that's missing a required part (e.g., no final assessment), the system blocks publishing and lists what's missing.	

Table 7: Fully Dressed Use Case: Create/Manage Modules

Use Case Name:	Manage Platform
Scenario:	An Admin from the IT Department keeps the platform running smoothly — checking that servers are healthy, security settings are correct, and the AI Chatbot connection isn't running into usage limits.
Triggering Event:	Routine system checks, a reported technical issue, or a scheduled maintenance/configuration update.
Brief Description:	Covers the Admin's responsibility for the technical health, security, and configuration of the platform itself, separate from the content or user accounts on it.
Actors:	Admin (IT Department)

Related Use Cases:		
Stakeholders:	IT Department (responsible for uptime and security), PNP Trainee and Supervisor (rely on the platform being available), PNP Organization (needs the platform compliant with data privacy requirements)	
Preconditions:	Admin is logged in with system-administration permissions.	
Postconditions:	The platform remains secure, available, and properly configured; any issues found are resolved or logged.	
Flow of Activities:	Actor	System
	1. Admin logs in and opens the system administration panel. 3. Admin reviews the information and makes any needed adjustments (e.g., security settings, server configuration). 5. Admin confirms the platform is stable.	2. System displays platform health, security status, and AI Chatbot usage stats. 4. System applies the changes. 6. System logs the administrative action for audit purposes.
Exception Conditions:		
System Alert	If a critical issue is detected (e.g., server downtime, AI Chatbot exceeding its usage limit), the system alerts the Admin immediately instead of waiting for the next routine check.	

Table 8: Fully Dressed Use Case: Manage Platform

User Stories and Acceptance Criteria

Epic	Story ID	Actor	User Story	Acceptance Criteria (Given / When / Then)
1: Account & Access Management	US1.1	PNP Trainee	As PNP Trainee, I want to register and log in directly on the SPT Platform, so that I can access the specialized training through my own dedicated account.	Given a new user navigates to the SPT Platform, When they submit valid registration details, Then the system creates their account and grants them access to their role-appropriate dashboard.
	US1.2	Admin	As an Admin, I want to securely manage user registration, authentication, and account credentials through the platform's backend, so that Trainee identities are validated and their accounts are properly	Given a user attempts to log in, When their credentials are submitted, Then the system validates them against the platform's database and grants or denies access accordingly.

			secured within the platform's own database.	
US1.3	Admin		As an Admin, I want to lock, disable, or reinstate individual user accounts, so that I can keep access to the platform secure in response to security concerns or account issues.	Given an Admin selects a specific user account, When they choose to lock, disable, or reinstate it and confirm, Then the system updates the account's status and applies the access change immediately.
US1.4	PNP Trainee		As PNP Trainee, I want to update my account information (such as my name, contact number, or office/unit), so that my profile stays accurate and current without having to touch my password.	Given a user submits updated account information, When the system validates the changes, Then the account record is updated and confirmed, independently of any password change.
US1.5	PNP Trainee		As PNP Trainee, I want to change my password while logged in, so that I can keep my credentials secure on my own schedule, separately from updating my other account details.	Given a user submits a valid current password and a new password, When the system verifies the current password, Then the account is updated with the new password and the change is confirmed.
US1.6	PNP Trainee		As PNP Trainee, I want to reset my password if I forget it, so that I can regain access to my account without needing Admin intervention.	Given a user requests a password reset with their registered email, when they follow the reset link and submit a new password, then their account password is updated and they can log in with the new credentials.
US1.7	PNP Trainee		As PNP Trainee, I want to set up two-factor authentication using Google Authenticator when I register, so that my account has an added layer of	Given a new user completes registration and verifies their email, When they scan the displayed QR code and enter the generated 6-digit code, Then the system links their Google Authenticator app to

			protection beyond my password.	their account and activates 2FA.
	US1.8	PNP Trainee	As PNP Trainee, I want to enter a time-based code from Google Authenticator each time I log in, so that my account stays protected even if my password is compromised.	Given a returning user submits valid login credentials, When they enter the current 6-digit code from their linked Google Authenticator app, Then the system verifies the code and grants access to their role-appropriate dashboard.
	US1.9	PNP Trainee	As PNP Trainee, I want to recover my account through my registered email if I lose access to my Authenticator app, so that I'm not permanently locked out of my account.	Given a user can no longer generate a valid 2FA code (e.g., lost or reset device), When they request account recovery through their registered email, Then the system verifies their identity and allows them to re-link a new Authenticator app before continuing to log in.
2: Training Delivery & Progress Tracking	US2.1	PNP Trainee	As PNP Trainee, I want to engage in a self-paced, multimedia-driven 7-module curriculum, so that I can fulfill the required 40-hour Program of Instruction at my own convenience.	Given a PNP Trainee is enrolled in a module, When they open a lesson, Then the system loads the multimedia content and allows them to progress at their own pace.
	US2.2	PNP Trainee	As PNP Trainee, I want to track and save my course progression asynchronously in the background, so that I can safely exit the application and resume my modules exactly where I left off without losing progress.	Given a PNP Trainee exits the platform mid-module, When they log back in, Then the system resumes their progress from the last completed step.
	US2.3	Admin	As an Admin, I want to isolate curriculum configurations and learning metrics within the platform's dedicated separate database, so that the	Given curriculum and metrics data is generated, When it is stored, Then the system keeps it isolated within the SPT

			system keeps its data footprint minimal, preventing server strain and optimizing performance for massive traffic spikes.	Platform's dedicated database, separate from POA records.
	US2.4	PNP Trainee	As PNP Trainee, I want to view my own module completion status, so that I have visibility into my training progress and what's left to complete.	Given a PNP Trainee opens their progress view, When the system retrieves their saved progress data, Then it displays their completion percentages, passed/failed assessments, and certificate status.
	US2.5	PNP Trainee	As PNP Trainee, I want lessons within a module to unlock in sequence as I complete each one, so that I follow a structured learning path and can access the final assessment only once all lessons are done.	Given a PNP Trainee completes a lesson, When they mark it complete, Then the system unlocks the next lesson or, if all lessons are done, unlocks the module's final assessment.
	US2.6	PNP Trainee	As PNP Trainee, I want to be notified of my result and allowed to retake a module's final assessment if I don't pass, so that I can review the material and try again without losing my overall progress.	Given a PNP Trainee does not meet the passing score on a final assessment, When the system grades their submission, Then it informs them of the result and allows them to review the material before retaking the assessment.
3: AI Learning Assistant	US3.1	PNP Trainee	As PNP Trainee, I want to process my text inputs through a real-time virtual AI tutor panel, so that I can receive immediate, context-aware educational guidance on the coursework materials.	Given a PNP Trainee submits a question in the AI tutor panel, When the system processes the request, Then the AI assistant returns guidance relevant to the current module's content.

4: Certification & Compliance	US3.2	Admin	As an Admin, I want to handle all third-party AI requests through the platform's backend layer, which only has access to approved educational materials, so that no personally identifiable information is ever transmitted to the external AI provider, ensuring accuracy of responses within the scope of the topic at hand.	Given a request is forwarded to the third-party AI provider, When the backend processes it, Then only course-related educational content is transmitted, with no personally identifiable information included.
	US3.3	Admin	As an Admin, I want to limit the AI tutor's responses strictly to the current module's approved educational materials, so that the virtual assistant never responds to questions outside the scope of the training curriculum.	Given a PNP Trainee asks a question outside the current module's content, When the AI assistant evaluates the request, Then it declines to answer and redirects the user back to the lesson's material.
	US4.1	PNP Trainee	As PNP Trainee, I want to receive a digital certificate featuring a unique, scannable, verifiable cryptographic hash immediately upon final module completion, so that I receive instant, official proof of course completion without waiting for manual processing or administrative sign-offs.	Given a PNP Trainee completes and passes the final assessment of the 7-module curriculum, When the system detects successful completion, Then it automatically generates a certificate with a scannable, verifiable cryptographic hash.
	US4.2	Supervisors	As a Supervisor, I want to cross-reference a scannable, verifiable hash against the platform's completion logs, so that I can instantly verify a Trainee member's credentials and completely	Given a Supervisor scans or enters a certificate's hash, When it is checked, Then the system matches it against the completion logs and confirms or denies its validity.

			eliminate manual document review overhead.	
	US4.3	Supervisors	As a Supervisor, I want to aggregate and display workforce completion rates on an automated dashboard, so that I can monitor real-time organizational compliance metrics without needing to manually collect or grade physical submissions.	Given a Supervisor opens the progress dashboard, When they view or filter by Trainee, unit, or module, Then the system displays up-to-date completion percentages and assessment results.
	US4.4	Admin	As an Admin, I want to securely synchronize internal platform transaction logs directly with the client's localized production infrastructure, so that all active operational data remains strictly under PNP ownership and complies with institutional data privacy mandates.	Given transaction logs are generated on the platform, When synchronization runs, Then the logs are securely transferred to the PNP's localized production infrastructure without exposure to external parties.
5: Content Management	US5.1	Supervisor	As a Supervisor, I want to add, update, or remove training content within the platform, so that the curriculum stays accurate and current as policies or materials change.	Given a Supervisor makes changes to training content and saves, When the system validates the changes, Then it updates the active content so PNP Trainee and the AI Chatbot see the latest version.
	US5.2	Supervisor	As a Supervisor, I want to build or restructure an individual module's lessons, activities, and final assessment, so that PNP Trainee have a complete, well-structured module to complete.	Given a Supervisor arranges a module's lessons and confirms it's complete, When the system checks that all required parts are present, Then it publishes the module, making it visible to PNP Trainee.

Table 9: User Stories and Acceptance Criteria

Test Cases based on Fully Dressed Use Case

MANAGE ACCOUNT				
TC ID	Test Steps	Test Data	Actual Result	Status
MA-01	Verify when a new PNP Trainee/Supervisor/Admin registers an account			
<i>Pre-requisite: User has a valid PNP email or identifier</i>				
<i>Pre-condition: Platform's account/authentication system is available</i>				
TC-001	1. Navigate to the SPT Platform URL 2. Click Register 3. Enter account details (email, password, role-relevant info) 4. Submit	Email: officer.delacruz@pnp.gov.ph Password: Valid123!	Account is created in the SPT database. User is logged into an active session and redirected to their role-appropriate dashboard.	
MA-02	Verify when a user logs in with valid existing credentials			
<i>Pre-requisite: User already has a registered SPT Platform account</i>				
TC-002	1. Navigate to the SPT Platform URL 2. Enter valid credentials 3. Click Login	Email: officer.delacruz@pnp.gov.ph Password: Valid123!	Credentials are verified against the SPT database. User is authenticated and redirected to their role-appropriate dashboard.	
MA-03	Verify registration is rejected when submitted details are invalid or already in use			
<i>Pre-requisite: N/A</i>				
TC-003	1. Navigate to Register 2. Enter an email already registered to another account 3. Submit	Email: officer.delacruz@pnp.gov.ph (already registered)	System rejects the registration and prompts the user to correct the entry.	
MA-04	Verify login is denied when credentials do not match any account			
<i>Pre-requisite: N/A</i>				
TC-004	1. Navigate to Login 2. Enter an unregistered email and any password 3. Click Login	Email: unknown.user@pnp.gov.ph Password: anything123	System displays an authentication error and denies access.	
MA-05	Verify login is denied when the account has been locked/disabled by Admin			

Pre-requisite: Account exists and has been locked/disabled by an Admin

TC-005	1. Navigate to Login 2. Enter valid credentials for a locked/disabled account 3. Click Login	Email: locked.officer@pnp.gov.ph Password: Valid123!	System denies login and displays the appropriate lockout/disabled notice.	
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MA-06	Verify user can update their account information			
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Pre-requisite: User is logged in with an active account

TC-006	1. Navigate to Account Settings/Profile 2. Select "Change Password" or any information that is updatable 3. Enter current password or new information 4. Enter new password or new information and confirm 5. Submit	Current Password: Valid123! New Password: NewValid456!	System verifies the current password or new information, updates the account with the new password or new information, and confirms the change was successful.	
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MA-07	Verify system rejects a password update with incorrect current password			
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Pre-requisite: User is logged in with an active account

TC-007	1. Navigate to Account Settings/Profile 2. Select "Change Password" 3. Enter incorrect current password 4. Enter new password and confirm 5. Submit	Current Password: WrongPass1! New Password: NewValid456!	System rejects the change and displays an error indicating the current password is incorrect.	
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MA-08	Verify system shows an error when Admin attempts to modify a non-existent account			
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Pre-requisite: Admin is logged in with account-management permissions

TC-008	1. Search for an account that no longer exists or was already removed 2. Attempt to select and modify it	Account: deleted.user@pnp.gov.ph	System shows an error and cancels the action.	
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MA-09	Verify Admin can lock, disable, or reinstate a user account			
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Pre-requisite: Admin is logged in with account-management permissions; the account exists

TC-009	1. Search for or select a specific user account 2. Choose an action: lock, disable, or reinstate 3. Confirm the action	Account: officer.delacruz@pnp.gov.ph Action: Lock	System updates the account status and applies the access change immediately. Action is logged for audit purposes.	
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MA-10	Verify user can recover their password via "Forgot Password"			
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Pre-requisite: User has a registered SPT Platform account with a valid email

TC-010	1. Navigate to Login 2. Click "Forgot Password" 3. Enter registered email 4. Submit 5. Open the reset link sent to email 6. Enter and confirm new password 7. Submit	Email: officer.delacruz@pnp.gov.ph New Password: NewValid456!	System sends a password reset link to the registered email. Once submitted, the account password is updated and the user can log in with the new password.	
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MA-11	Verify system does not reveal account existence for unregistered email in "Forgot Password"			
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Pre-requisite: N/A

TC-011	1. Navigate to Login 2. Click "Forgot Password" 3. Enter an unregistered email 4. Submit	Email: unknown.user@pnp.gov.ph	System displays a generic confirmation message (e.g., "If this email is registered, a reset link has been sent") without confirming whether the account exists.	
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MA-12	Verify expired password reset link is rejected			
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Pre-requisite: User has requested a password reset and the link has expired

TC-012	1. Open an expired password reset link 2. Attempt to enter a new password 3. Submit	Reset link: expired token	System rejects the request, informs the user the link has expired, and prompts them to request a new reset link.	
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MA-13	Verify user can log out and the session is ended			
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Pre-requisite: User is logged in with an active account

TC-013	1. Click Logout 2. Press the browser Back button 3. Attempt to open a dashboard page directly	N/A	Session is terminated and user is redirected to the Login page. Protected pages cannot be accessed without logging in again.	
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MA-14	Verify users cannot access pages outside their role			
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Pre-requisite: A PNP Trainee is logged in

TC-014	1. Enter the URL of a Supervisor or Admin page directly in the browser 2. Press Enter	Account: Trainee URL: /admin	System denies access and shows an authorization error or redirects the Trainee to their own dashboard.	
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MA-15	Verify user can enable two-factor authentication (2FA) using an authenticator app			
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Pre-requisite: User is logged in with an active account; 2FA is not yet enabled; user has an authenticator app installed on their device

TC-015	1. Navigate to Account Settings/Profile 2. Select "Two-Factor Authentication" and click Enable 3. Scan the displayed QR code (or enter the setup key) in the authenticator app 4. Enter the 6-digit code shown in the authenticator app 5. Confirm	Account: officer.delacruz@pnp.gov.ph Code: valid 6-digit authenticator code	System verifies the code, enables 2FA on the account, and confirms the change. The action is logged for audit purposes.	
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MA-16	Verify login requires an authenticator app code when 2FA is enabled			
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Pre-requisite: User has a registered account with 2FA enabled and linked to an authenticator app

TC-016	1. Navigate to Login 2. Enter valid credentials and click Login 3. Enter the current 6-digit code from the authenticator app 4. Submit	Email: officer.delacruz@pnp.gov.ph Password: Valid123! Code: valid 6-digit authenticator code	System asks for the authenticator code after the credentials are accepted. Once the code is verified, the user is authenticated and redirected to their role-appropriate dashboard.	
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MA-17	Verify login is denied when an incorrect authenticator code is entered			
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Pre-requisite: User has a registered account with 2FA enabled; credentials were accepted

TC-017	1. Navigate to Login and enter valid credentials 2. Enter an incorrect 6-digit code 3. Submit	Code: 000000 (incorrect)	System rejects the code, displays an error, and does not grant access. The user may try again.	
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MA-18	Verify an outdated (expired) authenticator code is rejected			
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Pre-requisite: User has a registered account with 2FA enabled; credentials were accepted

TC-018	1. Navigate to Login and enter valid credentials 2. Note the code shown in the authenticator app and wait for it to refresh 3. Enter the previous (now expired) code 4. Submit	Code: previous 6-digit code from an earlier time window	System rejects the code and prompts the user to enter the current code from the authenticator app.	
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MA-19	Verify user can log in with a recovery code when the authenticator app is unavailable			
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Pre-requisite: User has a registered account with 2FA enabled; user has an unused recovery code

TC-019	1. Navigate to Login and enter valid credentials 2. Select "Use a recovery code" 3. Enter an unused recovery code 4. Submit	Recovery code: valid unused recovery code	System accepts the recovery code, logs the user in, and invalidates that code so it cannot be used again.	
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MA-20	Verify the account is temporarily locked after repeated failed 2FA attempts			
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Pre-requisite: User has a registered account with 2FA enabled; credentials were accepted

TC-020	1. Navigate to Login and enter valid credentials 2. Enter an incorrect code repeatedly until the attempt limit is reached	Code: incorrect code, repeated (e.g., 5 attempts)	System blocks further attempts, temporarily locks the login, and shows an appropriate notice. The event is logged for audit purposes.	
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MA-21	Verify user can disable two-factor authentication			
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Pre-requisite: User is logged in with an active account; 2FA is enabled

TC-021	1. Navigate to Account Settings/Profile 2. Select "Two-Factor Authentication" and click Disable 3. Confirm with the current password and a current authenticator code	Current Password: Valid123! Code: valid 6-digit authenticator code	System verifies the password and code, then disables 2FA. Subsequent logins no longer ask for a code. The action is logged for audit purposes.	
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Table 10: Test Cases: Manage Account

LEARN MODULE				
TC ID	Test Steps	Test Data	Actual Result	Status
LM-01	Verify PNP Trainee can launch and progress through a module's lessons			

Pre-requisite: PNP Trainee is logged in and enrolled in the module

TC-001	1. Select an enrolled module from My Learning 2. Click Launch 3. Select a lesson 4. Read/view the lesson content 5. Mark the lesson complete 6. Repeat for remaining lessons	Module: Module 1 - Fundamentals of Artificial Intelligence	Lesson content loads correctly. Each completed lesson is marked done and progress bar updates. Next lesson unlocks. Final assessment unlocks once all lessons are complete.	
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LM-02	Verify PNP Trainee can consult the AI Chatbot for in-scope guidance during a lesson			
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Pre-requisite: PNP Trainee is actively viewing a lesson

TC-002	1. Open the AI Chatbot panel 2. Type a question related to the current lesson's content 3. Submit the question	Question: "What is the difference between narrow AI and general AI?"	AI Chatbot responds with guidance limited to the current module's approved educational content.	
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LM-03	Verify progress is preserved if connectivity is lost mid-lesson			
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Pre-requisite: PNP Trainee is mid-lesson with at least one completed step

TC-003	1. Complete at least one lesson step 2. Simulate loss of connectivity 3. Reconnect and reopen the module	N/A	System preserves the last completed step. Trainee resumes from that point without losing progress.	
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LM-04	Verify AI Chatbot declines to answer questions outside the module's approved scope			
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Pre-requisite: PNP Trainee is actively viewing a lesson

TC-004	1. Open the AI Chatbot panel 2. Type a question unrelated to the current module's content 3. Submit the question	Question: "What's the weather like today?"	AI Chatbot declines to answer and redirects the Trainee back to the current lesson's content.	
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LM-05	Verify PNP Trainee cannot open a lesson that is still locked			
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Pre-requisite: PNP Trainee is enrolled in a module and has not completed the previous lesson

TC-005	1. Open the module from My Learning 2. Attempt to select a lesson that comes after the current incomplete lesson	Module: Module 1 - Fundamentals of Artificial Intelligence Lesson: Lesson 3 (locked)	System keeps the lesson locked and informs the Trainee to complete the previous lesson first.	
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LM-06	Verify system shows a notice when the AI Chatbot is unavailable			
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Pre-requisite: PNP Trainee is actively viewing a lesson; AI Chatbot service is unreachable or its usage limit is exceeded

TC-006	1. Open the AI Chatbot panel 2. Type a question related to the current lesson 3. Submit the question	Simulated condition: AI Chatbot service unavailable	System displays a clear notice that the assistant is temporarily unavailable. Lesson content and progress remain usable.	
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Table 11: Test Cases: Learn Module

TAKE ASSESSMENT				
TC ID	Test Steps	Test Data	Actual Result	Status
TA-01	Verify PNP Trainee can pass a module's final assessment			

Pre-requisite: PNP Trainee has completed all lessons in the module

Pre-condition: The module's final assessment has been unlocked

TC-001	1. Start the module's final assessment 2. Answer each question 3. Submit the assessment 4. View the result	Assessment: Module 1 Final Assessment (20 questions) Passing score: 70%	System grades the submission against the answer key. Module is marked Passed. Certificate generation is unlocked if this is the final module.	
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TA-02	Verify PNP Trainee is notified and allowed to retake when failing an assessment			
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Pre-requisite: PNP Trainee has started the module's final assessment

TC-002	1. Start the module's final assessment 2. Answer questions with a score below the passing threshold 3. Submit the assessment	Score achieved: 50% (below 70% passing threshold)	System informs the Trainee of the failing result and allows the Trainee to review material before retaking the assessment.	
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TA-03	Verify an incomplete assessment submission is not graded			
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Pre-requisite: PNP Trainee has started the module's final assessment

TC-003	1. Start the module's final assessment 2. Answer some but not all required questions 3. Exit the assessment without submitting	N/A	System does not grade the attempt and prompts the Trainee to complete it later.	
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TA-04	Verify the final assessment stays locked until all lessons are complete			
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Pre-requisite: PNP Trainee is enrolled in a module with at least one incomplete lesson

TC-004	1. Open the module from My Learning 2. Attempt to start the final assessment	Module: Module 1 - Fundamentals of Artificial Intelligence	System keeps the final assessment locked and indicates which lessons still need to be completed.	
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TA-05	Verify PNP Trainee can claim a certificate after completing all modules			
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Pre-requisite: PNP Trainee has passed the final assessment of all required modules

TC-005	1. Open the progress tracking view or My Learning 2. Click Claim Certificate 3. View/download the generated certificate	N/A	System generates the certificate with a barcode and the Trainee's details, and the Trainee can view or download it. Certificate status is updated to Claimed.	
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TA-06	Verify the certificate barcode can be verified			
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Pre-requisite: A certificate has been generated for a Trainee

TC-006	1. Scan or enter the barcode/ID from the certificate in the verification page 2. Submit	Barcode: valid certificate barcode	System confirms the certificate is authentic and displays the matching Trainee and completion details. An invalid barcode is reported as not found.	
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Table 12: Test Cases: Take Assessment

MONITOR PROGRESS				
TC ID	Test Steps	Test Data	Actual Result	Status
MP-01	Verify PNP Trainee can view their own module completion status			

Pre-requisite: PNP Trainee is logged in

TC-001	1. Open the progress tracking view	N/A	System retrieves and displays own completion percentages, passed/failed assessments, and certificate status.	
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MP-02	Verify Supervisor can view and filter progress data for personnel under them			
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Pre-requisite: Supervisor is logged in; progress data exists for at least one personnel account

TC-002	1. Open the progress dashboard 2. Filter/search by personnel, unit, or module 3. Review the displayed data 4. Export a report (optional)	Filter: Unit = "Region 4A"	System displays completion percentages, passed/failed assessments, and certificate status for the filtered personnel. Report generates successfully if exported.	
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MP-03	Verify system displays a 'No activity' status when no progress data exists			
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Pre-requisite: User account has not started any module

TC-003	1. Open the progress tracking view for a user with no started modules	N/A	System displays a 'No activity' status instead of an error.	
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MP-04	Verify PNP Trainee cannot view other personnel's progress data			
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Pre-requisite: PNP Trainee is logged in

TC-004	1. Open the progress tracking view 2. Attempt to open another Trainee's progress (e.g., via URL or search)	Account: Trainee Target: another Trainee's progress page	System shows only the Trainee's own progress and denies access to other personnel's data.	
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Table 13: Test Cases: Monitor Progress

MANAGE TRAINING CONTENT				
TC ID	Test Steps	Test Data	Actual Result	Status
MTC-01	Verify Supervisor can add, update, or remove training content			

Pre-requisite: Supervisor is logged in with content-management permissions

TC-001	1. Go to the content management section 2. Select content to add, update, or remove 3. Make the changes 4. Save	Change: Update Lesson 3 slide content in Module 2	System validates and saves the changes. Updated content is immediately reflected to Trainees and to what the AI Chatbot can reference.	
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MTC-02	Verify system rejects incomplete or invalid content changes			
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Pre-requisite: Supervisor is logged in with content-management permissions

TC-002	1. Go to the content management section 2. Attempt to save a change with a required field left blank 3. Click Save	Missing field: Lesson title	System rejects the save and shows an error message asking the Supervisor to correct the entry.	
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MTC-03	Verify removed content is no longer shown to Trainees			
<i>Pre-requisite: Supervisor is logged in with content-management permissions; the content exists</i>				
TC-003	1. Go to the content management section 2. Select a lesson to remove 3. Confirm the removal 4. Log in as a Trainee and open the module	Content: Lesson 5 in Module 2	System removes the lesson after confirmation. It no longer appears for Trainees or in what the AI Chatbot can reference.	

Table 14: Test Cases: Manage Training Content

CREATE / MANAGE				
TC ID	Test Steps	Test Data	Actual Result	Status
CMM-01	Verify Supervisor can create a new module with lessons and a final assessment			
<i>Pre-requisite: Supervisor is logged in and inside the Manage Training Content section</i>				
TC-001	1. Choose to create a new module 2. Add lessons, activities, and final assessment questions 3. Arrange the order of lessons 4. Confirm the module is complete	Module: "Module 3: Responsible AI, Ethics, and Human Rights"	System checks all required parts are present (lessons + assessment), publishes the module, and makes it visible to Trainees for enrollment.	
CMM-02	Verify system blocks publishing a module missing a required part			
<i>Pre-requisite: Supervisor is building or editing a module</i>				
TC-002	1. Build a module with lessons but no final assessment 2. Attempt to publish	Missing: Final assessment questions	System blocks publishing and lists the missing required part(s).	
CMM-03	Verify Supervisor can edit an existing published module			
<i>Pre-requisite: Supervisor is logged in; a published module exists</i>				
TC-003	1. Open an existing module 2. Edit the module title, lesson order, or assessment questions 3. Save the changes	Module: "Module 3: Responsible AI, Ethics, and Human Rights"	System saves the changes and shows the updated module to Trainees. Existing Trainee progress is not lost.	
CMM-04	Verify Supervisor can unpublish or delete a module			
<i>Pre-requisite: Supervisor is logged in; a module exists</i>				
TC-004	1. Select a module 2. Choose Unpublish or Delete 3. Confirm the action	Module: "Module 3: Responsible AI, Ethics, and Human Rights"	System asks for confirmation, then removes the module from Trainee view (and from enrollment) according to the chosen action.	

Table 15: Test Cases: Create/Manage Modules

MANAGE PLATFORM				
TC ID	Test Steps	Test Data	Actual Result	Status
MPL-01	Verify Admin can review platform health, security status, and AI Chatbot usage stats			
<i>Pre-requisite: Admin is logged in with system-administration permissions</i>				
TC-001	1. Log in and open the system administration panel 2. Review platform health, security status, and AI Chatbot usage stats 3. Make any needed adjustments (e.g., security settings, server configuration)	N/A	System displays current health/security/usage data. Adjustments are applied and the action is logged for audit purposes.	
MPL-02	Verify Admin is alerted immediately when a critical issue is detected			
<i>Pre-requisite: Admin is logged in; a critical issue occurs (e.g., server downtime, AI usage limit exceeded)</i>				
TC-002	1. Simulate a critical issue (e.g., AI Chatbot usage limit exceeded) 2. Observe system behavior	Simulated condition: AI Chatbot usage limit exceeded	System immediately alerts the Admin rather than waiting for the next routine check.	
MPL-03	Verify Admin can manage the AI Chatbot connection settings			
<i>Pre-requisite: Admin is logged in with system-administration permissions</i>				
TC-003	1. Open the AI Chatbot integration settings in the administration panel 2. Update a setting (e.g., usage limit or connection key) 3. Save	Setting: AI Chatbot usage limit	System validates and saves the setting, the new value takes effect, and the change is logged for audit purposes.	

Table 16: Test Cases: Manage Platform

Gap Analysis

Product Backlog ID	Current System	Proposed Changes	Impact
US1.1	No dedicated training account exists; personnel rely solely on POA credentials.	PNP Personnel register and log in directly on the SPT Platform through their own dedicated account.	Enables personnel to access specialized AI training independently of POA account limitations.
US1.2	User authentication and credential management for AI training does not exist.	Admin manages user registration, authentication, and credentials through the platform's own backend and database.	Ensures personnel identities are validated and secured without relying on or duplicating POA's authentication system.

US1.3	No mechanism exists to manage AI training account status independently.	Admin can lock, disable, or reinstate individual accounts.	Allows the platform to respond to security concerns without PNP-wide account intervention.
US2.1	AI training is delivered via in-person lecture using printed handouts and slide presentations (Table 1, P002–P003).	Personnel engage in a self-paced, multimedia-driven 7-module curriculum.	Removes dependence on scheduled classroom sessions, allowing personnel to train at their own pace.
US2.2	Attendance and completion are tracked manually via sign-in sheets and spreadsheets	Course progression is tracked and saved asynchronously in the background.	Eliminates lost progress and manual re-entry, allowing personnel to resume training exactly where they left off.
US2.3	No dedicated data infrastructure exists for AI training records.	Curriculum configurations and learning metrics are isolated within the platform's own dedicated database.	Keeps the platform's data footprint minimal and prevents server strain from PNP's broader user base.
US2.4	Personnel have no way to check their own training status; progress exists only on paper records held by Training Officers.	PNP Personnel can view their own module completion status.	Gives personnel direct visibility into their own progress without needing to request records manually.
US2.5	Lecture content is delivered linearly by an instructor with no enforced completion gating	Lessons unlock sequentially, with the final assessment gated until all lessons are completed.	Ensures personnel engage with all required material before being evaluated.
US2.6	Failed evaluations are handled ad hoc through verbal or paper-based re-testing	Personnel are notified of failing results and permitted to retake the assessment after reviewing material.	Standardizes the retake process and ensures consistent evaluation standards across all personnel.
US3.1	No interactive guidance is available;	Personnel can consult a real-time AI tutor panel for	Provides immediate clarification of concepts without

	personnel passively receive lecture content	contextual guidance during lessons.	requiring instructor availability.
US3.2	Not applicable — no AI integration exists in the current system.	All AI requests are routed through the platform's backend, transmitting only approved educational content with no personally identifiable information.	Ensures compliance with data privacy requirements while enabling AI-assisted learning.
US3.3	Not applicable — no AI integration exists in the current system.	The AI tutor's responses are limited strictly to the current module's approved content.	Prevents the AI assistant from providing inaccurate or out-of-scope guidance.
US4.1	Certificates of attendance are issued manually on paper.	A scannable, verifiable digital certificate is generated automatically upon passing a module's final assessment.	Eliminates administrative delay in issuing proof of training completion.
US4.2	Certificate authenticity cannot be verified; records are cross-checked manually against spreadsheets.	Supervisors cross-reference a certificate's scannable hash against the platform's completion logs.	Removes manual document review and reduces the risk of fraudulent or unverifiable certificates.
US4.3	Workforce completion rates are compiled manually from attendance sheets and grading records.	Supervisors view aggregated workforce completion rates on an automated dashboard.	Provides real-time compliance visibility without manual data collection.
US4.4	Training records exist only as physical documents or local spreadsheets, with no centralized digital storage.	Platform transaction logs are securely synchronized with the client's localized production infrastructure.	Keeps operational data under PNP ownership in compliance with institutional data privacy mandates.
US5.1	Training content (slides, handouts) is updated	Supervisors add, update, or remove training content	Ensures the curriculum can be kept current without

	manually and redistributed physically or via shared files.	directly within the platform.	reprinting or redistributing physical materials.
US5.2	No structured module-authoring process exists; instructional materials are assembled informally per session.	Supervisors build or restructure individual modules, with the system blocking publication until all required parts are complete.	Guarantees every published module is complete and consistent before personnel can access it.

Table 17: Gap Analysis

System Analysis and Design

System Context Diagram

C1 - System Context: SPT Platform

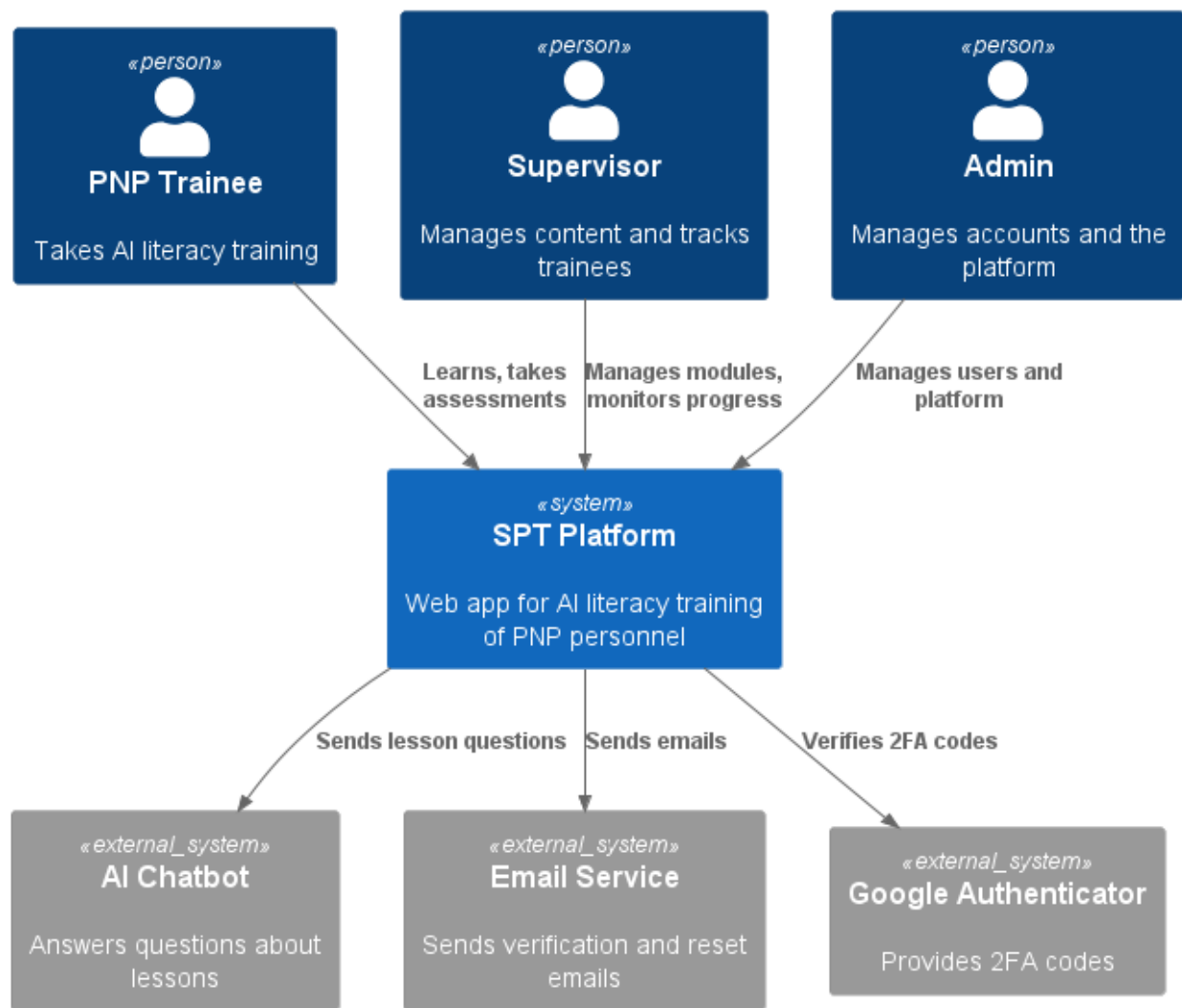


Figure 16: System Context Diagram

Container Diagram

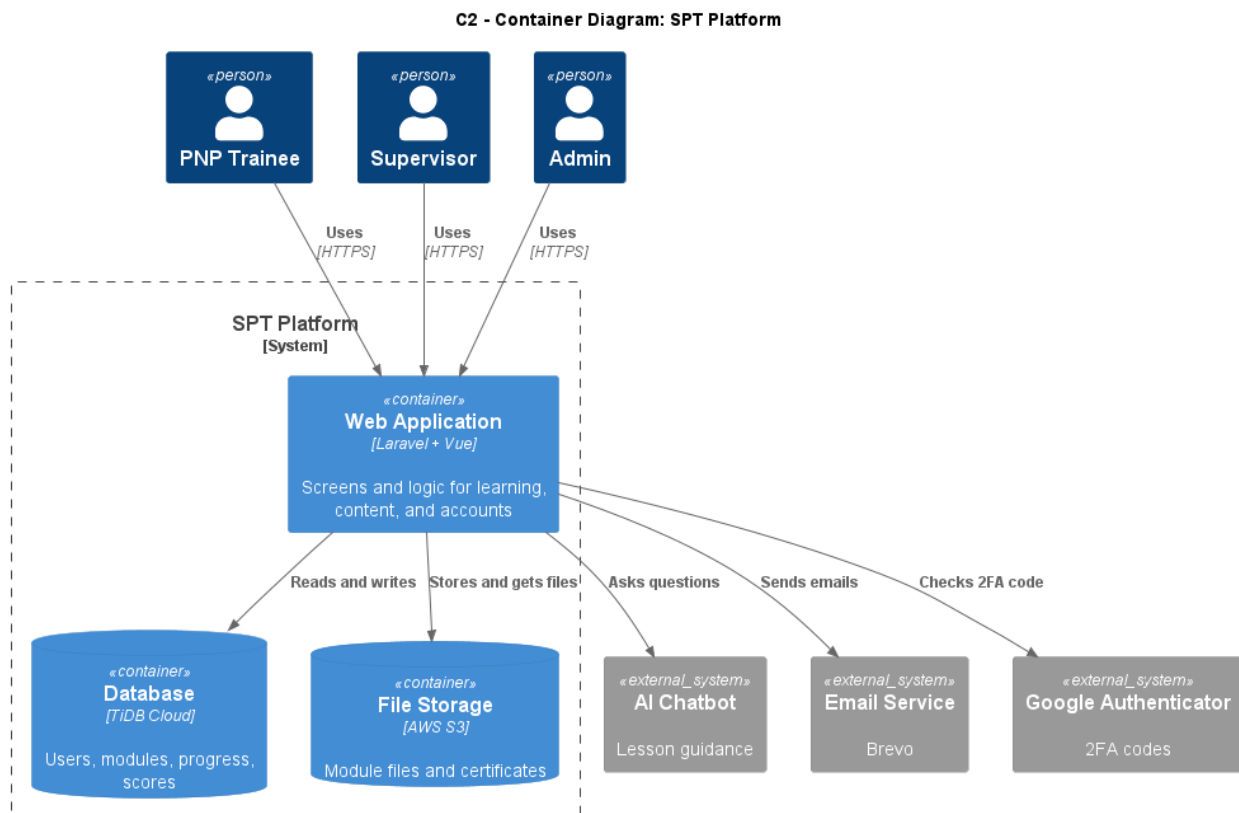


Figure 17: Container Diagram

Component Diagram

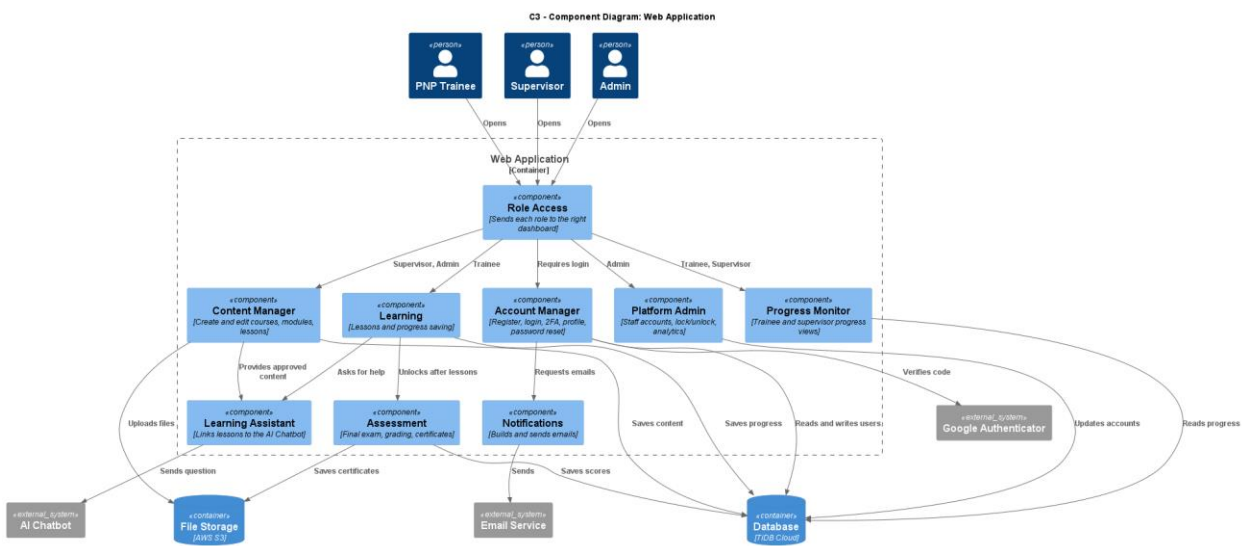


Figure 18: Component Diagram

Entity Relationship Diagram

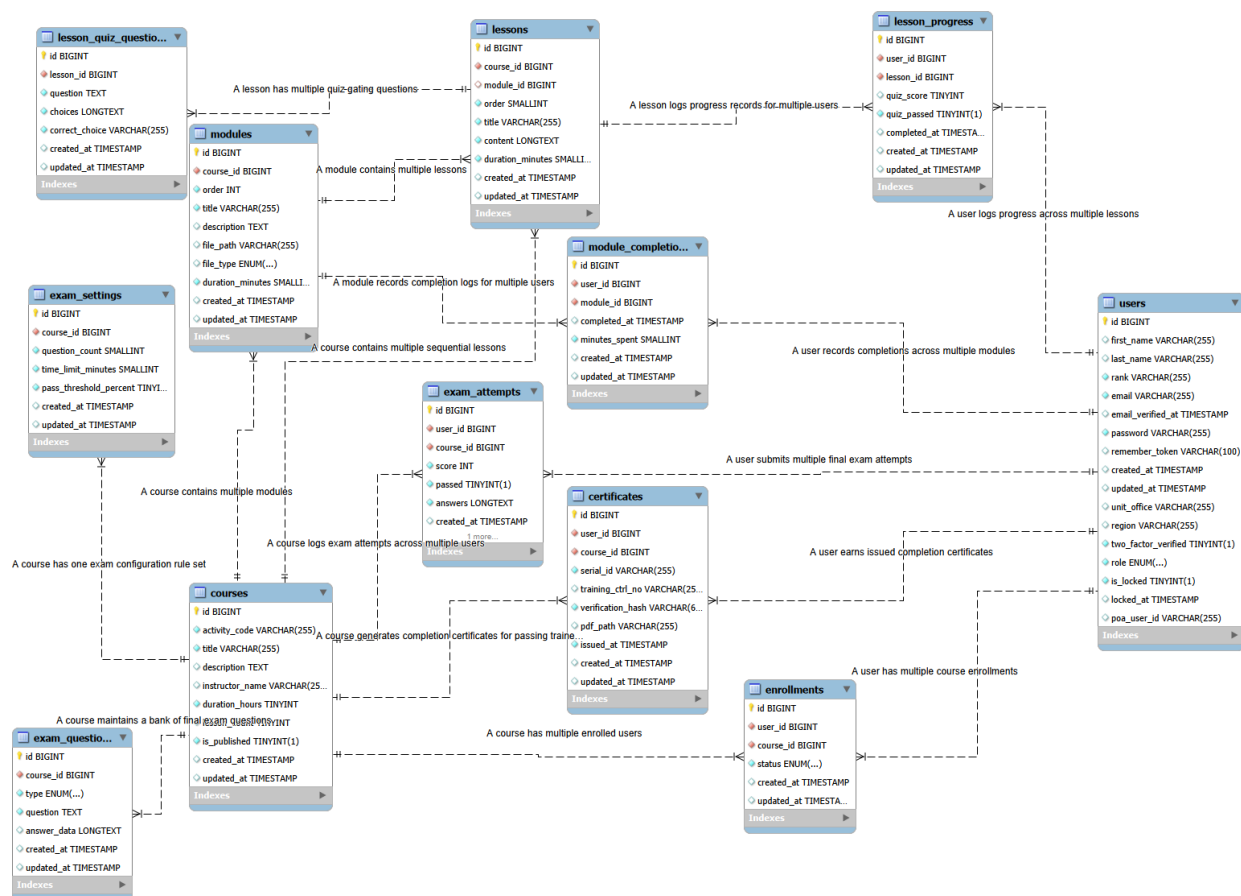


Figure 19: Entity Relationship Diagram

Results and Discussion

This section presents the results of the current iteration of the PNP Specialized Personnel Training (PP-SPT) Platform project. It reports progress against the release plan and the product backlog, presents the working prototype delivered for each user role, and describes the user classes the platform serves. Where results fall short of the objectives stated in the Introduction, the section explains the reasons and identifies the dependencies that remain open. The findings draw on the deployed prototype, the prototype showcase held for the client's DICTM Technical Working Group on 17 September 2026, and the client and adviser consultations documented in Appendix D.

Release Plan

(

Include here your updated release plan and discuss whether you are on/ahead/behind schedule. Explain also how much of the product backlog has been finished (it is up to you how you will present it, without being repetitive. You can say something like, “The complete product backlog is in Table ____”.

)

Prototype

Trainee User Interface


PNP LMS v2.4 SECURED
 Specialized Police Training & Education Portal

Portal Log In Officer Portal

Enter your PNP email and password to access the LMS portal.

SELECT ACCESS ROLE

Officer / Trainee

Supervisor

Administrator

PNP Email Address

✉ maria.cruz@pnp.gov.ph

Password Forgot password?

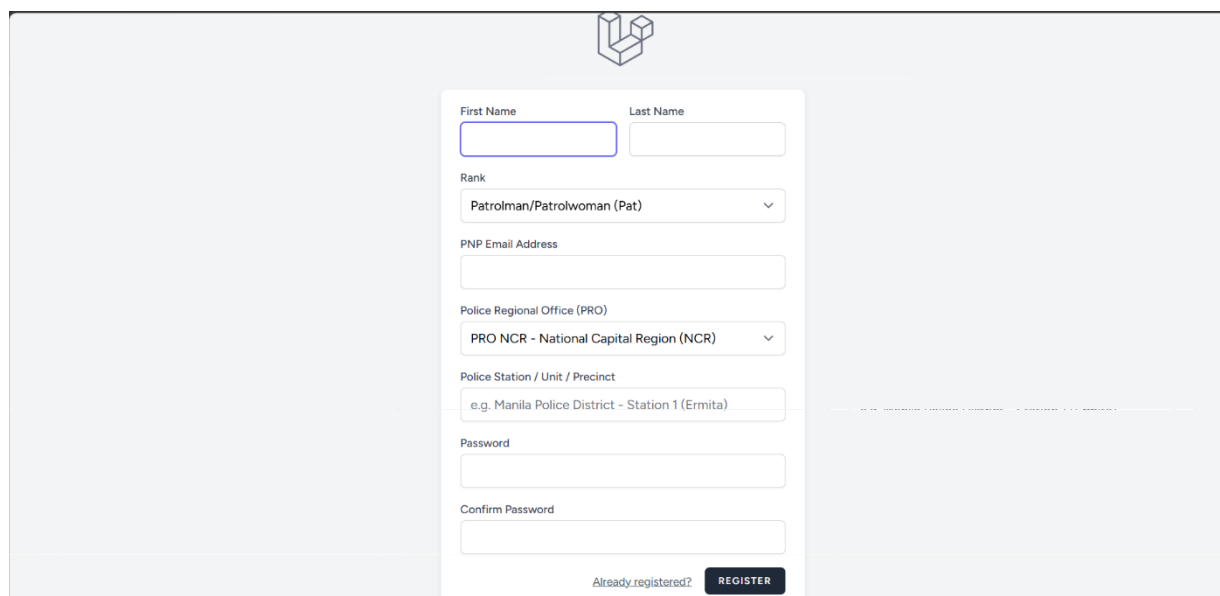
🔒

→ Log In to Officer Portal

Don't have an account yet? [Register New Account](#) →

🔒 Encrypted 256-Bit SSL Connection 🛡️ PNP Cybercrime Compliant
 © 2026 Philippine National Police — Directorate for Human Resource & Doctrine Development

Figure 20: Trainee Interface – Login Page



The image shows a web interface for registering a new account. At the top center is a logo consisting of three stacked cubes. Below the logo is a registration form with the following fields:

- First Name: A text input field.
- Last Name: A text input field.
- Rank: A dropdown menu with the selected option "Patrolman/Patrolwoman (Pat)".
- PNP Email Address: A text input field.
- Police Regional Office (PRO): A dropdown menu with the selected option "PRO NCR - National Capital Region (NCR)".
- Police Station / Unit / Precinct: A text input field with the placeholder text "e.g. Manila Police District - Station 1 (Ermita)".
- Password: A text input field.
- Confirm Password: A text input field.

At the bottom of the form, there is a link "Already registered?" and a dark blue button labeled "REGISTER".

Figure 21: Trainee Interface – Register New Account

PNP LMS
Officer Portal

maria.cruz@pnp.gov.ph **TRAINER** Log out

Manila Police District - Station 1 (Ermita)

2 new modules available this week

Welcome Officer

Continue your professional development with our training modules and certification programs.

My Progress

COURSES ENROLLED	HOURS SPENT	COMPLETED MODULES	CERTIFICATES EARNED
2	0	0	1

My Learning

Resume your enrolled courses and track your progress.

[Go to My Learning →](#)

Find Learning

Browse the catalog and discover new training modules.

[Find Learning →](#)

Certificates

View and download your earned module credentials.

[View Certificates →](#)

My Learning

Resume your enrolled courses and track your progress.

[Go to My Learning →](#)

Find Learning

Browse the catalog and discover new training modules.

[Find Learning →](#)

Certificates

View and download your earned module credentials.

[View Certificates →](#)

Recent Exams (History)

Read-only records of your completed assessments.

[View History Archive →](#)

Exam Title	Final Grade	Status
Fundamentals of Artificial Intelligence Final Exam Completed on 2026-09-23	100%	✓ Passed
Fundamentals of Artificial Intelligence Final Exam Completed on 2026-09-23	100%	✓ Passed
Fundamentals of Artificial Intelligence Final Exam Completed on 2026-09-23	100%	✓ Passed

Recommended Next Steps

- Complete Police Ethics modules
- Review Community Policing certificate
- Explore new AI Law Enforcement course

Figure 22: Trainee Interface – Officer Dashboard

PNP LMS
Officer Portal

Dashboard
Courses
Certificates
Verification

MC

Back to Home

Course Catalog

My Learning (2)

Course Catalog

Browse and enroll in available learning modules (5 to 6 hours each).

Search catalog by title or code...

ACTIVITY CODE	MODULE DETAILS	STATUS	ACTION
SRAIU-2026-M1	Fundamentals of Artificial Intelligence 6 Hours - 2 Lessons (Includes Final Assessment)	Completed	Completed
SRAIU-2026-M2	Generative AI and AI Tools for Law Enforcement 6 Hours - 1 Lessons (Includes Final Assessment)	Enrolled	Drop
SRAIU-2026-M3	Responsible AI, Ethics, and Human Rights 6 Hours - 1 Lessons (Includes Final Assessment)	Available	Enroll
SRAIU-2026-M4	Data Privacy, Cybersecurity, and AI Risks 6 Hours - 1 Lessons (Includes Final Assessment)	Available	Enroll

PNP LMS
Officer Portal

Dashboard
Courses
Certificates
Verification

MC

Back to Home

Course Catalog

My Learning (2)

My Learning

Continue your active course modules and take knowledge check quizzes.

MODULE TITLE	PROGRESS	LAUNCH	FINAL EXAM
Fundamentals of Artificial Intelligence 2 of 2 modules completed	100%		Passed
Generative AI and AI Tools for Law Enforcement 0 of 0 modules completed	0%		Locked

Figure 23: Trainee Interface – Courses (Officer)

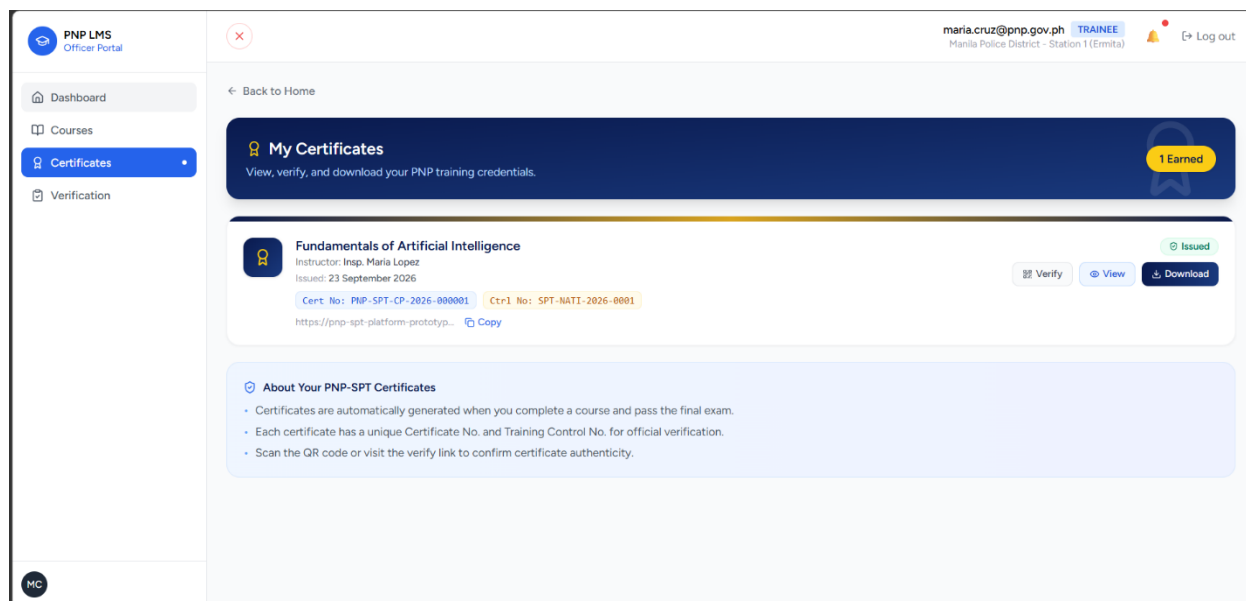


Figure 24: Trainee Interface – My Certificates (Officer)

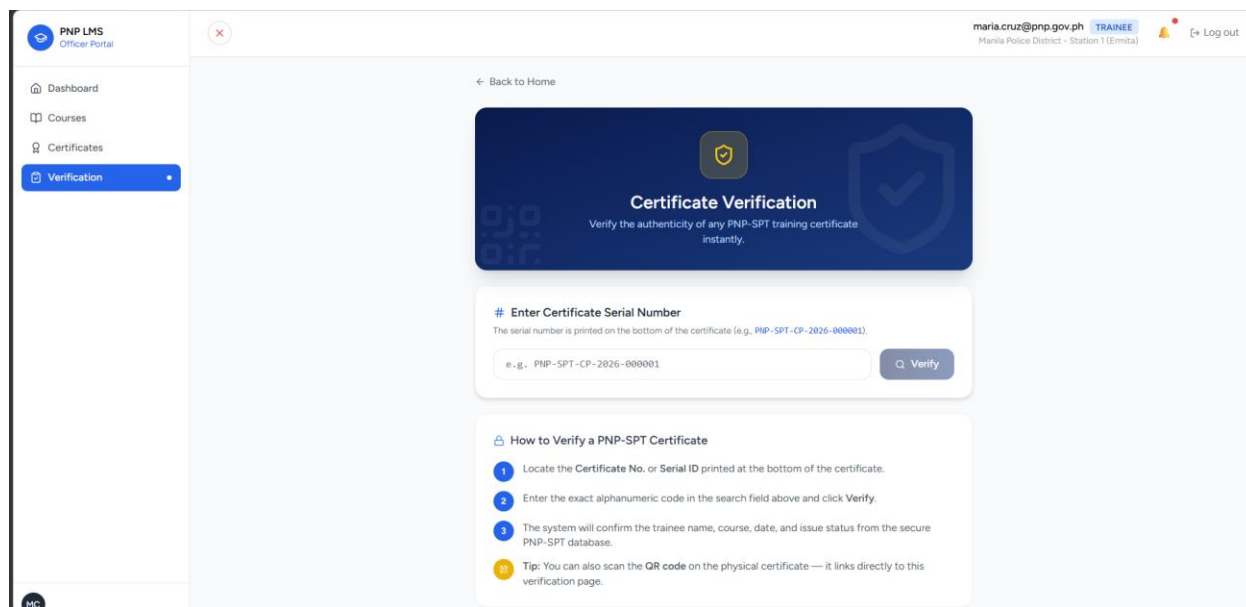


Figure 25: Trainee Interface – Certificate Verification (Officer)

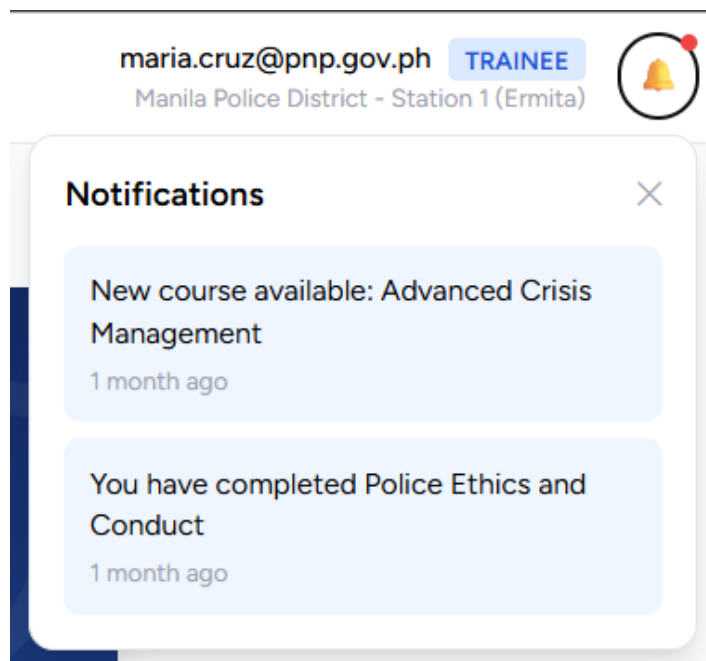


Figure 26: Trainee Interface – Notification

Supervisor User Interface

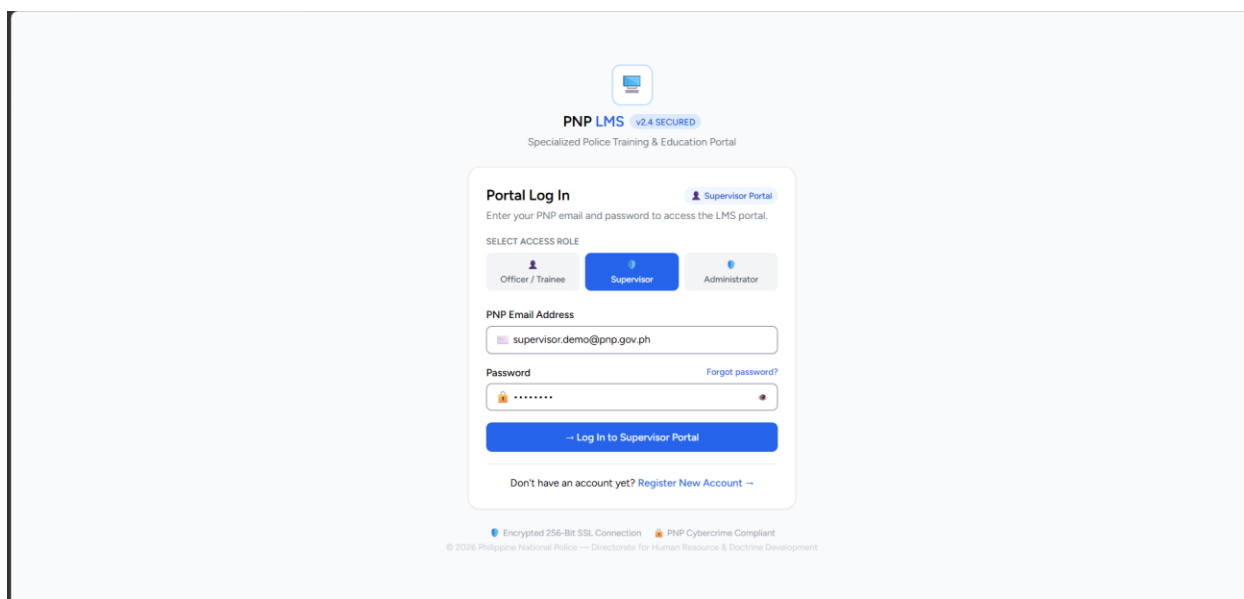
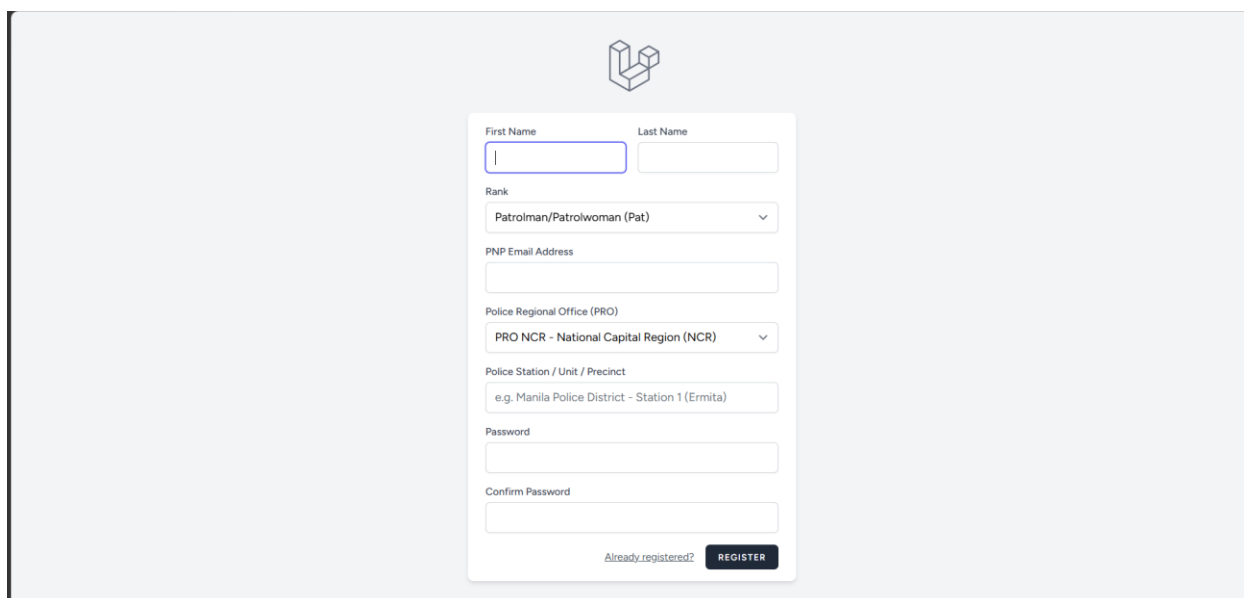


Figure 27: Supervisor Interface – Login Page



Registration form for a new Supervisor account. The form includes fields for First Name, Last Name, Rank, PNP Email Address, Police Regional Office (PRO), Police Station / Unit / Precinct, Password, and Confirm Password. A 'REGISTER' button is at the bottom right, and a link for 'Already registered?' is at the bottom left.

First Name Last Name

Rank
Patrolman/Patrolwoman (Pat) ▼

PNP Email Address

Police Regional Office (PRO)
PRO NCR - National Capital Region (NCR) ▼

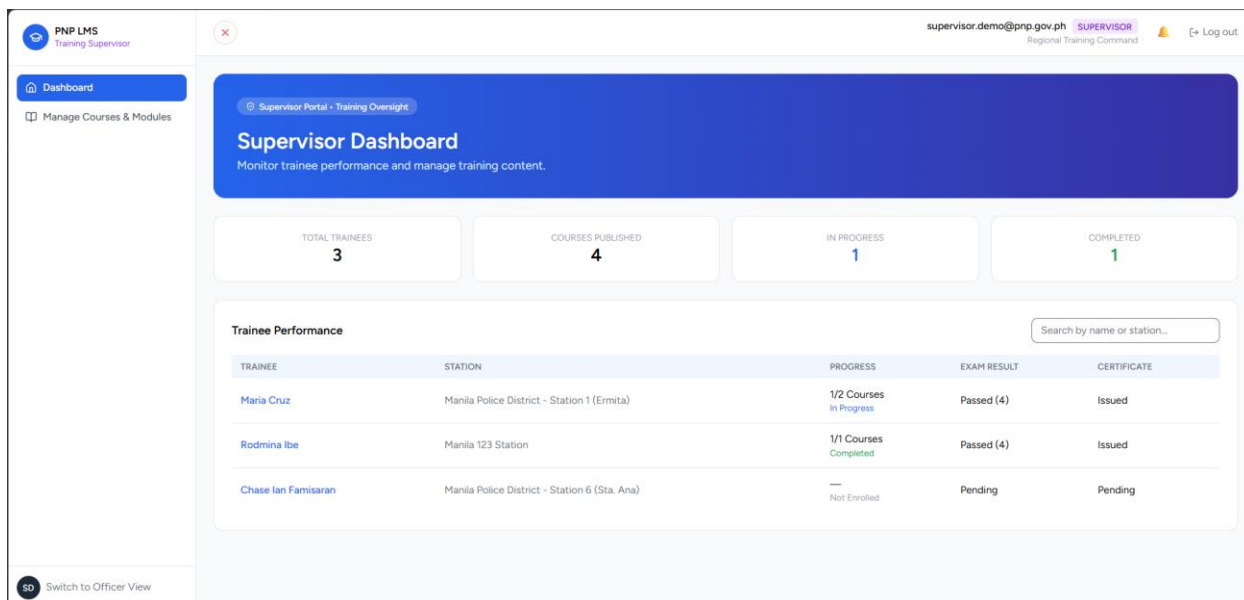
Police Station / Unit / Precinct
e.g. Manila Police District - Station 1 (Ermita)

Password

Confirm Password

[Already registered?](#) **REGISTER**

Figure 28: Supervisor Interface – Register New Account



Supervisor Dashboard interface showing trainee performance and training content management. The dashboard includes a sidebar with navigation links (Dashboard, Manage Courses & Modules) and a main content area with a summary of trainee statistics and a table of trainee performance.

Supervisor Dashboard
Monitor trainee performance and manage training content.

Summary Statistics:

- TOTAL TRAINEES: 3
- COURSES PUBLISHED: 4
- IN PROGRESS: 1
- COMPLETED: 1

Trainee Performance

TRAINEE	STATION	PROGRESS	EXAM RESULT	CERTIFICATE
Maria Cruz	Manila Police District - Station 1 (Ermita)	1/2 Courses In Progress	Passed (4)	Issued
Rodmina Ibe	Manila 123 Station	1/1 Courses Completed	Passed (4)	Issued
Chase Ian Famisaran	Manila Police District - Station 6 (Sta. Ana)	Not Enrolled	Pending	Pending

Search by name or station...

Switch to Officer View

Figure 29: Supervisor Interface – Supervisor Dashboard

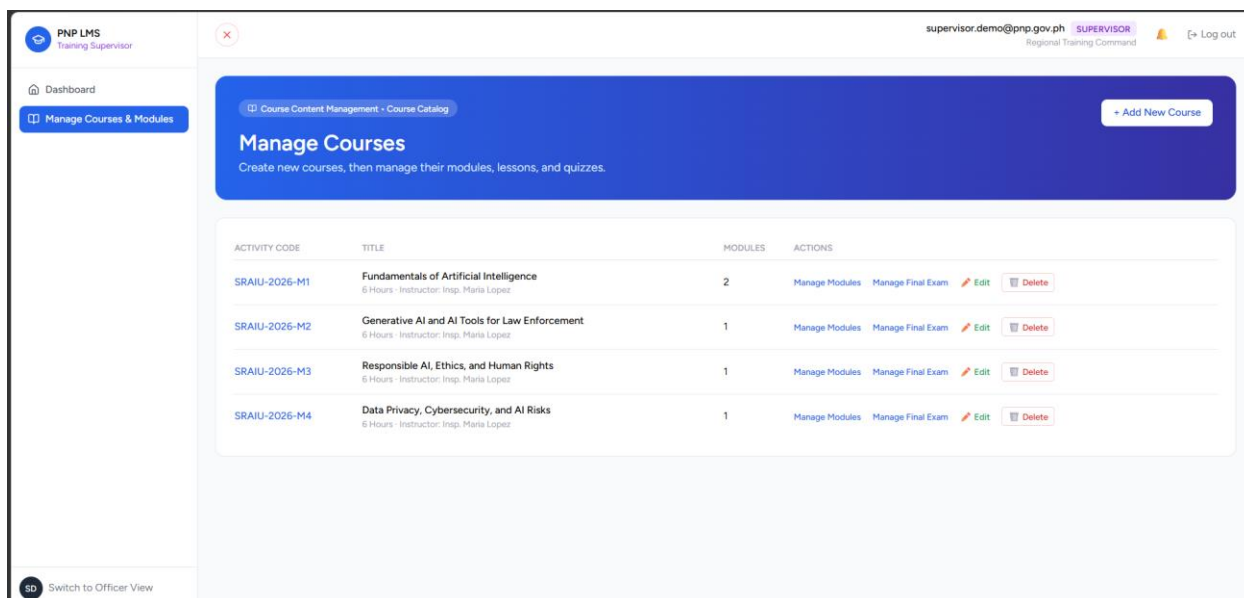


Figure 30: Supervisor Interface – Manage Courses

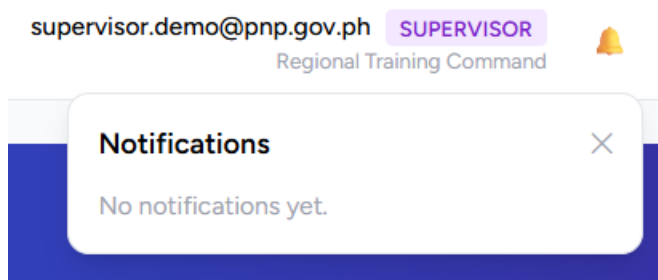


Figure 31: Supervisor Interface – Notification

Admin User Interface

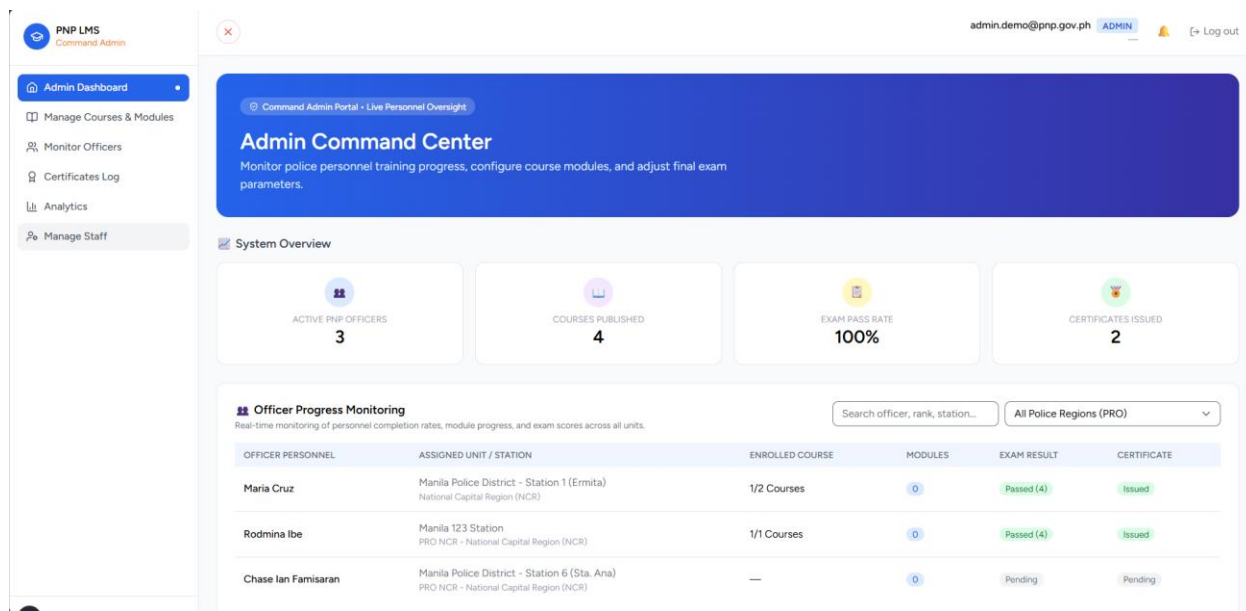


Figure 32: Admin Interface – Admin Dashboard

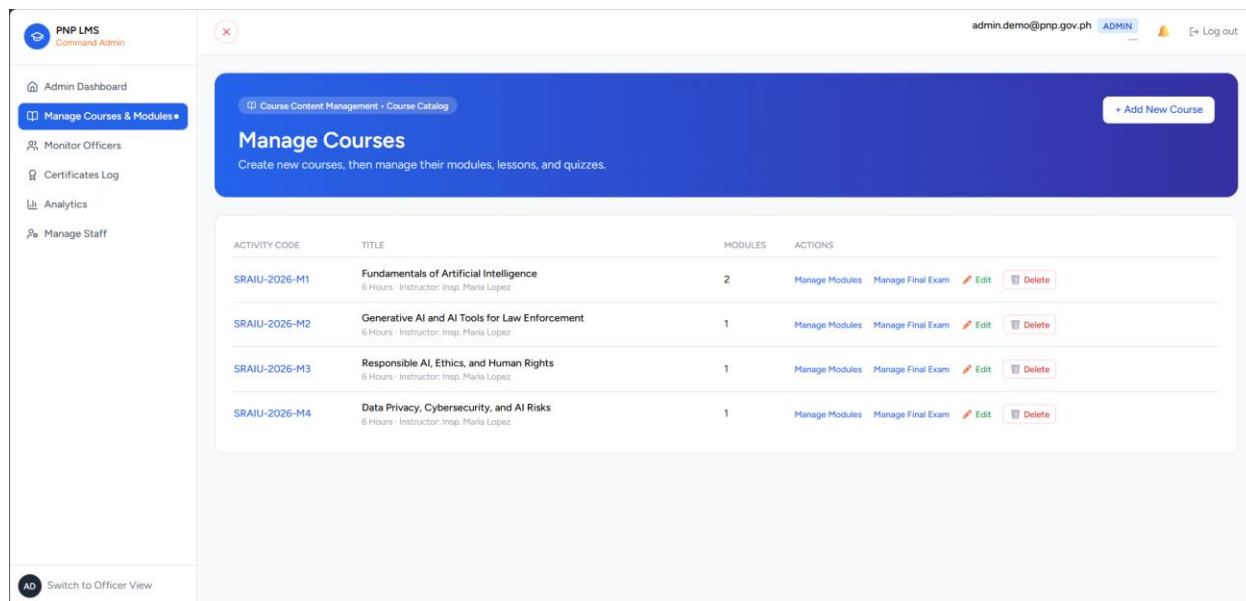


Figure 33: Admin Interface – Manage Courses

PNP LMS
Command Admin

admin.demo@pnp.gov.ph ADMIN [+] Log out

Admin Dashboard
Manage Courses & Modules
Monitor Officers
Certificates Log
Analytics
Manage Staff

Monitor Officers

Track enrolled police trainees, module progress percentages, exam results, and certification status.

Search by name, email, or organization...

Total Students: 3
Total Enrollments: 3
Courses Completed: 2
Avg Progress: 50%

STUDENT	ORGANIZATION	COURSES	PROGRESS	CERTIFICATES	ACTIONS
Maria Cruz maria.cruz@pnp.gov.ph	Manila Police District - Station 1 (Ermita)	1/2 complete	50% complete	1	View Details Edit Lock Delete
Rodmina Ibe raibe@student.apc.edu.ph	Manila 123 Station	1/1 complete	100% complete	1	View Details Edit Lock Delete
Chase Ian Famisanan chase.famisanan@gmail.com	Manila Police District - Station 6 (Sta. Ana)	0/0 complete	0% complete	0	View Details Edit Lock Delete

Switch to Officer View

Figure 34: Admin Interface – Manage Officers

PNP LMS
Command Admin

admin.demo@pnp.gov.ph ADMIN [+] Log out

Admin Dashboard
Manage Courses & Modules
Monitor Officers
Certificates Log
Analytics
Manage Staff

Monitor Certificates

Audit, verify, and download official PNP certificates issued to personnel across regional commands.

Search certificate by officer name or police station...

Monitored Units: 2 Stations
Trainee Personnel: 2 Officers
Total Issued Certificates: 2
Verification Status: 100% Validated

Manila Police District - Station 1 (Ermita) NCR	1 Officers Certified
Manila 123 Station NCR	1 Officers Certified

Switch to Officer View

Figure 35: Admin Interface – Monitor Certificates

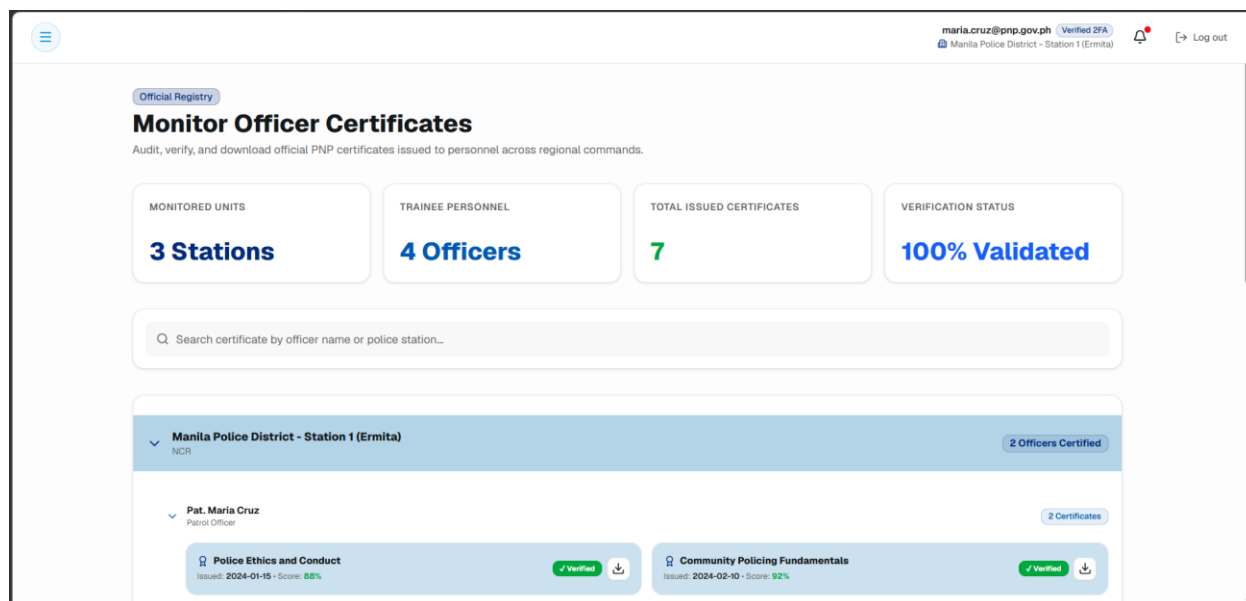


Figure 36: Admin Interface – Monitor Officers Certificates

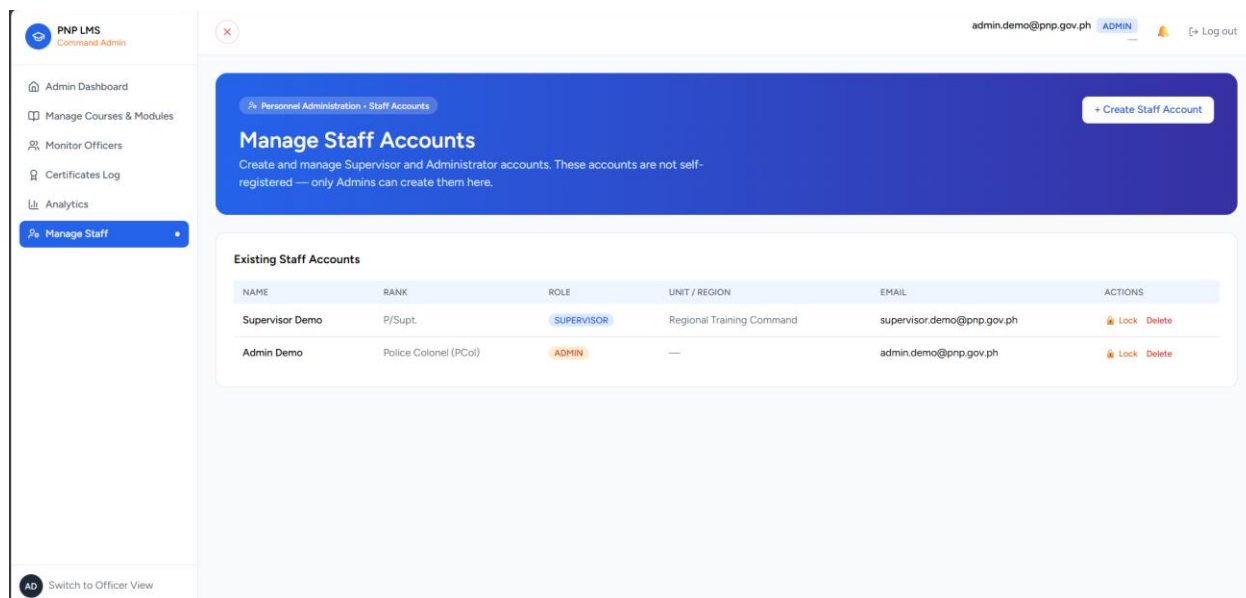


Figure 37: Admin Interface – Manage Staffs

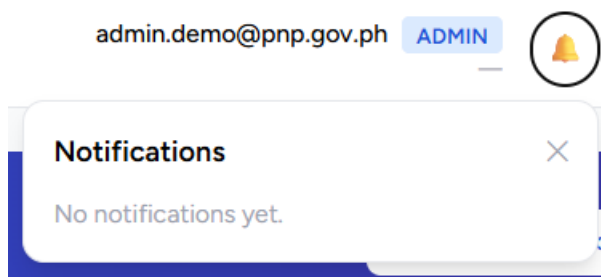


Figure 38: Admin Interface – Notification

Use Classes and Characteristics

Roles	Description
PNP Personnel	The primary users who register and log in directly on the SPT Platform to complete the standardized 7-module AI curriculum and interact with the AI learning assistant.
Supervisors	Users responsible for monitoring workforce training compliance and maintaining the training curriculum, utilizing the system's automated progress tracking and scannable, verifiable certificates to monitor module completion rates without manually grading submissions, and managing and updating training content and modules as needed.
Admin	Technical personnel (IT Department) responsible for maintaining the platform's compatibility with the existing POA platform for future integration, managing user accounts and access, and monitoring the health, security, and usage limits of the third-party AI API connection.

Table 18: Use Classes and Characteristics

Conclusion

This iteration of the PNP Specialized Personnel Training (PNP-SPT) Platform project advanced the work from the requirements and design outputs of the previous term into a working, deployed prototype that was demonstrated to the client's DICTM Technical Working Group on 17 September 2026. The project set out five specific objectives. Of these, two have been achieved, two have been partially achieved, and one has not yet been started because it depends on infrastructure decisions that remain with the client. Each objective is summarized below, together with how it was pursued and what remains to be done.

Objective 1: Develop a self-paced, web-based training platform that removes dependence on classroom instruction. This objective has been achieved at the level of core functionality. The platform was implemented as an interoperable web application using Laravel 11 and PHP 8.2, matching the Police Officer Academy (POA) technology stack as agreed with the PNP ITMS Office, with a Vue front end delivered through Inertia. It maintains its own dedicated registration and login system, a separate MySQL-compatible database limited to training records, and role-based access control enforced at the route level for the Trainee, Supervisor, and Admin roles. Registration captures the name, rank, Police Regional Office, and unit or office requested

by the client on 22 July 2026 for certificate generation. The prototype is hosted on a cloud environment for testing purposes, with uploaded materials and generated certificates stored in a private object-storage bucket. What remains is the pilot testing called for in the objective; the platform has been demonstrated to the client but has not yet been exercised by a pilot group of PNP personnel.

Objective 2: Design and implement an interactive, multimedia-driven training interface across all seven curriculum modules. This objective has been partially achieved. The training interface was built around a structured, block-based lesson editor that supports headings, paragraphs, bulleted lists, uploaded videos, and YouTube embeds, alongside a legacy module-file upload path for PDF, PowerPoint, and MP4 materials with server-side file-type verification. Each lesson carries its own knowledge check, lessons unlock sequentially once the preceding quiz is passed, and progress is saved per lesson so that a trainee can leave and resume at lesson granularity. Administrators can author lessons directly or import text from existing PowerPoint files, which addresses the client's materials being designed for classroom delivery. Test cases were derived from the fully dressed use cases, but their formal execution and the documented 100% pass rate required by the objective have not yet been completed, and the seven modules cannot be fully populated until the curriculum is finalized under Objective 3. The client also advised during the prototype showcase that failing an initial review quiz should not block succeeding modules; the current gating rule will be revised accordingly.

Objective 3: Research and finalize a standardized 40-hour Program of Instruction divided into seven modules. This objective is in progress but not yet achieved. During the 17 September 2026 meeting, the client presented its Program of Instruction for the Seminar on Responsible Artificial Intelligence Utilization for PNP Personnel (SRAIU-PNP), structured as a five-day, 40-hour seminar, and committed to providing the updated POI and a project timeline template. The platform's course, module, and lesson structure is designed to accommodate this POI, but the adaptation of the client's materials into the seven-module format and the formal submission of the finalized curriculum structure for client validation remain outstanding.

Objective 4: Integrate an approved third-party AI API as a real-time virtual tutor. This objective has not yet been achieved. Consultations with the client progressively tightened the constraints on this component, from the initial prohibition of external services such as ChatGPT to the requirement that the AI model be hosted on PNP premises, and most recently to the client's suggestion of Meta's open-source Llama model for local hosting. The platform architecture reserves this function for a backend-mediated service that will transmit only approved course content and no personally identifiable information, consistent with user stories US3.1 to US3.3, but implementation was deferred until the hosting environment and model are confirmed by the client. The team has committed to completing the AI chatbot integration by February 2027.

Objective 5: Implement automated progress tracking and certificate generation. This objective has been largely achieved. Lesson progress and final exam attempts are recorded automatically. The final examination supports multiple-choice, true/false, matching, and identification questions, with a configurable time limit and an 80% default passing threshold in line with the assessment rules set by the client. When a trainee passes, the system immediately marks the enrollment complete, generates a certificate bearing a unique serial number, a training control number, and a SHA-256 verification hash, renders it as a PDF with a QR code, and stores it in private storage. Supervisors and the public can verify a certificate by scanning the QR code or

entering its serial number on a verification page, which satisfies user story US4.2. The supervisor monitoring dashboard and administrator analytics for completion by region, pass rates, and certificates issued are also in place. What remains is the adoption of the official certificate numbering format that the PNP office will provide, and full test-case coverage across all seven modules once the curriculum is loaded.

Beyond the stated objectives, this iteration delivered the architectural and security decisions documented in the Technical Background and Methodology chapters: a block-based content model that stores lesson content as structured data rather than raw HTML to prevent stored cross-site scripting by any compromised authoring account, immediate session termination when an account is locked, validation of YouTube embeds by video identifier only, and mitigations for known framework advisories affecting signed URLs, email header injection, and file validation. Two deviations from the original design should also be acknowledged. First, content management (US5.1 and US5.2) is currently restricted to the Admin role rather than the Supervisor role described in the use cases; the role assignment will be confirmed with the client before it is finalized. Second, the test deployment uses third-party cloud infrastructure, which conflicts with user story US4.4 if read literally; this was a deliberate choice for the testing phase, and migration to PNP-controlled infrastructure is planned for handover.

Plans for the Next Iteration (MCSPROJ)

The next course in the project-based learning sequence will focus on completing the remaining objectives and preparing the platform for pilot testing and handover, in coordination with the PNP ITMS Office. Specifically, the team will:

1. **Finalize the curriculum.** Adapt the client's SRAIU-PNP Program of Instruction into the seven-module structure, populate all modules with approved materials, and submit the finalized curriculum structure to the client for formal validation (Objective 3).
2. **Integrate the AI learning assistant.** Implement the backend-mediated virtual tutor using the locally hosted model agreed with the client, scoped strictly to the current module's approved content, and verify through test cases that no personally identifiable information is transmitted and that out-of-scope questions are declined (Objective 4).
3. **Apply the client's feedback from the prototype showcase.** Revise the module prerequisite rules to allow open, self-paced progress with certificate eligibility granted on completion of all modules and tests; add copy-paste restrictions and camera-based proctoring to complement the existing tab-switching detection during assessments; introduce administrator approval of registrations with PNP email checks and two-factor authentication; and apply the official certificate numbering format.
4. **Complete testing and pilot deployment.** Execute the documented test cases across all seven modules to the required 100% pass rate, configure real email delivery for account verification and password reset, and conduct pilot testing with a group of PNP personnel, using the results to drive a feedback and iteration cycle (Objectives 1, 2, and 5).
5. **Prepare for handover.** Migrate the platform and its database from the cloud testing environment to PNP-controlled on-premise infrastructure, finalize the user manual and source code appendices, and turn the system over to the PNP ITMS Office, which has committed on-site workspace and internal programming support, while leaving future integration with the POA to the PNP IT team as advised by the client.

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Appendices

Appendix A: Project Vision

The PNP Specialized Personnel Training (PNP-SPT) Platform envisions a future where every one of the Philippine National Police's 250,000 personnel is equipped with standardized AI literacy, not through logistically unsustainable classroom setups, but through a seamless, self-paced digital experience that meets them where they already are.

The platform is designed as an interoperable web application, built to match the existing Police Officer Academy (POA)'s technology stack so that, upon handover, the PNP IT Department retains the option to integrate or synchronize the two systems without requiring a full redevelopment. Personnel register and log in directly on the PNP-SPT Platform through its own dedicated account system, bringing them into a fully interactive 7-module AI curriculum with minimal friction, keeping the barrier to training as low as possible across all ranks and regions.

At its core, the PNP-SPT Platform replaces passive, static learning with a dynamic environment powered by an approved third-party AI assistant that guides each learner in real time. The AI assistant is restricted to approved educational materials, ensuring its responses remain within the scope of the current topic.

The end goal is an organization that can verify, track, and scale its technological upskilling automatically, where certificates are generated by the SPT Platform instantly, progress is saved seamlessly, and supervisors can monitor workforce readiness without a single manual submission. The PNP-SPT Platform is not just a training tool; it is the foundation for a digitally prepared and institutionally compliant police force.

Appendix B: Schedule

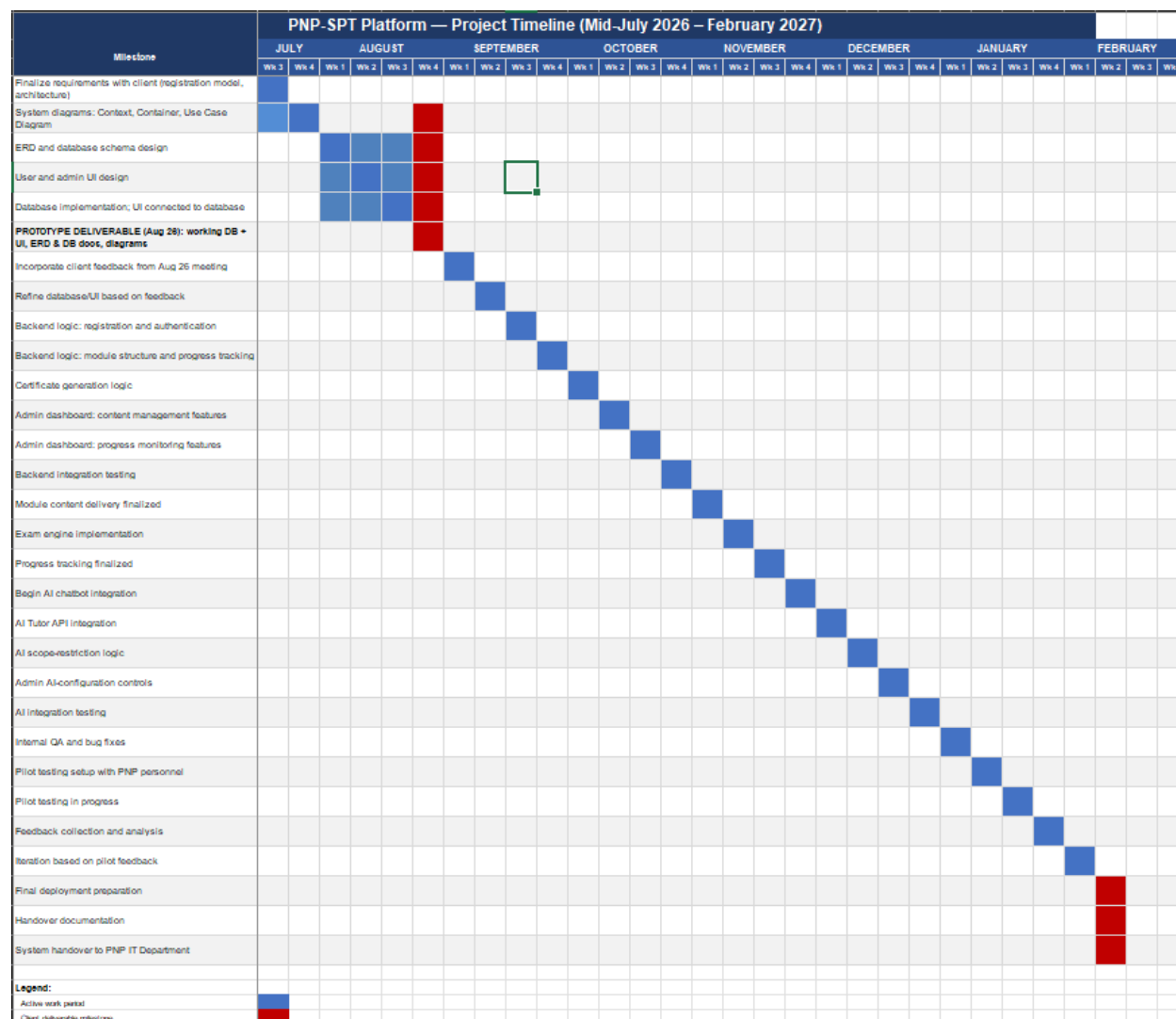


Figure 39: Project Schedule

Appendix C: Product Roadmap

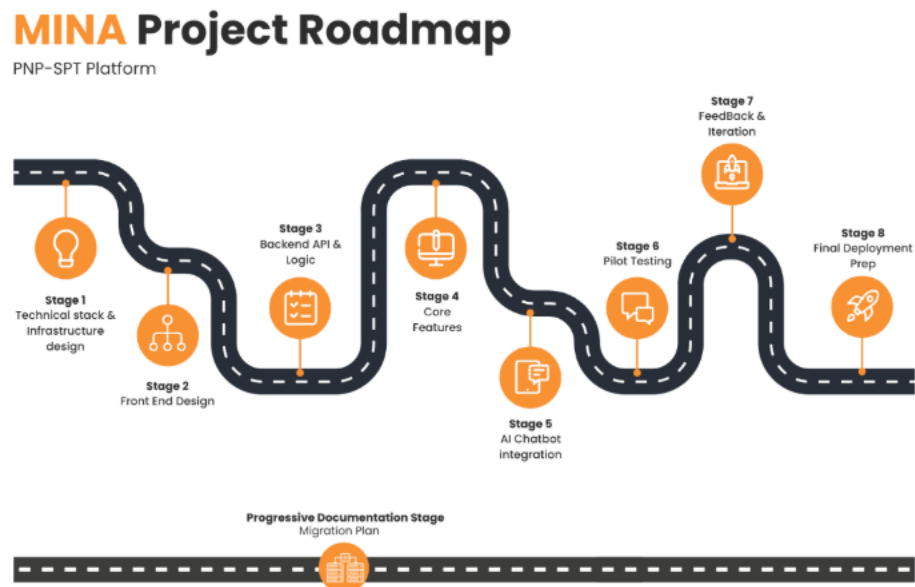


Figure 40: Product Roadmap

Appendix D: Team Meetings



Figure 41: MINA Online Meeting with Client

Meeting Title: PNP AI Training LMS Project - Initial Client Discussion

Date: March 11, 2026

Location: Online/Virtual Meeting

Attendees:

- **Client Representative:** Philippine National Police (PNP)
- **Project Team (APC Students):** Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To discuss the project proposal, specifications, and feasibility of developing an AI Training Learning Management System for the PNP.

Key Discussion Points:

1. **Project Scope Change:** The client proposed shifting from the initially discussed small scale membership system to a large scale Learning Management System (LMS) designed to train 250,000 PNP personnel on AI tools.
2. **Syllabus and Training Structure:** The PNP team is currently developing a syllabus that will be structured into multiple tiers. This includes Level 0/1 for Governance, Level 2 for Field Specific training across four operational quadrants, and Level 3 specifically for IT Developers.

3. **LMS Design Requirements:** The client requested that the new LMS replicate the interactivity and design of the UN mission training platform. Required features include self paced learning, topic time limits, per topic exams, and automatic certificate generation.
 4. **AI Integration and Data Security:** The client explicitly stated that external platforms like ChatGPT cannot be used due to critical data security concerns regarding sensitive police intelligence. The agreed approach is to integrate a locally trained, self hosted AI chatbot directly into the LMS.
 5. **Implementation Strategy:** The rollout will follow a phased approach, starting with a pilot program and training of trainers, followed by a full LMS implementation to cover approximately 80% of the workforce.
 6. **Technical and Resource Support:** The client confirmed that the PNP will provide budget support for the necessary storage and infrastructure to handle the massive user base.
-

Meeting Title: AI Training LMS Project: Advisor and Team Consultation

Date: March 13, 2026

Location: Asia Pacific College (On site)

Attendees:

Academic Adviser: Sir Jose Quesada

Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To review the client's project requirements, discuss technical feasibility, and secure formal advisory support for the AI Training Learning Management System project.

Key Discussion Points:

1. **Project Scope and Target Audience:** The project involves developing an AI literacy and certification LMS for approximately 250,000 users. The program is structured into five tiers: Level 1 for general AI awareness, Levels 2 and 3 for advanced users, and Levels 4 and 5 for specialized IT personnel. The student team will focus on implementing modules 2 through 5.
2. **Platform Modernization:** The client's current setup is basic, relying heavily on standard PDFs and PowerPoint presentations. The team proposed building the new platform on an Open Learning foundation (such as Moodle) to modernize the user experience, integrating features like AI powered Q&A and automated practice test generation based directly on the syllabus.
3. **Infrastructure and Security Restrictions:** Due to strict data privacy requirements, the client explicitly prohibits the use of external cloud services or subscriptions (e.g., Gemini or ChatGPT). The LMS and AI tools must be hosted on a local, on premise server. The client has committed to fully funding and providing the necessary hardware and IT infrastructure for this setup.

4. **Development and Testing Strategy:** The team will utilize open source tools for the initial development and small scale testing phases. Once the core platform is delivered, the client's internal IT department will assume responsibility for stress testing the system to ensure it can handle the full 250,000 user load.
5. **Academic Timeline and Recognition:** The project will span one full academic year across three terms. Upon successful completion, the client will provide formal recognition and certification to the student team. The academic advisor formally agreed to mentor the group for the duration of the project.

Meeting Title: PNP-LMS: A Learning Management System for Philippine National Police Personnel On Site Consultation

Date: April 10, 2026

Location: Asia Pacific College (On site)

Attendees:

Academic Adviser: Sir Jose Quesada

Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To define the project's scope and limitations, clarify the technical architecture, and establish strict data security requirements for the proposed learning management system.

Key Discussion Points:

1. **Core Project Objectives:** The primary goal is to address the operational training gap by transitioning 250,000 PNP personnel from traditional classroom based training to a secure, interactive digital platform.
2. **Strict On Premise Architecture:** To guarantee data security, it was established that the entire system, not just the localized AI chatbot, must be hosted on premise. The LMS, AI model, and all associated databases must reside on physical servers within the Philippines to comply with government data residency regulations.
3. **Security and Compliance:** The project cannot utilize any external or third party cloud services (such as ChatGPT) due to the risk of exposing sensitive intelligence. The system must strictly adhere to Philippine data privacy and security laws, which includes operating under the oversight of a designated data protection officer.
4. **Scope and Known Limitations:** The target audience is strictly limited to PNP personnel. The project scope focuses solely on delivering the LMS and the integrated chatbot. A major technical limitation identified is that the system's scalability and final capacity will be entirely dependent on the physical hardware, GPUs, and IT resources the client can provide.
5. **Development and Scaling Strategy:** The team discussed leveraging adaptable open source LMS frameworks to avoid building from scratch. For the AI component, Small Language Models (SLMs) will be investigated as a viable local alternative to massive

commercial models. The architecture will need to support dynamic scalability, potentially utilizing tools like Kubernetes.



Figure 42: MINA Onsite Meeting with Client

Meeting Title: PNP-LMS: A Learning Management System for Philippine National Police Personnel On Site Client Discussion

Date: May 11, 2026

Location: PNP National Headquarters, ITMS Office

Attendees:

- **Client Representative:** Philippine National Police (PNP)
- **Project Team (APC Students):** Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Client Representatives: PNP Officer, Carolina (ITMS Programmer), and PNP ITMS Staff

Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To evaluate the feasibility of the LMS proposal, review existing PNP IT infrastructure, and address data privacy, security, and system integration challenges.

Key Discussion Points:

1. **Academic Timeline and Client Background:** The student team has two weeks before their final defense. The client shared insights from their UN deployment in South Sudan and publications on cross cultural challenges and blockchain. The client encouraged the team to reference the PNP GPT master plan proposal in their academic paper.
2. **Existing PNP Infrastructure:** The PNP currently operates the Police Officer Academy (POA) platform, which already serves over 200,000 personnel. This existing system successfully handles enrollment and exams, and it recently integrated anti cheating features such as camera monitoring and screen navigation tracking.
3. **Technical and Policy Constraints:** Developing a duplicate standalone LMS is discouraged by ICT Management (DICTM) policies. Furthermore, housing sensitive PNP personnel data on external databases (such as a school provided server) raises severe concerns with the Data Privacy Commission (PCC).
4. **Integration and Authentication Challenges:** Connecting a new system to the existing POA via API is complex due to strict data sharing limitations. Additionally, the PNP no longer issues domain names to external systems, complicating user authentication and log in procedures.
5. **Proposed Alternative Solutions:** The most viable option is to integrate the course content directly into the existing POA platform. While this restricts UI customization (the team originally wanted to replicate the UN training platform interface), the project team could potentially pivot to developing specific add on features, such as the localized chatbot, without duplicating core functions.
6. **Support and Workspace Allocation:** The PNP offered the students an on site workspace at the ITMS office with internet access. They are also willing to assign one or two internal programmers to provide development guidance and direction.



Figure 43: MINA Onsite Meeting with Client

Meeting Title: PNP LMS Project - Client Discussion & Requirements Alignment

Date: July 22, 2026

Location: ITMS, Camp Crame, Quezon City

Attendees:

- Client Representative: Philippine National Police (PNP) & PNP IT
- Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To present the updated user interface (UI) to the PNP and PNP IT team onsite at Camp Crame ITMS, discuss system features, and align on administrative controls and assessment rules.

Key Discussion Points:

1. **UI Presentation & Feedback:** The team presented the updated user interface (UI) to the PNP and PNP IT teams for review and feedback.
 2. **Deliverables Requested by Client:** The client asked the team to send an editable copy of the final paper and the presentation slides via Viber.
 3. **Login & Authentication Fields:** The client requested login credentials including Name, Email, Password, and a 2-layer authentication process specifying the user's unit or office (Region, PRO, PPO, Police Station) for certificate generation.
 4. **Admin Controls & Module Management:** The admin account must have full control to add, update, delete, and modify time settings for course materials (PDFs, PowerPoint slides, and videos).
 5. **Assessment & Passing Rules:** The system will include 1 to 3 questions after every module, followed by a final 80-question course exam with an 80% passing grade.
 6. **Certificate Numbering:** The PNP office will provide the official certifying numbering format to be placed on generated completion certificates.
 7. **Learning Delivery Mode Alignment:** Discussion focused on the delivery mode, where the PNP Viber contact confirmed an online self-paced format, while Miss Damaso suggested balancing self-paced learning with face-to-face sessions and phased module rollouts.
 8. **Integration Guidance:** PNP representatives suggested keeping login credentials in the system and letting the POWA team handle future system integration.
 9. **Action Items for Next Meeting:** The team must prepare user/admin diagram flows, the admin UI, module content structure, a backend code review, and a project timeline.
-

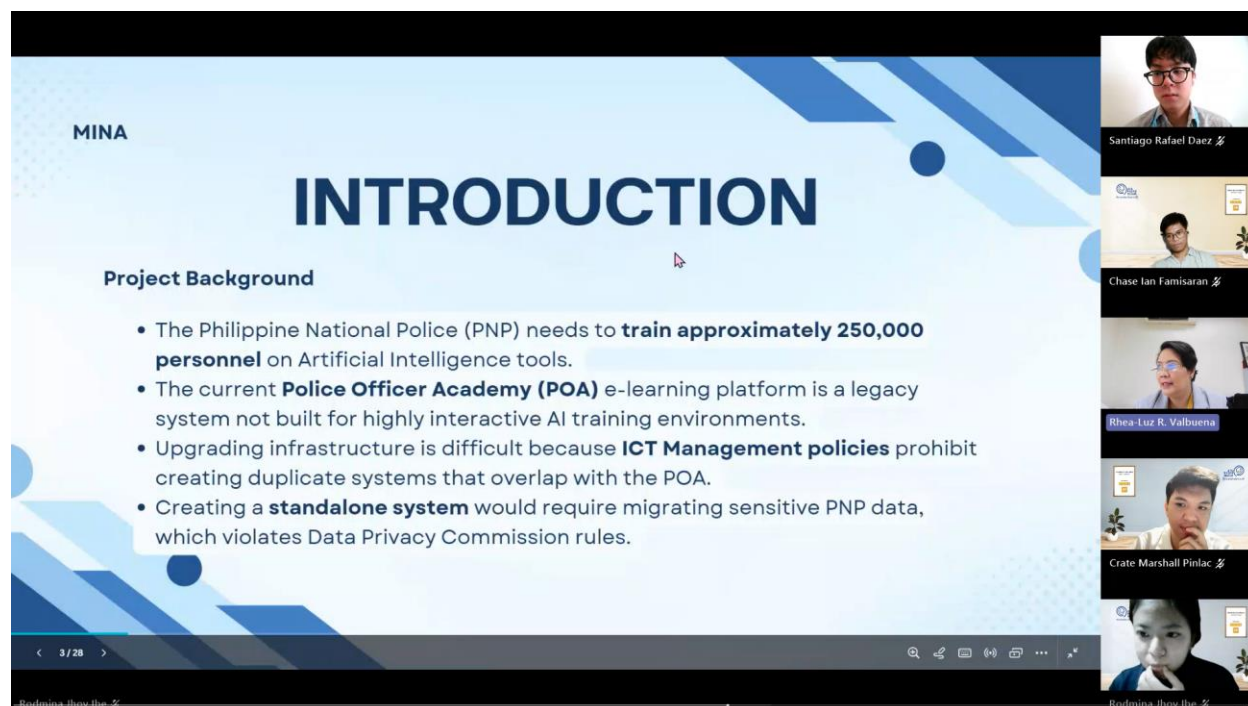


Figure 44: MINA Online Meeting with Advisor

Meeting Title: PNP LMS Project - Initial Advisement & Scope Alignment

Date: July 24, 2026

Location: Online/Virtual Meeting

Attendees:

- **Academic Adviser:** Ms. Rhea-Luz R. Valbuena
- **Project Team (APC Students):** Chase Ian Famisan, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To hold the initial advisement meeting with Ms. Rhea-Luz R. Valbuena, review project background, and establish foundational expectations for the MSYADD1.

Key Discussion Points:

1. **Initial Advisement Session:** This session marked the first official project advisement meeting with Ms. Rhea for the MSYADD1 term to align on project expectations and progress tracking.
2. **Review of Previous InStep Output:** The team presented an overview of the work done during InStep to provide context on the current system concept and client requirements.

3. **Identifying Gaps in Understanding:** Discussion revealed ambiguities regarding how the LMS connects to PNP's existing platform POA, highlighting the need to clarify system boundaries and data flow.
4. **Clarification on Data Privacy & Mock Data:** Ms. Rhea emphasized that developers must never handle live client data, setting the guideline that all development and testing must use simulated/mock data.
5. **Initial Guidance on System Diagrams:** The adviser highlighted the importance of accurate technical documentation and instructed the team to prepare draft high-level diagrams (Context Diagram and Use Case Diagram) for future review.
6. **Establishment of Regular Consultation Schedule:** The team and adviser agreed on holding regular weekly consultations (Fridays/Wednesdays) to ensure continuous monitoring and proper alignment before client presentations.

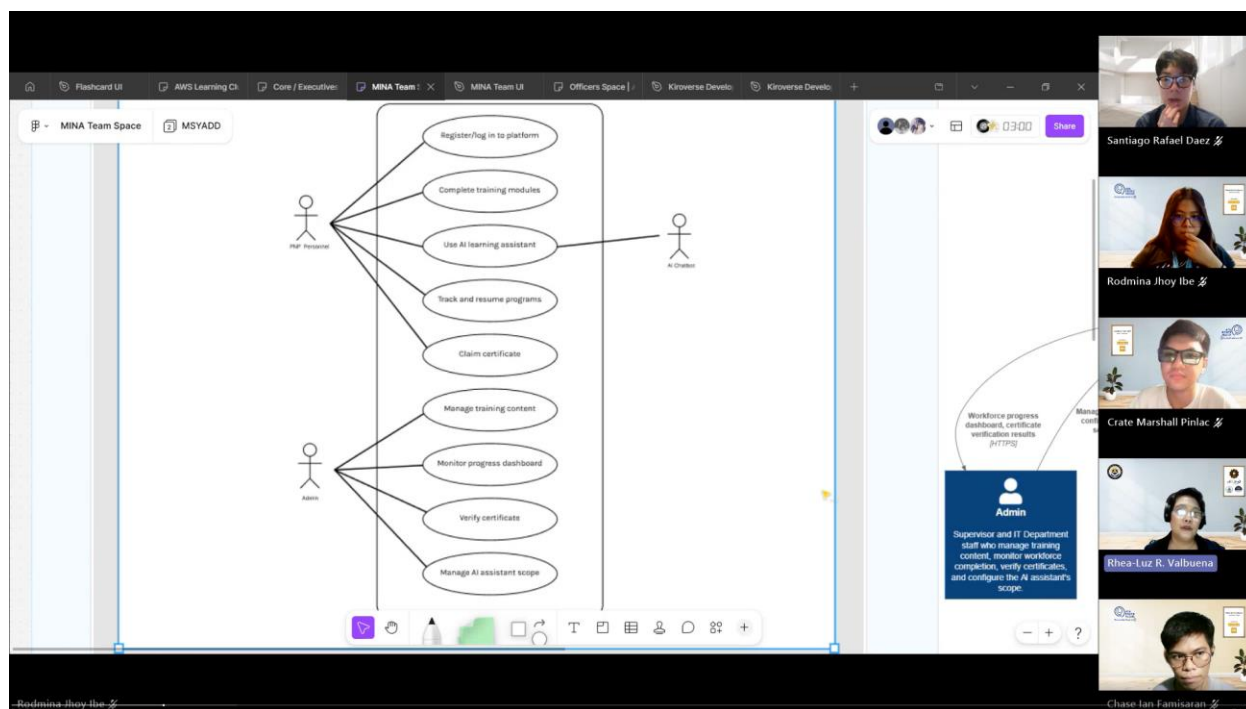


Figure 45: MINA Online Meeting with Advisor

Meeting Title: PNP LMS Project - System Diagrams & Use Case Review

Date: July 29, 2026

Location: Online/Virtual Meeting

Attendees:

- Academic Adviser: Ms. Rhea-Luz R. Valbuena
- Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To review the updated Use Case Diagram, Context Diagram, and Container Diagram, clear up actor roles, and get guidance on use case descriptions.

Key Discussion Points:

1. **Use Case Diagram Revisions:** The team presented a simplified Use Case Diagram created after the earlier consultation to align with MSYADD1 requirements.
2. **Actor Name Changes:** Ms. Rhea suggested changing the actor name from "PNP Personnel" to "PNP Trainees" for better clarity regarding learner roles.
3. **Consolidating Admin Roles:** The team combined supervisor, program owner, and IT staff responsibilities under one consolidated "Admin" actor for simplicity.
4. **Fixing Incorrect Actors:** Ms. Rhea pointed out that the database cannot be an external actor because it sits inside the system boundary. The external AI chatbot remains an external actor since it connects from outside.
5. **Clearing Up Diagram Types:** Ms. Rhea explained the differences between diagram types: Use Case Diagrams show main business processes, Context Diagrams show external users and connected systems, and Component/Container Diagrams detail internal modules and APIs.
6. **Use Case Descriptions & Team Discussion:** The team was instructed to write fully dressed use case descriptions (basic flows and alternative flows) with consistent naming. Ms. Rhea stressed that all members must understand every use case to answer panel questions effectively during presentations.

Meeting Title: PNP LMS Project - Onsite Use Case Review & Faculty Approval

Date: August 3, 2026

Location: Onsite Meeting

Attendees:

- Academic Adviser: Ms. Rhea-Luz R. Valbuena
- Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To review the updated Use Case Diagram of the Specialized Personnel Training Platform (SPT-Platform) onsite following prior approval from Sir Sean.

Key Discussion Points:

1. Onsite Advisement Consultation: The team met with Ms. Rhea in Room 206 after her class to get her opinion and feedback on the revised Use Case Diagram for the SPT-Platform.
2. Faculty Approval Status: The team informed Ms. Rhea that the updated Use Case Diagram had already been reviewed and approved by Sir Sean earlier that day.
3. Review of Diagram Actors and Processes: The team showed the updated diagram featuring PNP Trainees, Supervisor, Admin, and the external AI Chatbot connected to their respective system processes.
4. Client Verification Requirement: Ms. Rhea advised that even though the diagram was approved by faculty, the team must also show and confirm this updated Use Case Diagram with the client to make sure it fits their operational expectations.

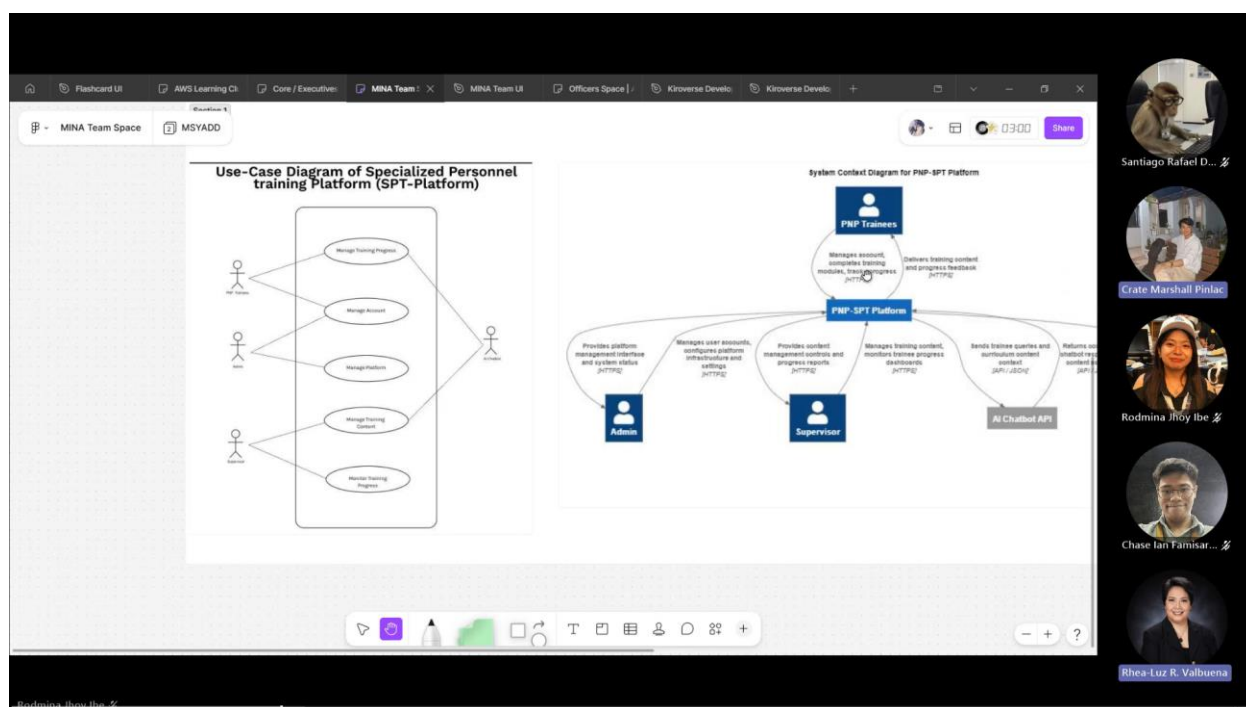


Figure 46: MINA Online Meeting with Advisor

Meeting Title: PNP LMS Project: Use Case Diagram & Full Description Review

Date: August 14, 2026

Location: Online/Virtual Meeting

Attendees:

- Academic Adviser: Ms. Rhea-Luz R. Valbuena

- Project Team (APC Students): Chase Ian Famisaran, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: To review the updated Use Case Diagram and fully dressed use case descriptions, clarify actor roles and process names, and set the schedule for mock presentation practice.

Key Discussion Points:

1. Use Case Diagram Updates: The team presented the revised diagram and separated the actual learning process ("Learn Module") from tracking ("Monitor Progress").
2. Role Differences in Progress Tracking: Ms. Rhea explained that tracking progress is one business process shared by both PNP Trainees (viewing their own progress) and Supervisors (viewing all trainees' progress), differing only by user access rights.
3. Fully Dressed Use Case Format: The adviser required a full description for each use case, including the name, trigger, basic steps, alternative steps, actors, and input/output data.
4. Learning and Assessment Steps: The team agreed that accessing learning modules and taking quizzes/assessments should be structured as connected steps within the learning flow.
5. Content Management and AI Actor: The "Manage Training Content" process links the Supervisor and the external AI Chatbot to handle course module creation, updates, and reviews.
6. Admin Account Management: The team discussed adding account management controls (lock, disable, and reinstate accounts) for the IT Admin role alongside login and registration.
7. Full Document Compilation: Ms. Rhea asked the team to submit the complete, compiled document (paper, diagrams, and descriptions) together for review instead of sending separate parts.

Presentation Practice Schedule: The team scheduled mock presentation practice with Ms. Rhea for Friday, August 21 at 10:00 AM to prepare for the midterm presentation on August 24.

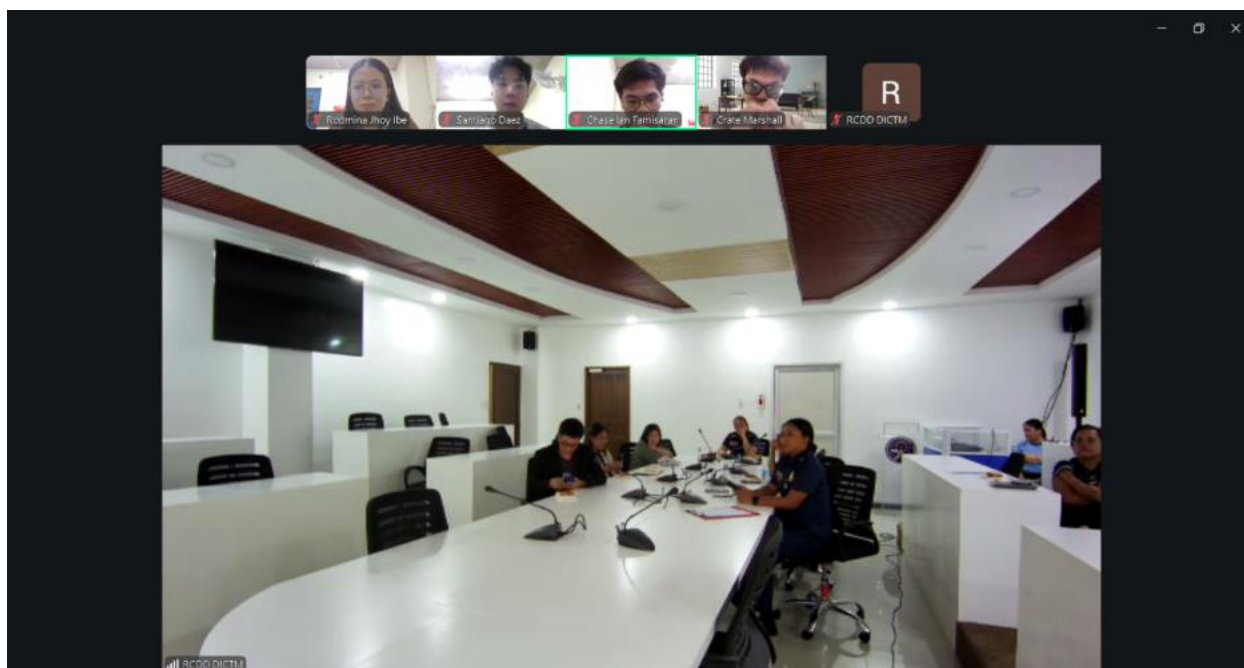


Figure 47: MINA Online Meeting with Client

Meeting Title: PNP LMS Project: Prototype Showcase & Seminar Development Updates

Date: September 17, 2026

Location: Online/Virtual Meeting

Attendees:

Client Representatives (DICTM / TWG):

- Presiding: PCOL Maribel M. Damaso (C, RCDD / Head)
- PLTCOL Jhoanna Gracia M. Fabro (Asst. Head)
- PLTCOL Teodoro B. Jose (Head Secretariat)
- PCPT Veronica Q. Tajon (Asst. Secretariat)
- PCPT Rogel Vergel Cabatin, Jr. (Member)
- PCPT Cristina Joy V. Cruz (Member)
- Pat Avegail C. Odero (Member)
- NUP Ma Goudelia M. Bustarde (Member)
- NUP Ronalyn S. Balongoy (Member)
- NUP Eric Navarro (Member)
- Project Team (APC Students):

Chase Ian Famisanan, Crate Marshall Pinlac, Rodmina Jhoy Ibe, Santiago Rafael Daez

Meeting Objective: Updates on the Development of the Seminar on Responsible Artificial Intelligence (AI) Utilization for the PNP Personnel (SRAIU-PNP) and prototype demonstration.

Key Discussion Points:

- **Seminar Scope & Delivery:** The client presented the Program of Instruction (POI) for the Seminar on Responsible Artificial Intelligence Utilization (SRAIU) for PNP personnel, structured as a 5-day seminar totaling 40 hours.
- **Functional Prototype Demonstration:** The project team presented the working prototype hosted on Render, showing user flows for trainees (accessing modules, embedded videos, and knowledge checks) and administrative tools (creating lessons, exam builders, progress monitoring, and account controls).
- **Module Access & Prerequisite Rules:** The client advised against blocking succeeding modules if a trainee fails an initial review quiz. They recommended allowing open or self-paced progress across modules, with certificate eligibility granted once all modules and tests are finished.
- **Assessment Security & Exam Proctoring:** The client recommended adding camera proctoring during exams and copy-paste restrictions to prevent cheating. The team confirmed that the system currently tracks and flags browser tab switching during assessments.
- **Real-Time Progress Saving:** The team confirmed that user answers are saved continuously so that trainees who experience connectivity drops can resume the test from where they stopped.
- **Verifiable Completion Certificates:** The client required verifiable completion certificates following industry standards. The team noted that certificates currently download as PDFs from AWS S3 and will soon feature a unique authentication code and scannable QR/barcode.
- **AI Chatbot Model Selection:** The team explained that the AI chatbot will be limited strictly to training topics. The client suggested using Meta's open-source Llama model for local on-premise hosting.
- **File Upload Security (FUDS Principle):** The team and client aligned on Functional, User-Friendly, Design, and Security (FUDS) principles, setting strict file upload controls to permit only safe document types (such as PDF and PowerPoint) while blocking dangerous formats like scripts, executables, and zip archives.
- **Trainee Account Management:** To avoid non-PNP personnel accessing the portal, the client and team discussed having admin accounts manage registrations or enforce admin approvals and PNP email checks, supported by Google Authenticator for login security.
- **Timeline Alignment:** The client will provide the updated POI and project timeline template. The team confirmed their commitment through February to complete the AI chatbot integration and finalize the system.

Appendix E: Source Code

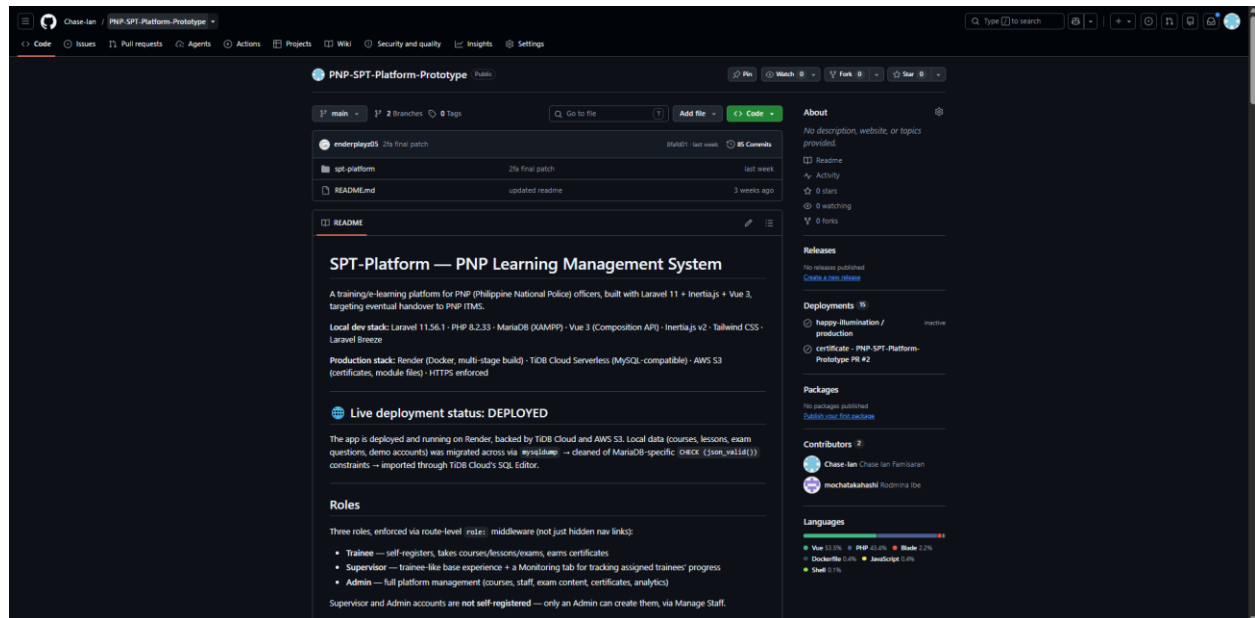


Figure 48: PNP-SPT Platform Prototype Source Code Repository

<https://github.com/Chase-Ian/PNP-SPT-Platform-Prototype.git>

Appendix F: Finalized Design Thinking Deliverables

Say • "No instructor to help when I don't understand the lesson" • "I barely have time to read long PowerPoint slides" • "I just need to pass this module and get certified" • "We're not allowed to put intelligence data into ChatGPT"	Think • "I hope this isn't just another boring seminar" • "The examples should reflect real PNP scenarios" • "Completing this could help my promotion" • "I need my progress saved in case I get called away"	Does • Balances demanding fieldwork with mandatory training • Skims/rushes through unengaging "flat" materials • Trains during odd hours, self-paced by necessity • Uses jargon/procedures specific to their unit	Feels • Relieved when a built-in chatbot is there to ask • Lost without support to figure out confusing parts • Frustrated that no one explains the hard parts • Bored by flat, traditional presentation
See • A 250,000-PNP Trainees with no 1-on-1 instructor time • Flat, text-heavy Word documents and PPT slides for training • Directives from the Chief of PNP mandating AI training	Hear • Warnings about data security risks • Warnings about leaking intelligence data on public AI apps • Colleagues struggling with tech topics without a teacher • Training is mandatory and affects their record	Pains • No instructor available for confusing AI jargon • Hard to fit reading in amid high-stress field duties • Struggling to stay focused on boring, non-interactive docs • Fear of exposing sensitive data while using AI tools	Gains • 24/7 built-in chatbot as a personal tutor • Fully self-paced system that saves progress automatically • Dynamic, engaging platform vs. flat documents • Instant certificate upon passing – useful for promotion

Figure 49: Empathy Map

Activity Diagram

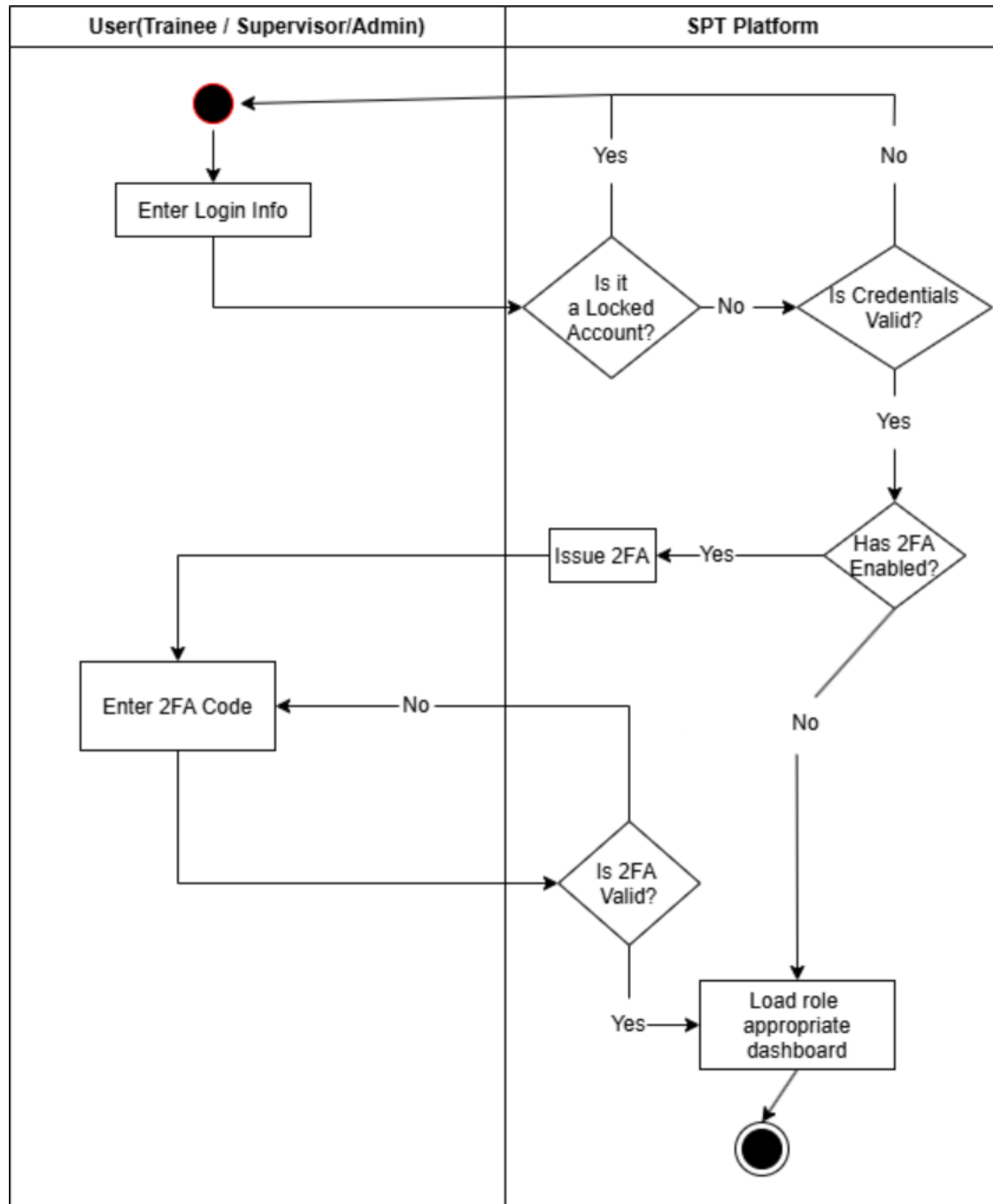


Figure 50: Activity Diagram – Manage Account: Logging into spt-platform

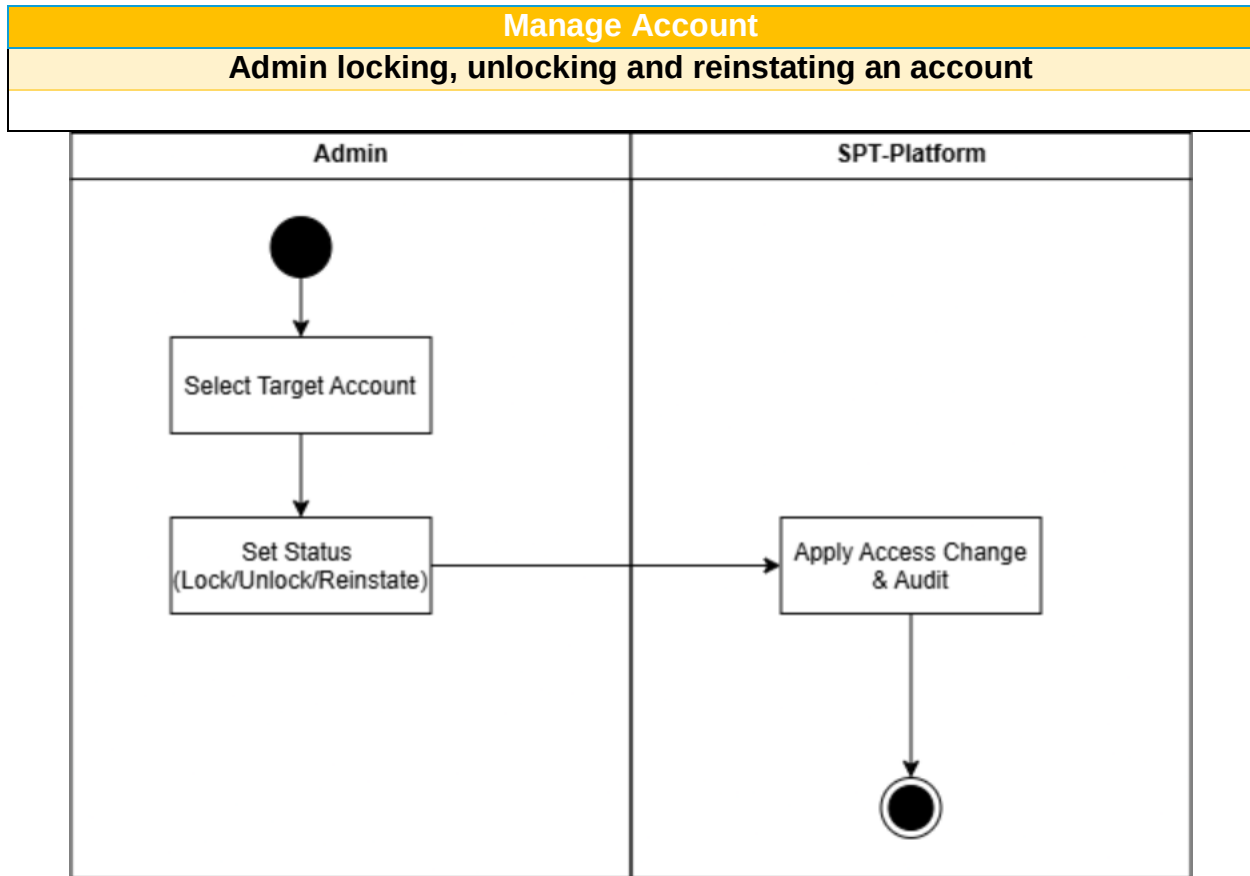


Figure 51: Activity Diagram – Manage Account: Admin locking, unlocking and reinstating an account

Manage Account
User updating account information

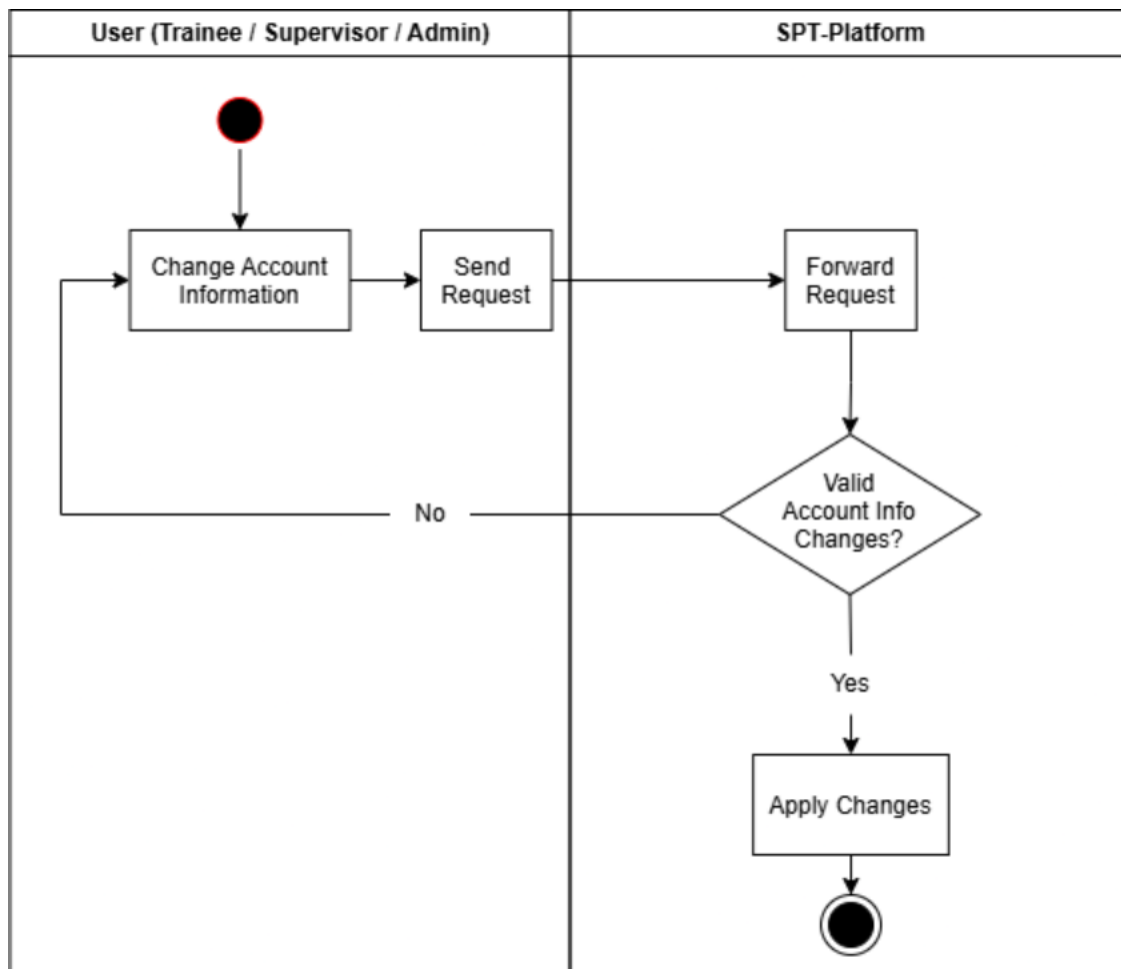


Figure 52: Activity Diagram – Manage Account: User updating account information

Manage Account

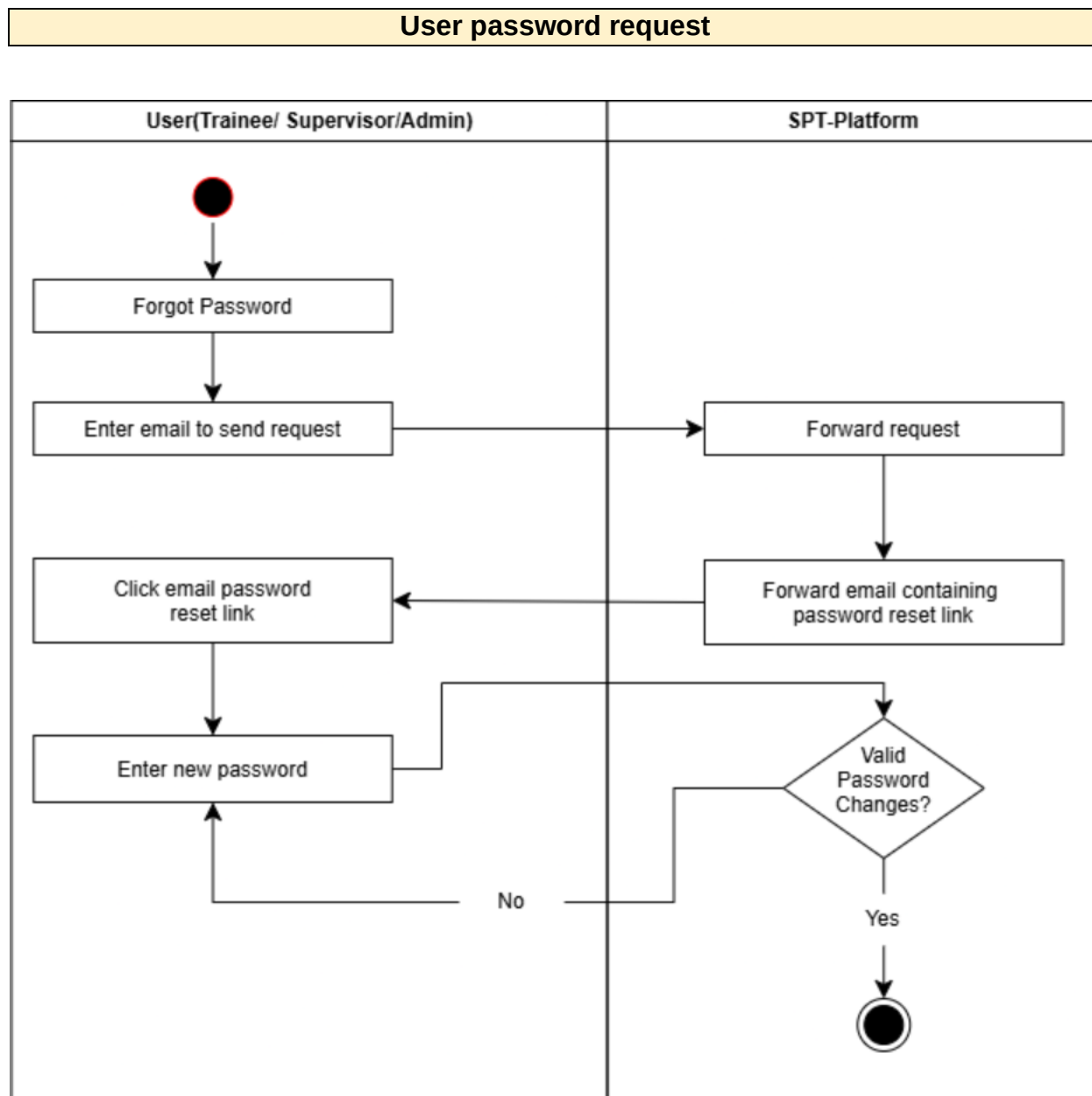


Figure 53: Activity Diagram – Manage Account: User password request

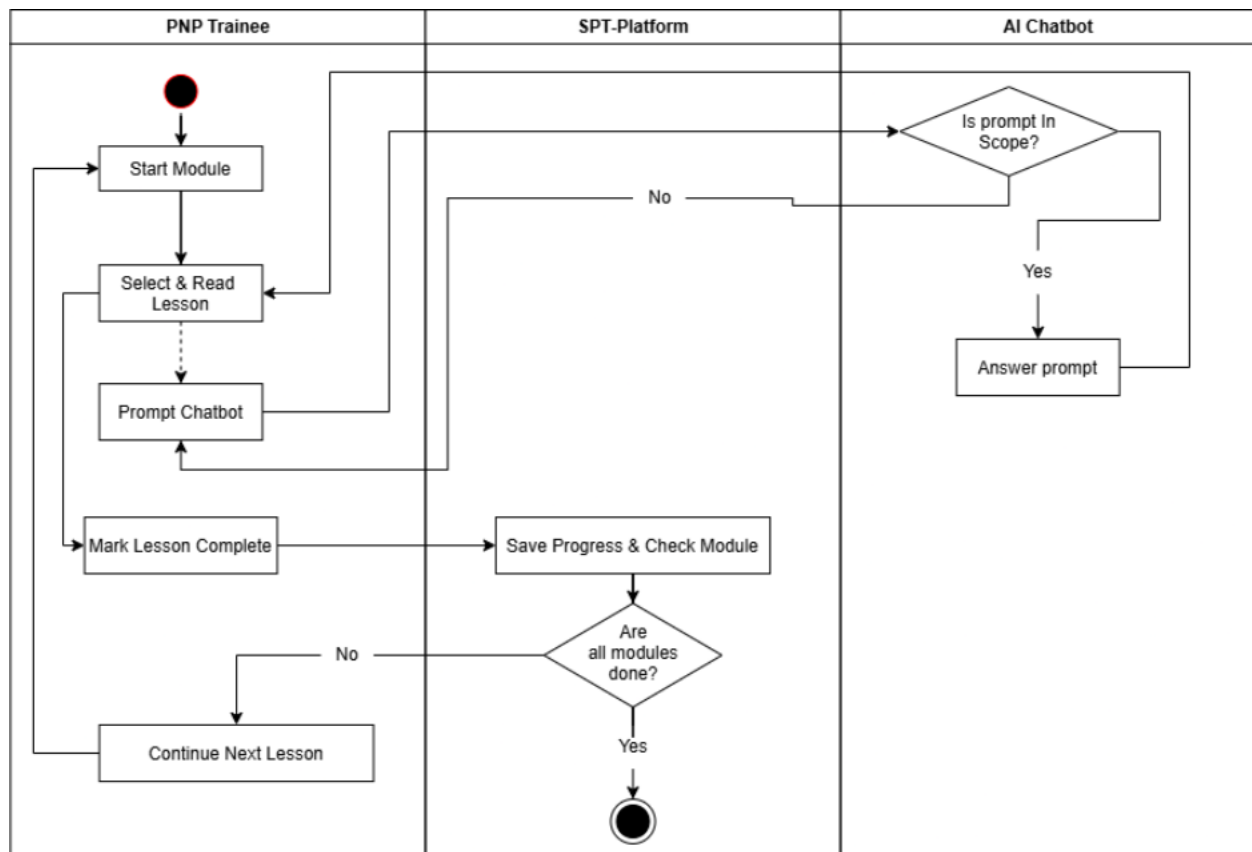


Figure 54: Activity Diagram – Learn Module

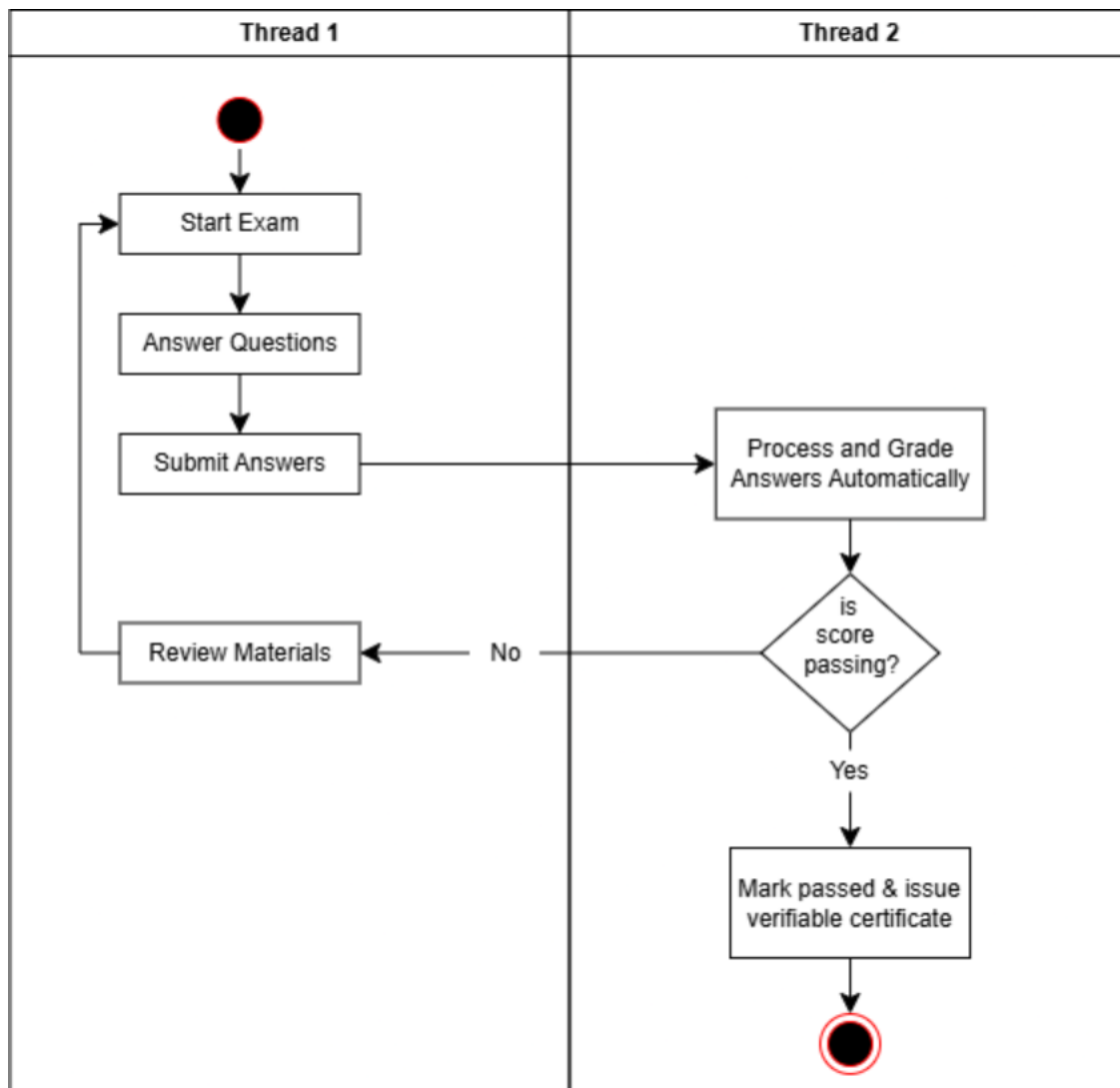


Figure 55: Activity Diagram – Take Assessment

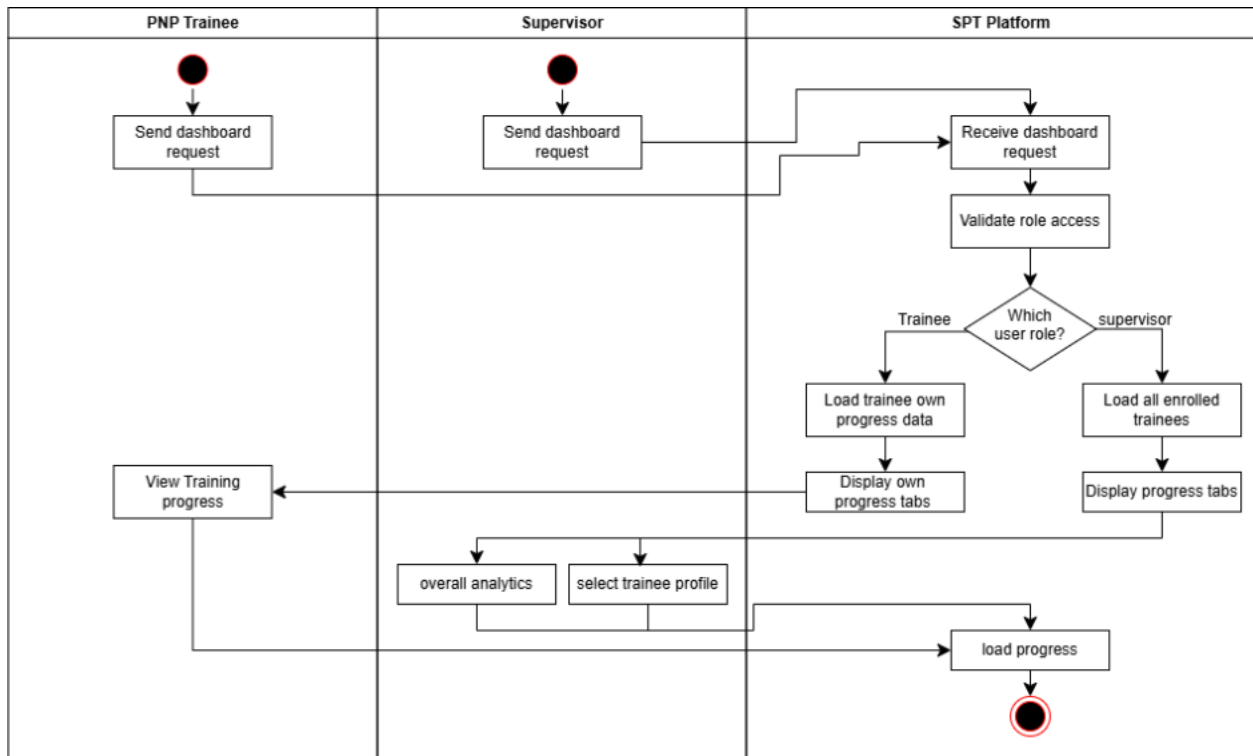


Figure 56: Activity Diagram – Monitor Progress

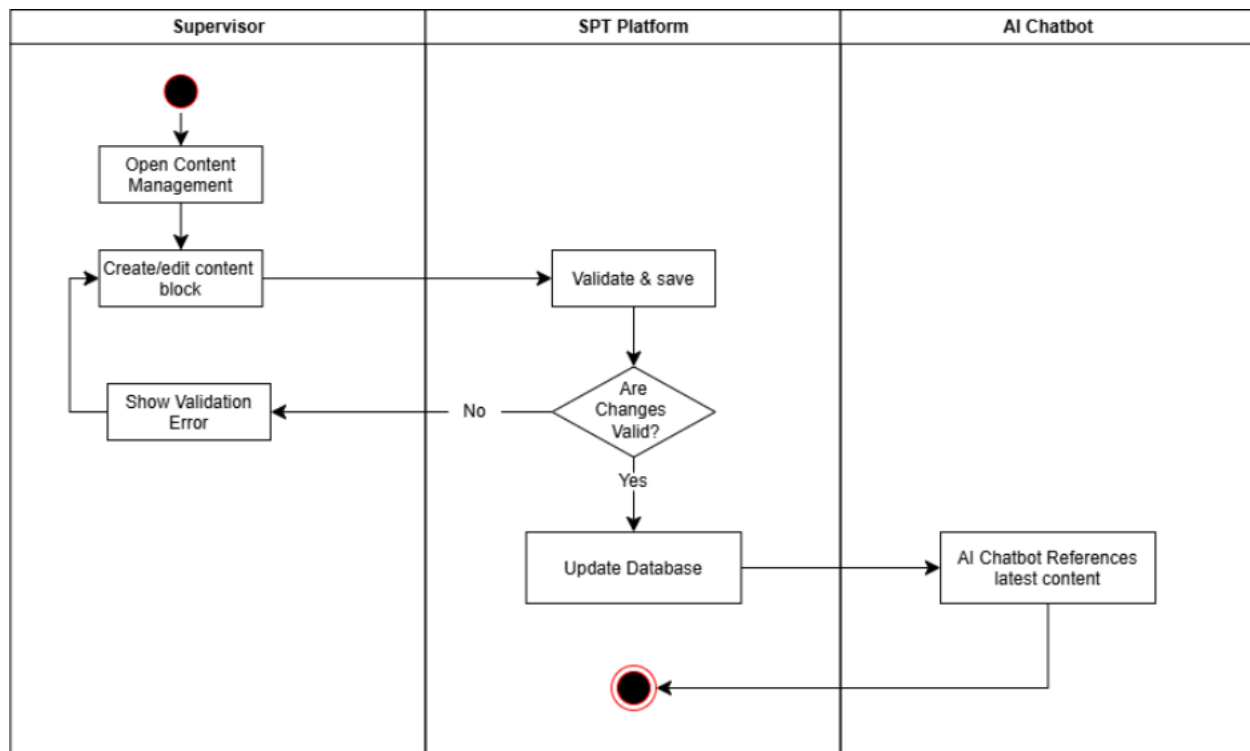


Figure 57: Activity Diagram – Manage Training Content

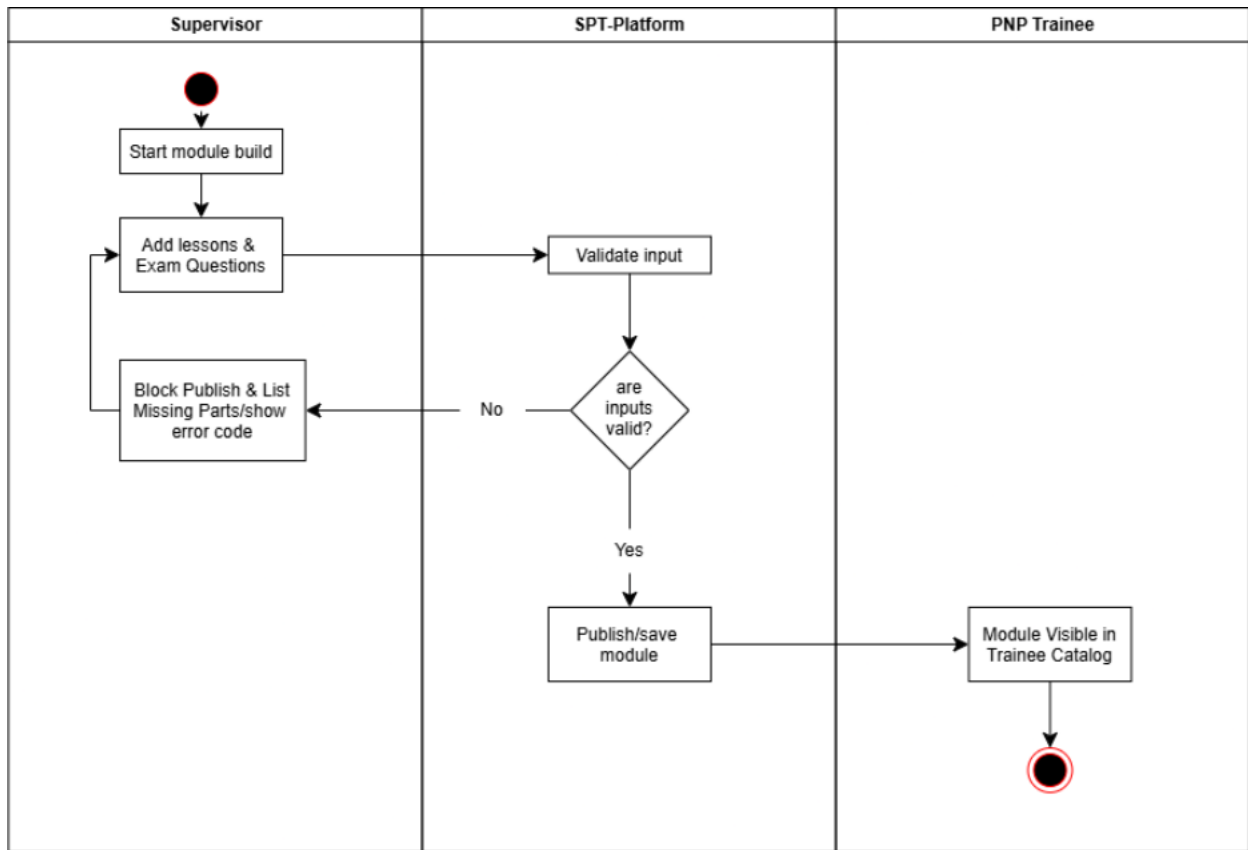


Figure 58: Activity Diagram – Create / Manage Modules

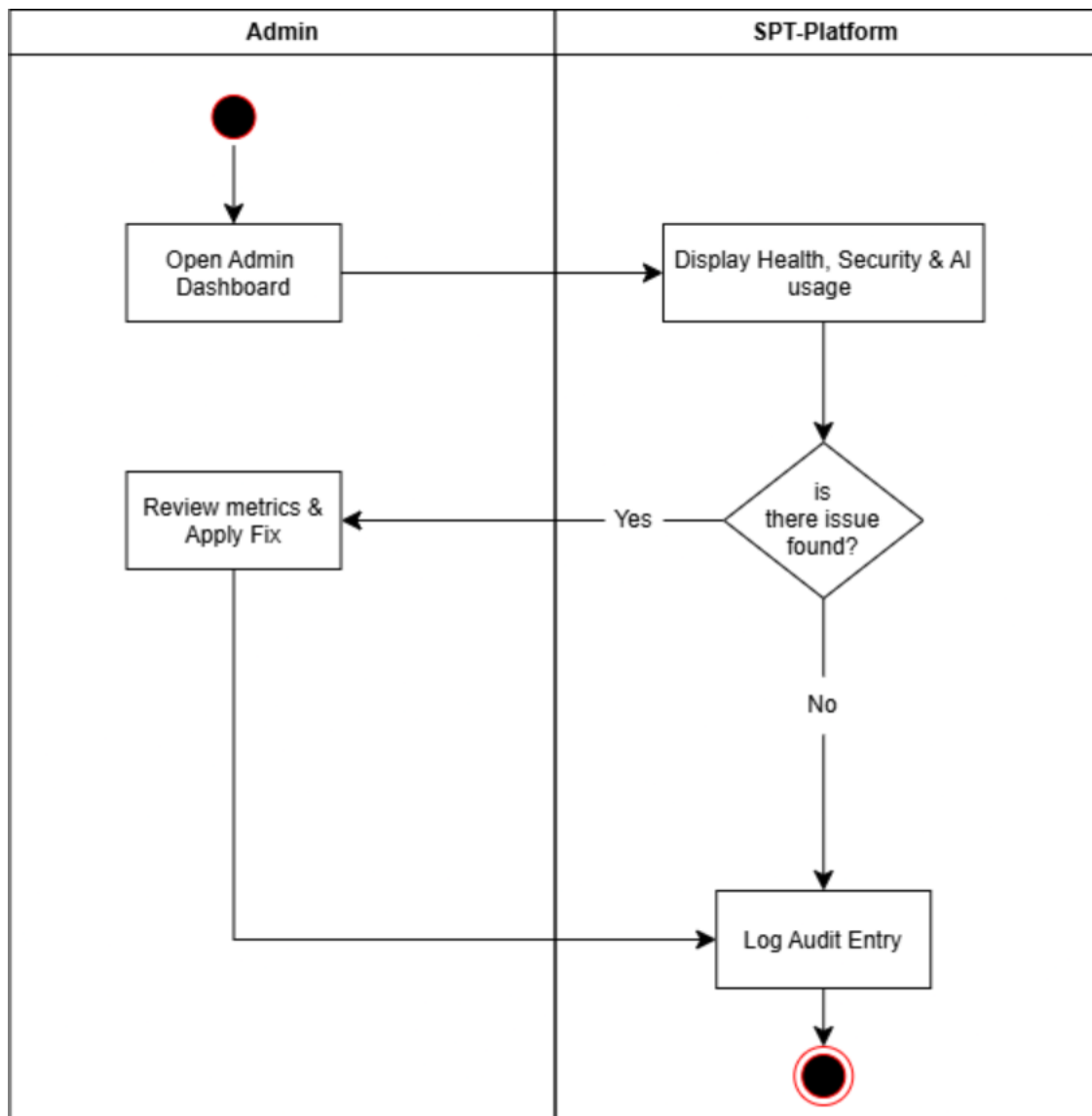


Figure 59: Activity Diagram – Manage Platform